Metodología y Estado Actual del Proyecto

Pasantías 2025-2026

Glosario de Términos

- **Evento:** Ventana temporal de aproximadamente 26 observaciones, centrada en un episodio de volatilidad significativa del WTI. Recopila el comportamiento conjunto de mercados de valores, divisas y materias primas antes y después de dicho choque de volatilidad.
- **Mercado de Divisas (Forex)**
  - **DEXJPUS / DEXCHUS / DEXHKUS / DEXSZUS:** Representan la cantidad de moneda extranjera (Yen japonés, Yuan chino, Dólar de HK y Franco suizo) necesaria para comprar 1 Dólar estadounidense.
  - **DEXUSEU / DEXUSUK:** Representan el valor inverso; cuántos Dólares estadounidenses se necesitan para comprar 1 Euro o 1 Libra esterlina.

(Federal Reserve Bank of St. Louis, s.f.)

- **Materias Primas y Energía (Commodities)**
  - **WTI, Brent y Dubai Crude:** Los tres principales "benchmarks" o precios de referencia del petróleo a nivel mundial (EE. UU., Europa/Global y Asia).
  - **Oro al contado Dólar onza troy:** Precio por onza troy para entrega inmediata; actúa como el principal activo de refugio frente a la inflación y crisis.

(EIA / LBMA, s.f.).

- **Índices Bursátiles y Volatilidad**
  - **S&P 500 / Nikkei 225 / STOXX 50:** Índices que representan el rendimiento de las mayores empresas de EE. UU., Japón y la Eurozona, respectivamente.
  - **VIX / Nikkei Vol / VSTOXX:** Índices que miden la "volatilidad implícita" o el miedo del mercado. A mayor valor, mayor incertidumbre y riesgo percibido por los inversores.

(S&P Dow Jones Indices / CBOE / Japan Exchange Group.s.f.).

- **Indicadores Macroeconómicos y Política Monetaria**
  - **Effective Federal Funds Rate:** La tasa de interés nominal a la que los bancos comerciales de EE. UU. se prestan dinero entre sí.
  - **Tasa de Interés Euro Area:**Tasa oficial fijada por el BCE
  - **Índice de Sentimiento (U. Michigan):**Encuesta que mide qué tan optimistas son los consumidores sobre su situación financiera y la economía general de EE. UU.
  - **Tasa de Interés Euro Area:**Tasa oficial fijada por el BCE

(Federal Reserve Board / Banco Central Europeo (BCE) / University of Michigan.s.f.).

Metodología

- Selección de Eventos de Volatilidad en WTI

Para la identificación de eventos se utiliza el precio del WTI como indicador de referencia, dada su relevancia como variable energética clave en la economía global.

1. Se calcula el cambio porcentual diario del precio del WTI:

$$\text{pct_change}_t = \frac{WTI_t - WTI_{t-1}}{WTI_{t-1}}$$

2. Se toma el valor absoluto del cambio porcentual para capturar la magnitud del movimiento, independientemente de su dirección.
3. Un día se clasifica como día de alta volatilidad si el cambio porcentual absoluto supera el 4% diario.

4. Los días de alta volatilidad se agrupan en episodios, y para cada episodio se selecciona como evento representativo el día con el mayor movimiento porcentual absoluto.

5. Para cada evento se construye una ventana de análisis de 21 observaciones, compuesta por 10 observaciones previas, el día del evento, y 10 observaciones posteriores.

- Normalización

Se aplicó Min–Max normalization a cada variable dentro de la ventana del evento.

Esta normalización reescala los valores al intervalo [0,1], donde 0 corresponde al valor mínimo y 1 al valor máximo observados en la ventana, facilitando la comparación visual entre variables con distintas unidades.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Wikipedia. (s. f.). Feature scaling. En Wikipedia, The Free Encyclopedia.

Fórmula Adaptada a Google Sheets

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        min_v; MIN(col_datos);
        max_v; MAX(col_datos);
        SI(ESNUMERO(col_datos); (col_datos - min_v) / (max_v - min_v); col_datos)
      )
    )
  )
))
```

En la hoja “**normalizacion-individual-eventos”** [Normalizacion-1-hoja - Hojas de cálculo de Google](#) se encuentra en las celdas resaltadas en amarillo.

Añadir Nuevos Eventos

Se debe editar la hoja “**Eventos-Originales**” [Normalizacion-1-hoja - Hojas de cálculo de Google]() ya que de ahí se toman los datos para la normalización y estadísticos

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Plots Normalizados

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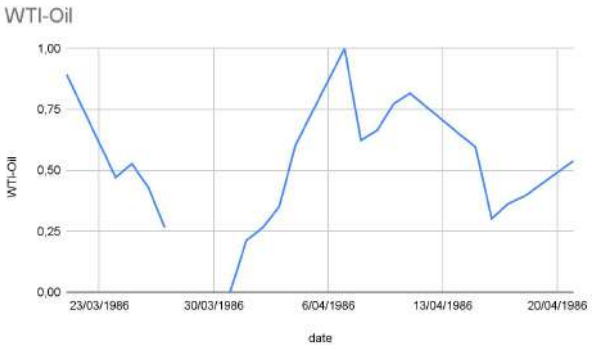
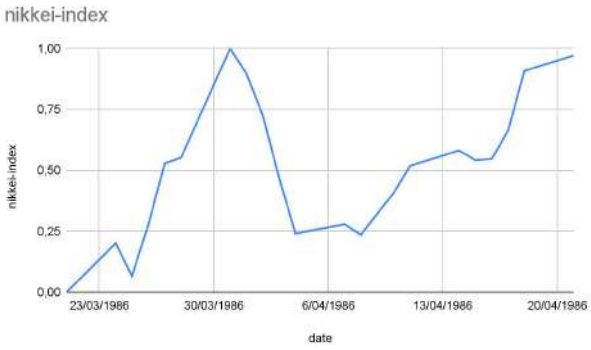
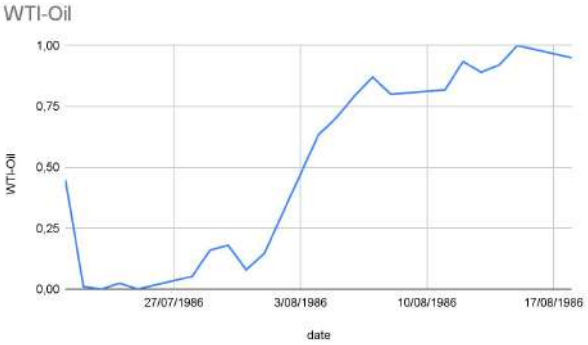
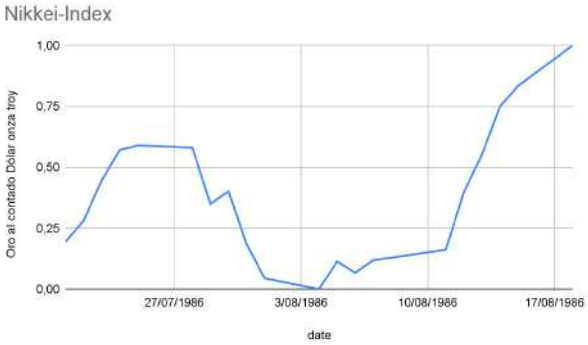


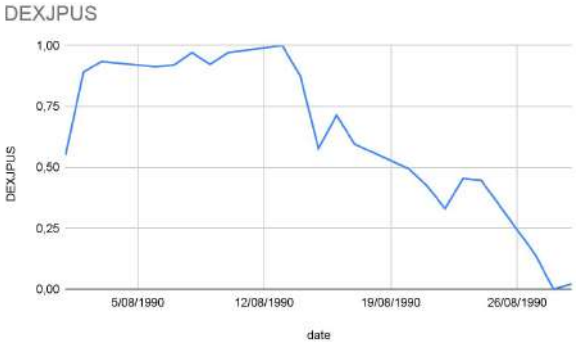


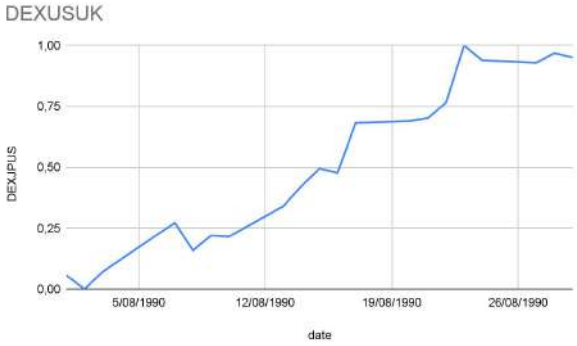
Evento 2 (21/07/1986 - 18/08/1986)

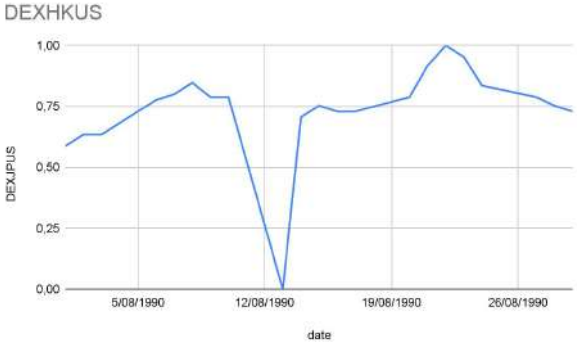


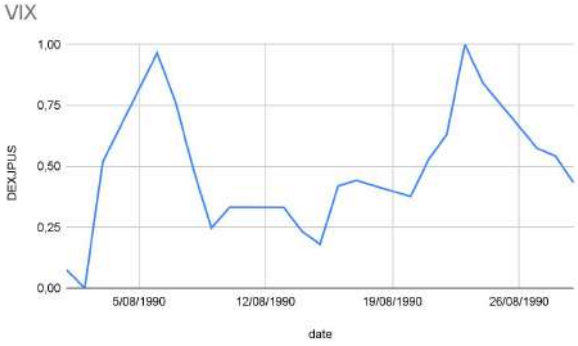


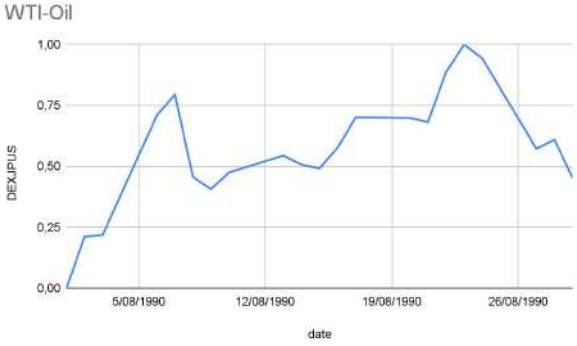
Evento 3 (1/08/1990 - 29/08/1990)

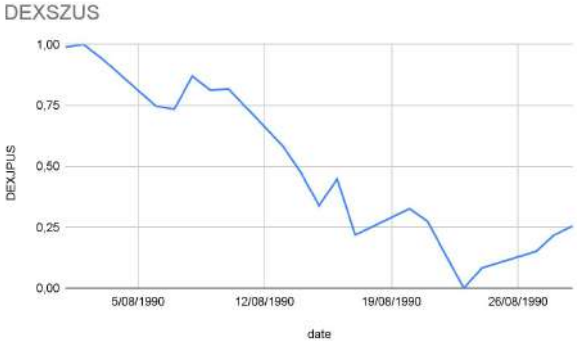












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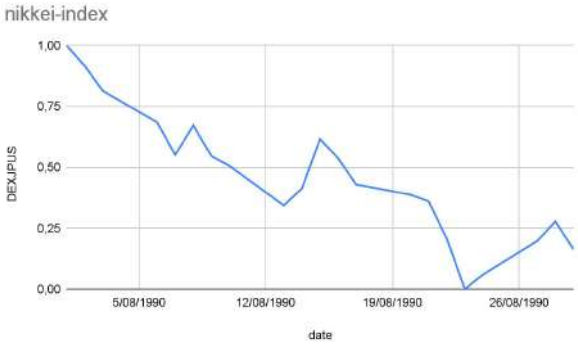

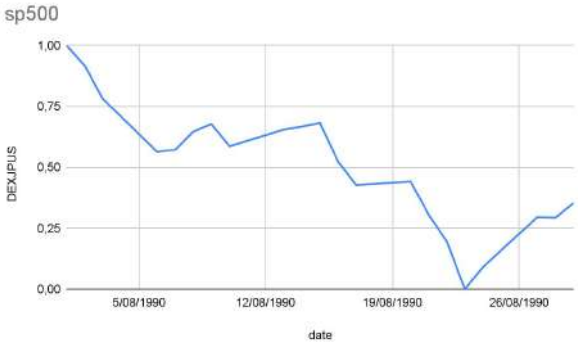

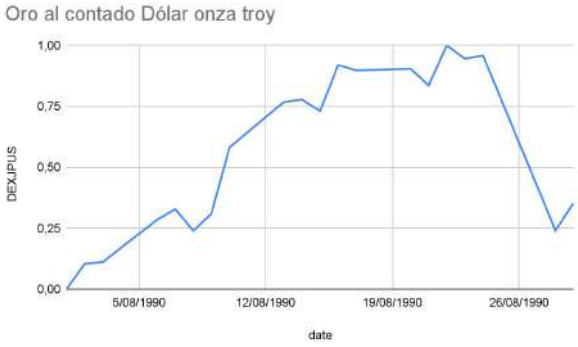

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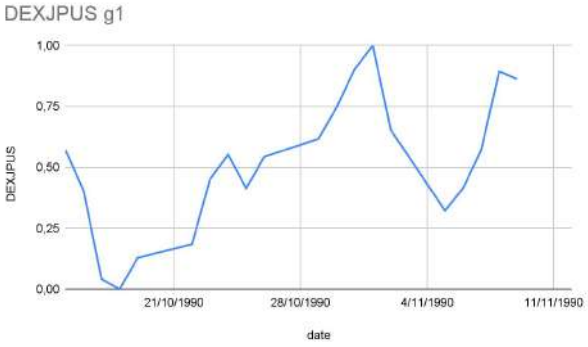

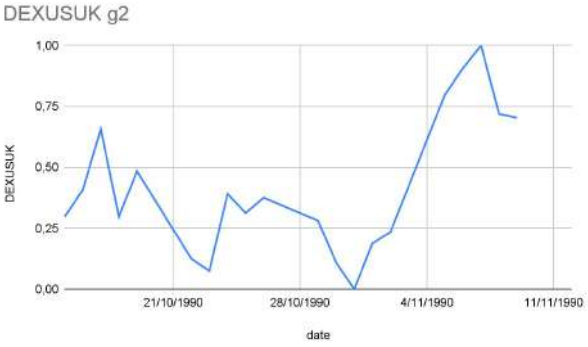

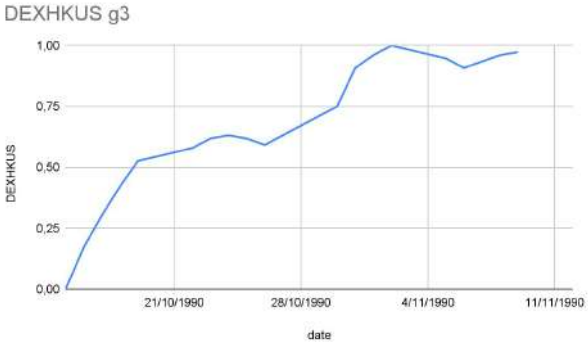

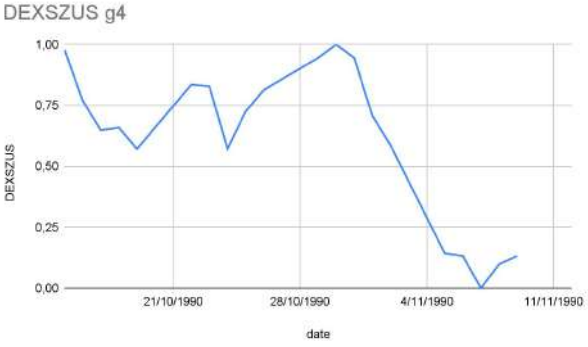

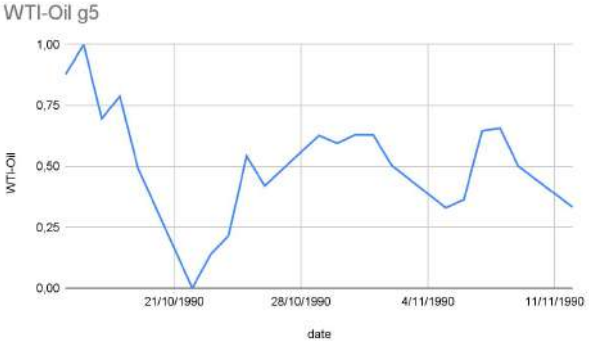

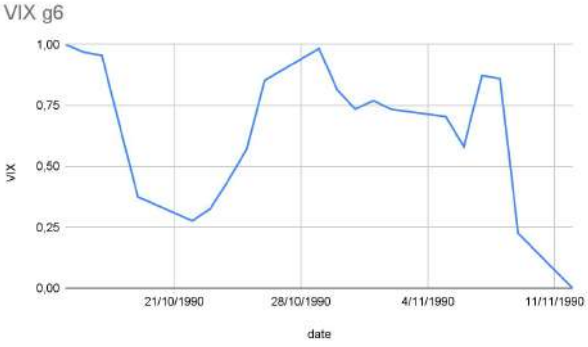

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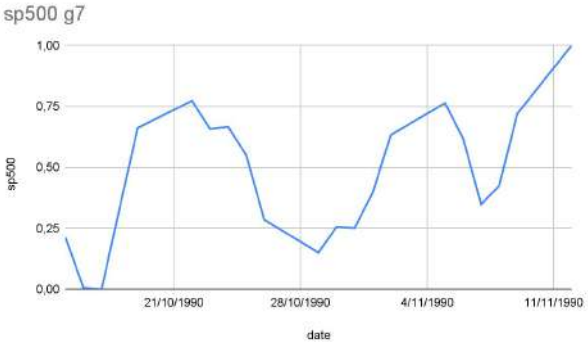
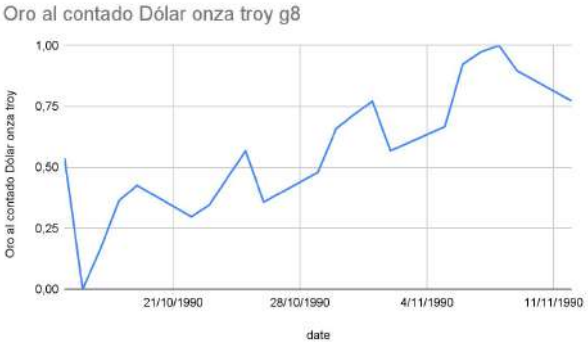
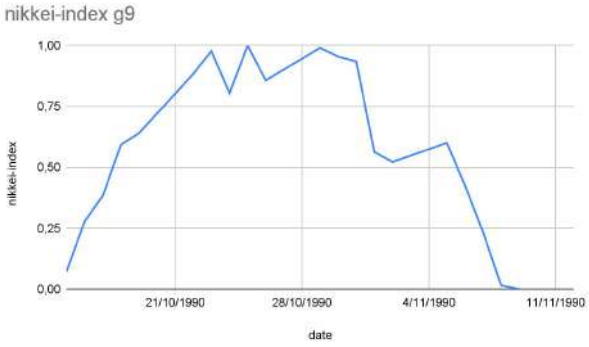
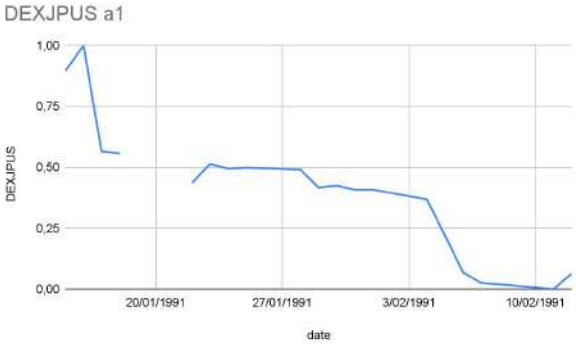
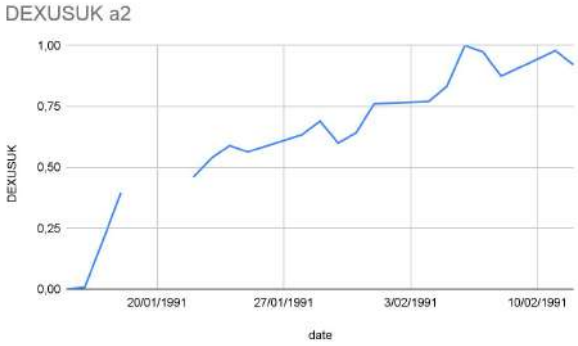


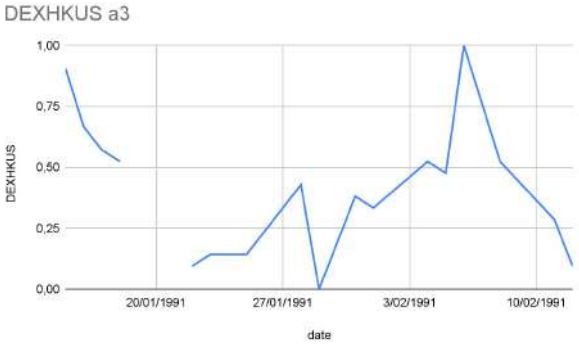


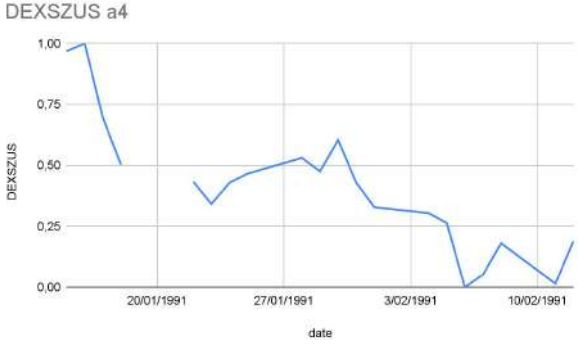


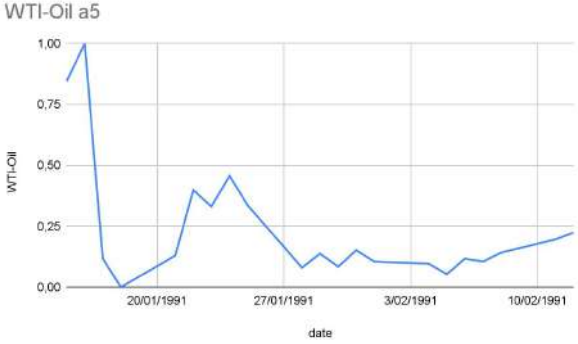
Evento 5 (15/01/1991 - 12/02/1991)

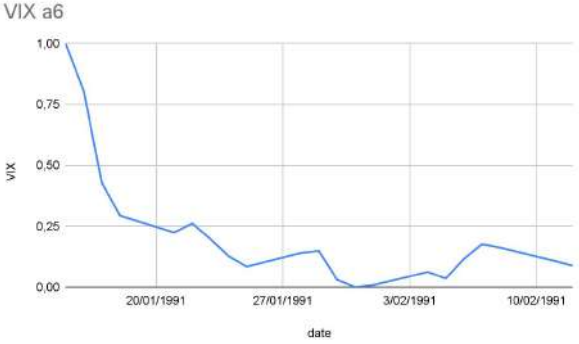












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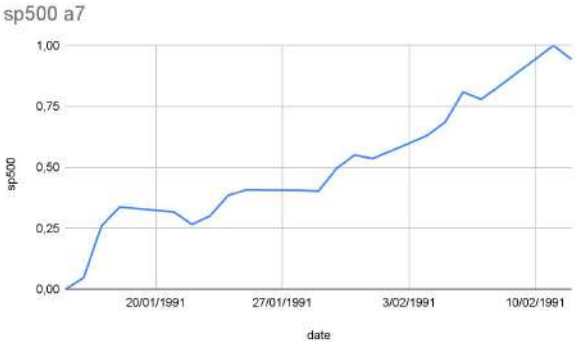
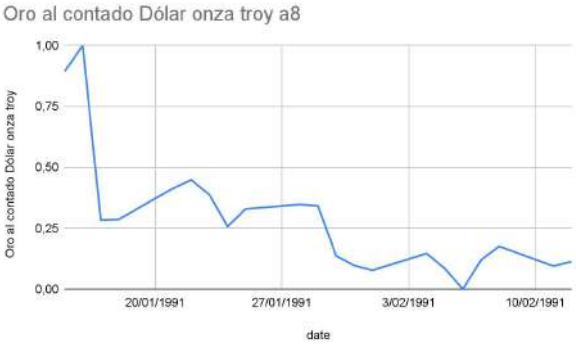
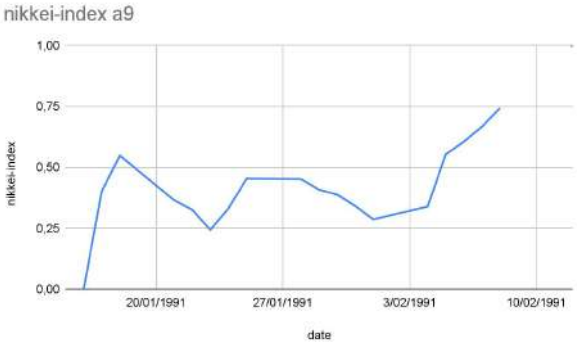
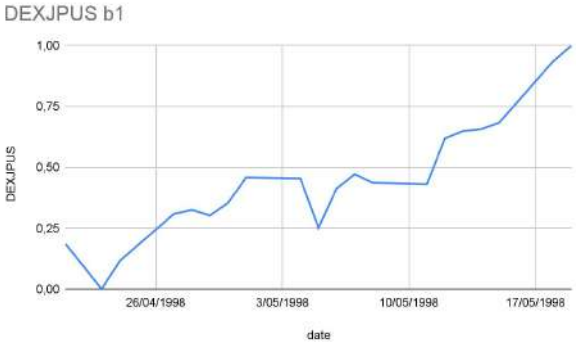
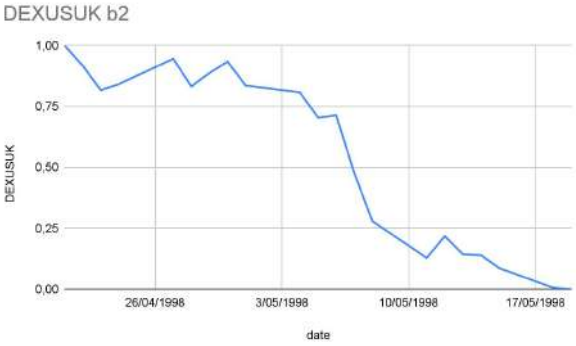


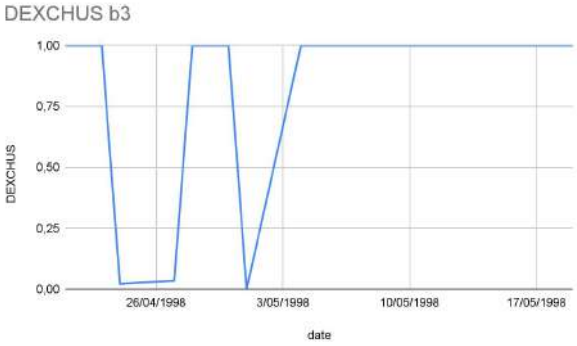


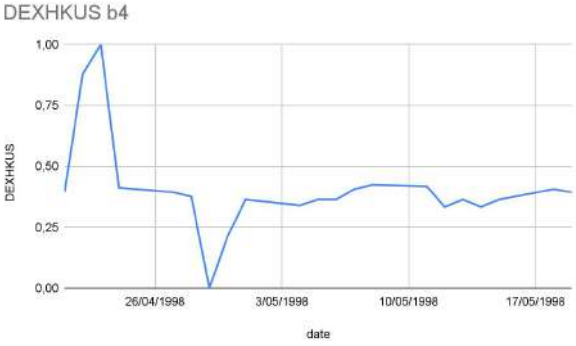


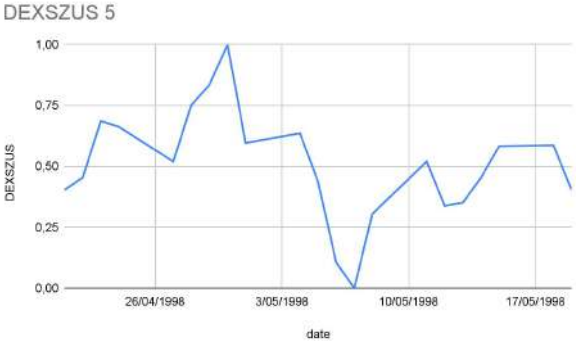
Evento 6 (21/04/1998 - 19/05/1998)

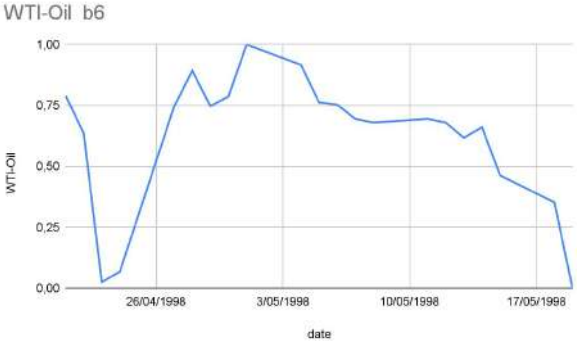




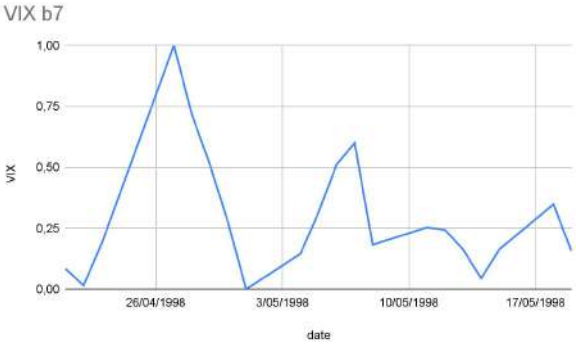


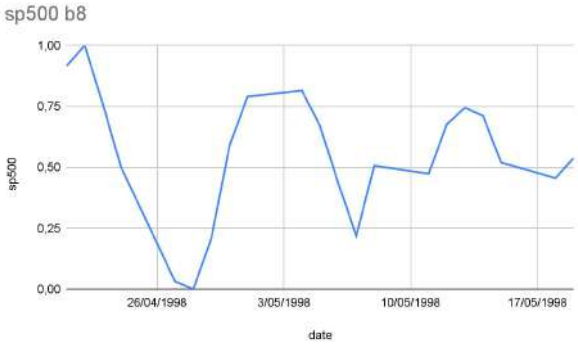


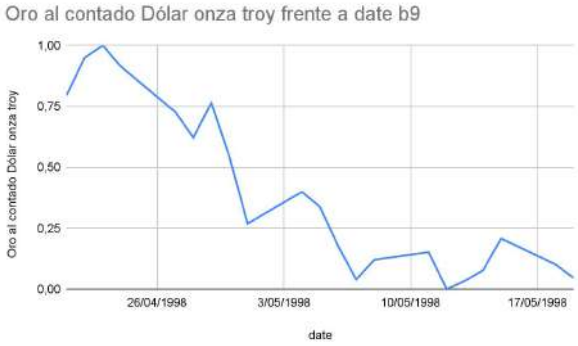


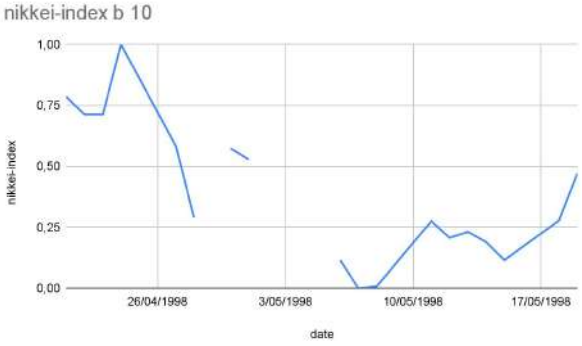


Evento 6 (21/04/1998 - 19/05/1998)

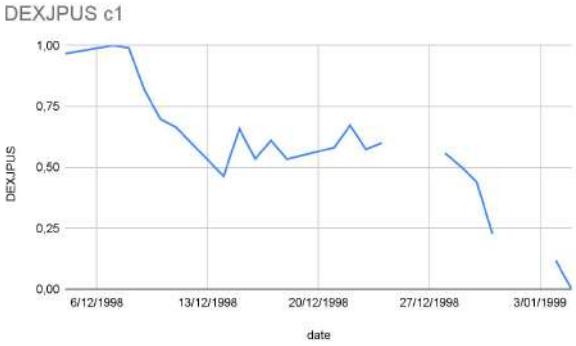


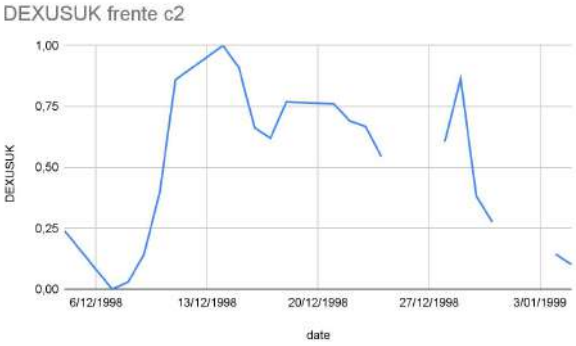


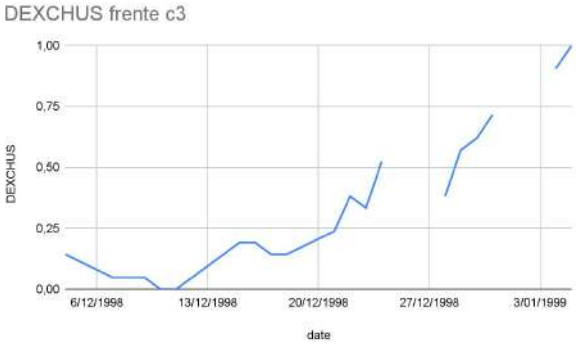


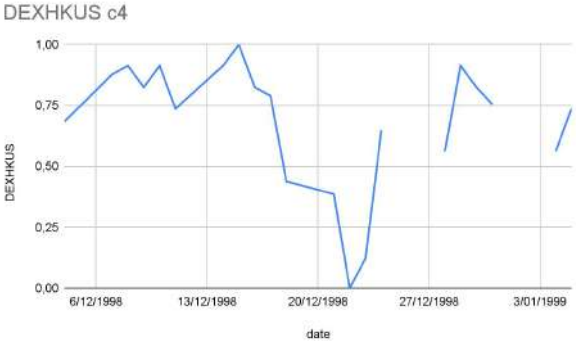


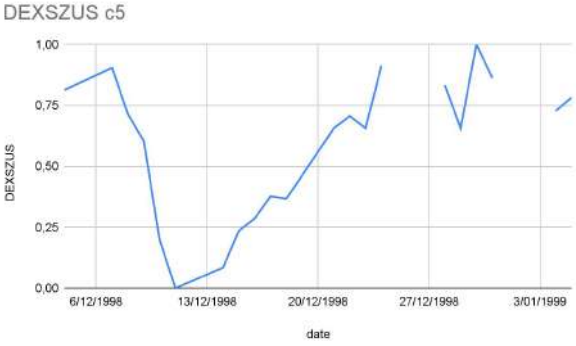
Evento 7 (4/12/1998 - 5/01/1999)

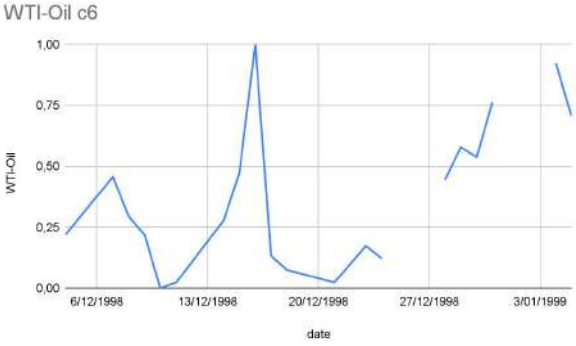




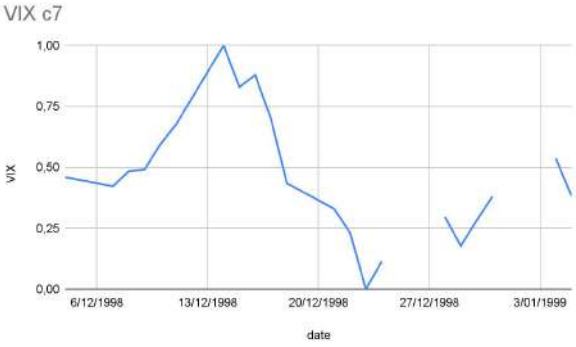


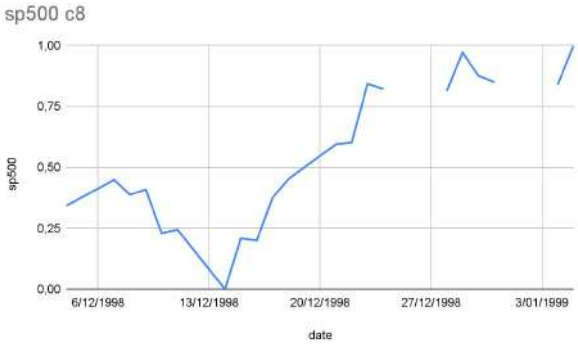


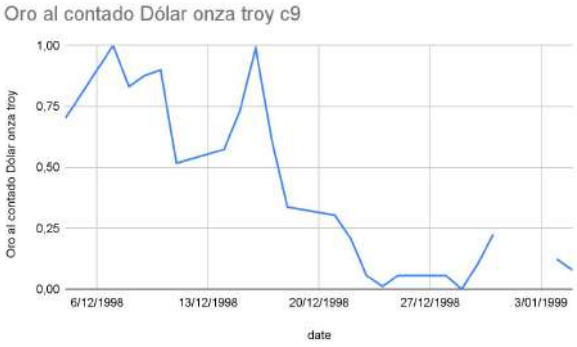


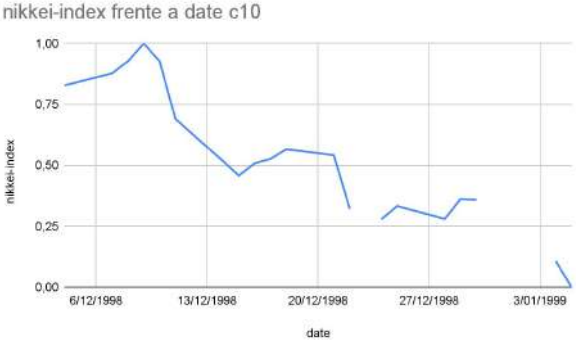


Evento 7 (4/12/1998 - 5/01/1999)

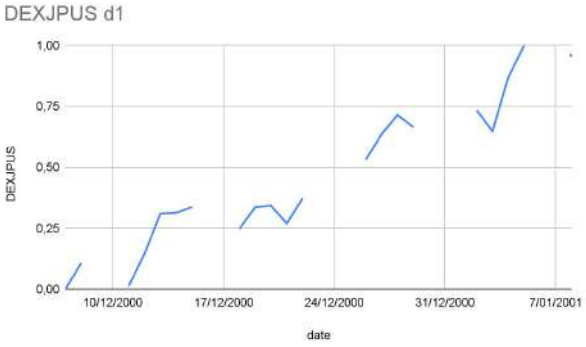


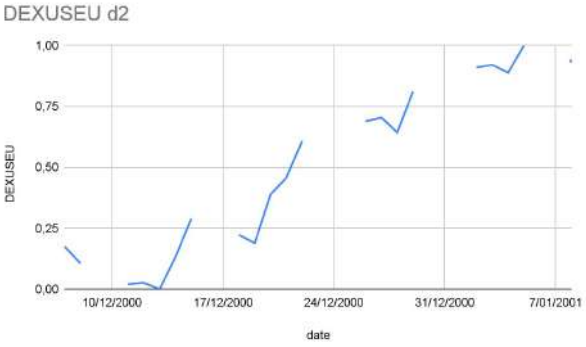


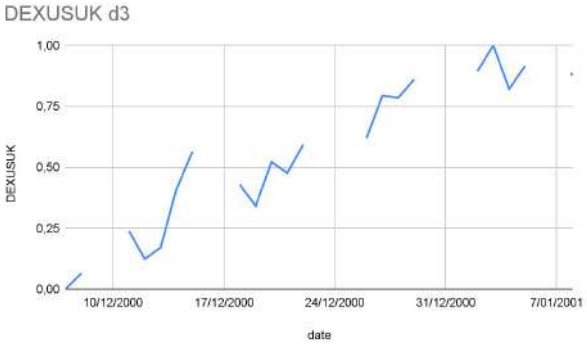


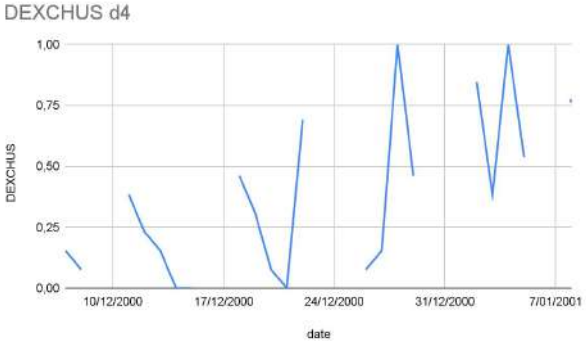


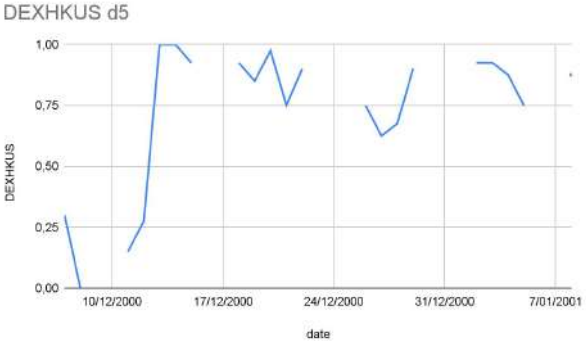
Evento 8 (7/12/2000 - 8/01/2001)

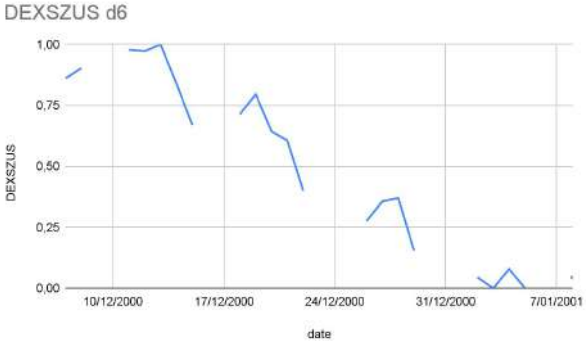




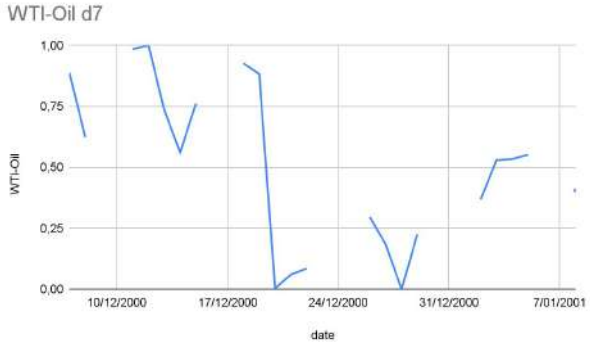


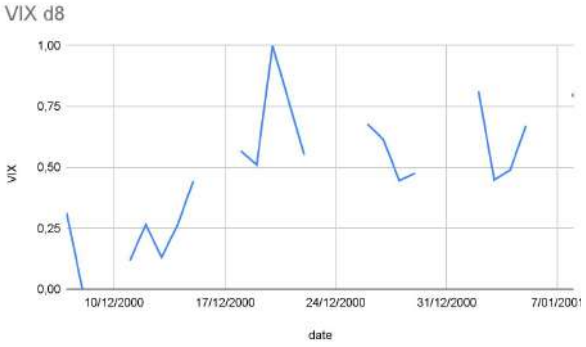


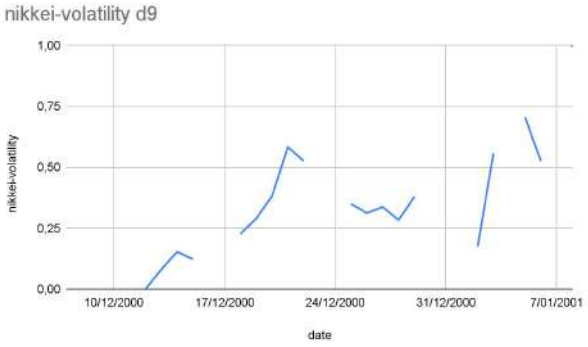


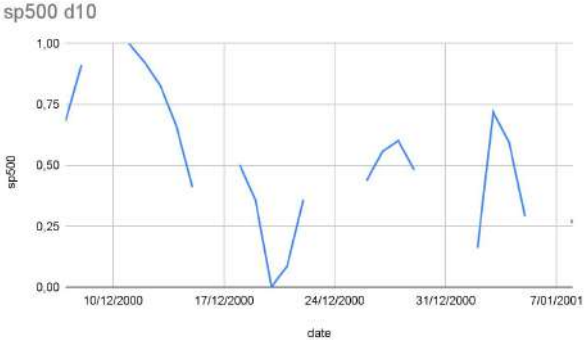


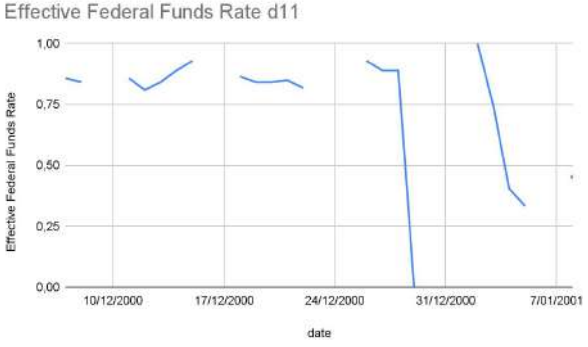
Evento 8 (7/12/2000 - 8/01/2001)



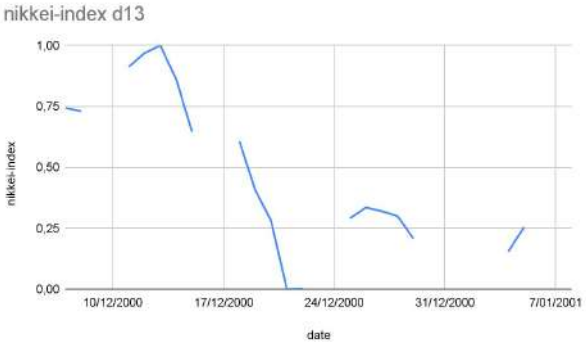


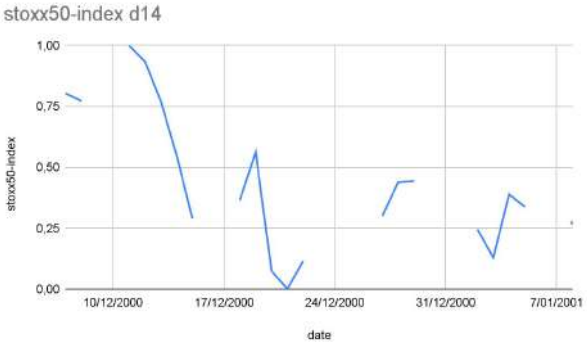




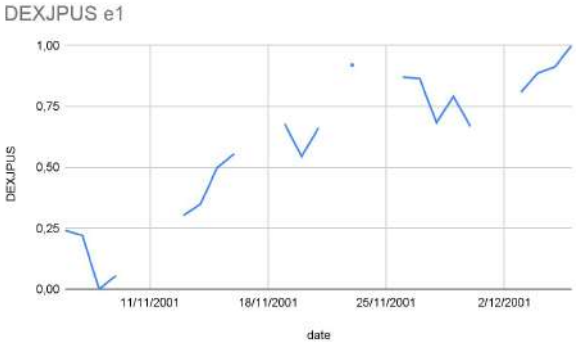


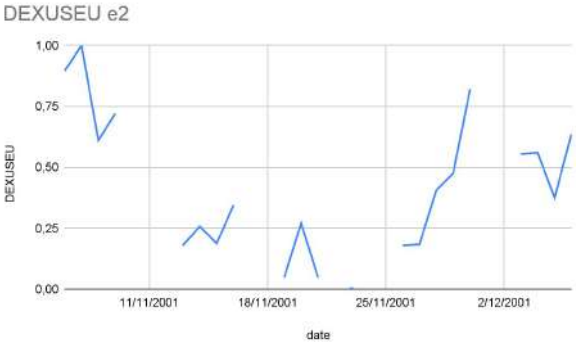
Evento 8 (7/12/2000 - 8/01/2001)

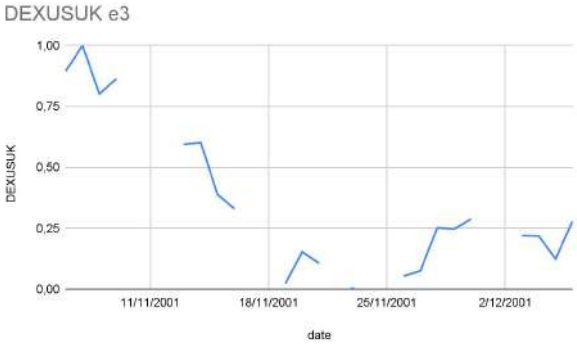


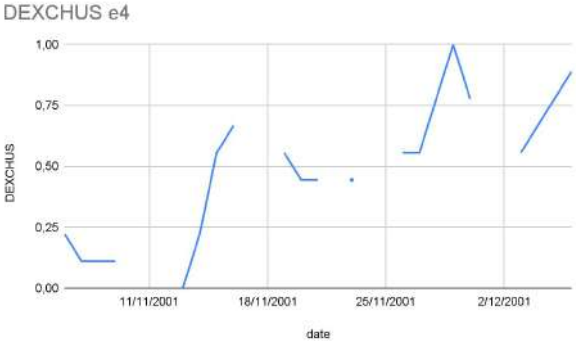


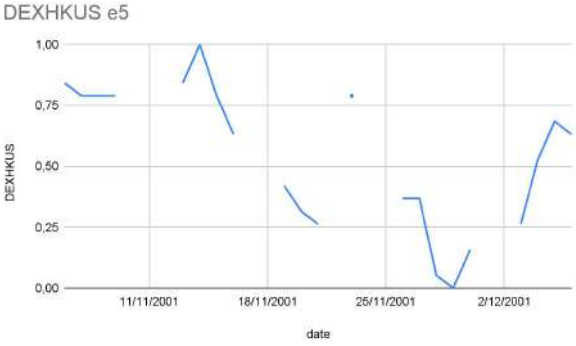
Evento 9 (6/11/2001 - 6/12/2001)

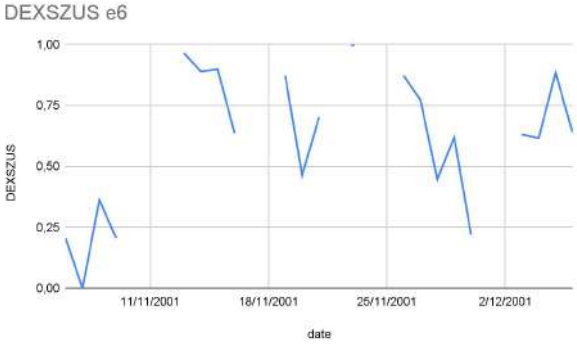




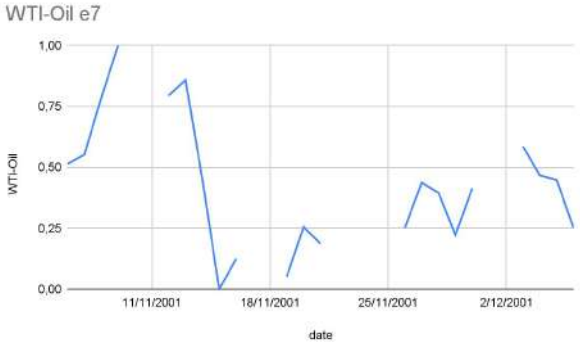


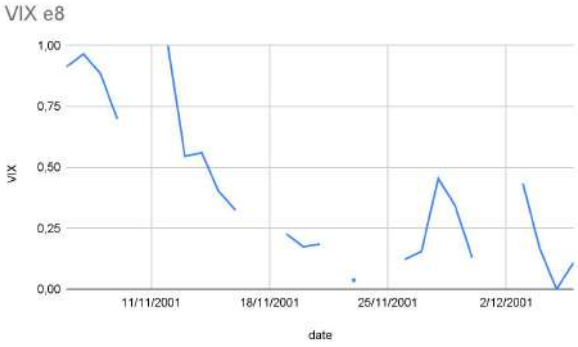


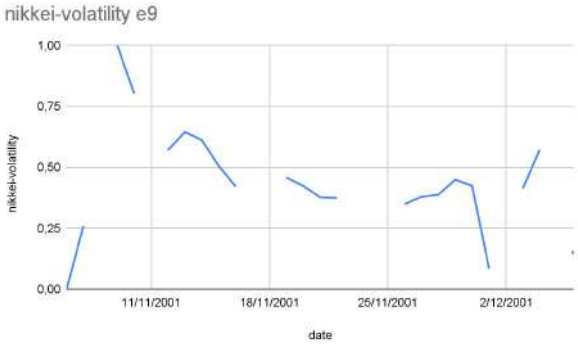


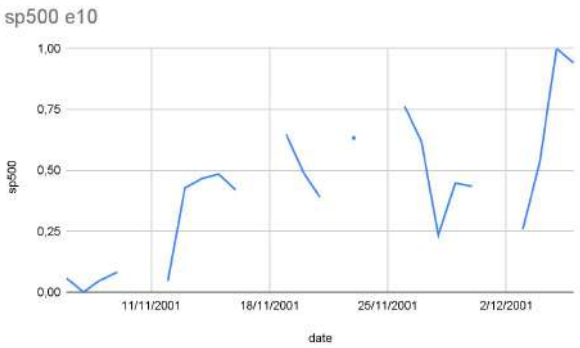


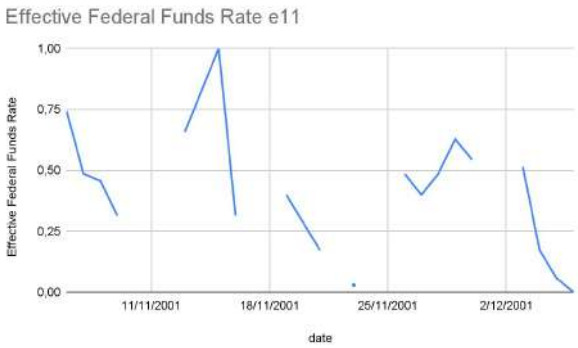
Evento 9 (6/11/2001 - 6/12/2001)

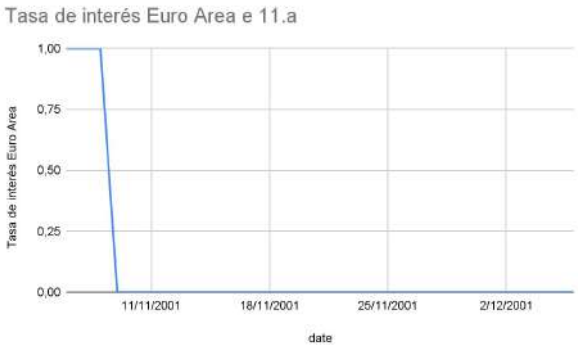




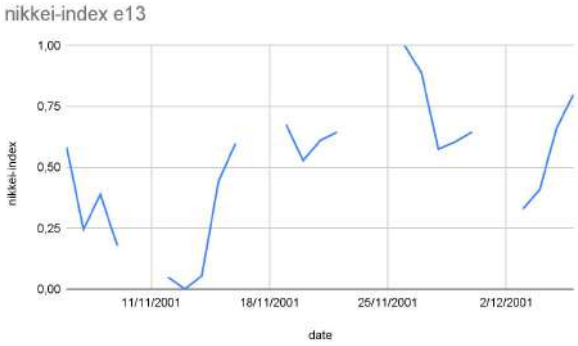


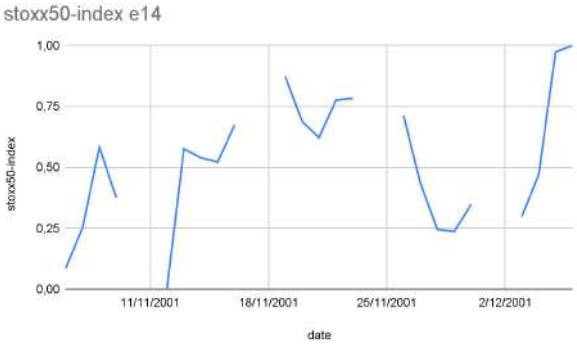




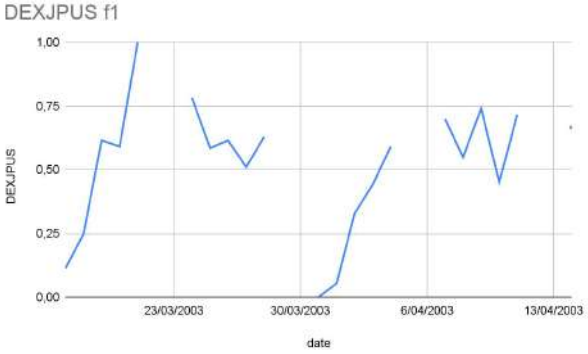


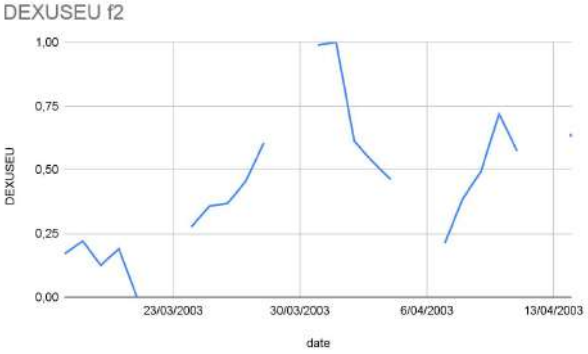
Evento 9 (6/11/2001 - 6/12/2001)

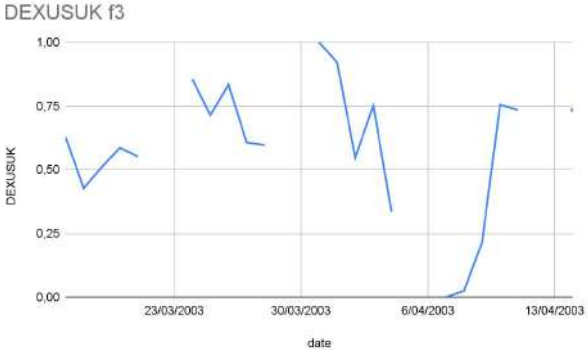


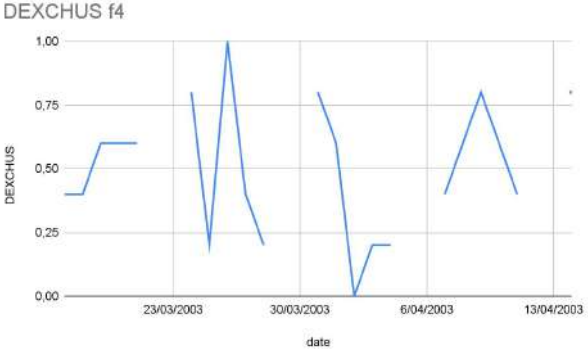


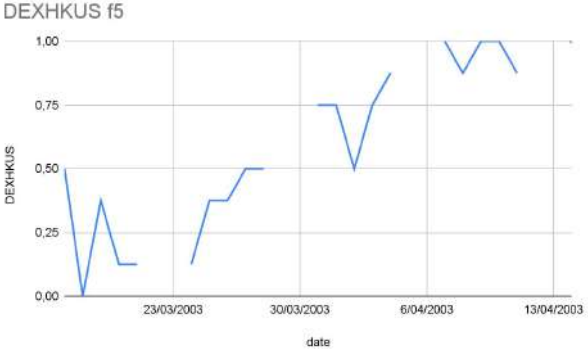
Evento 10 (17/03/2003 - 14/04/2003)

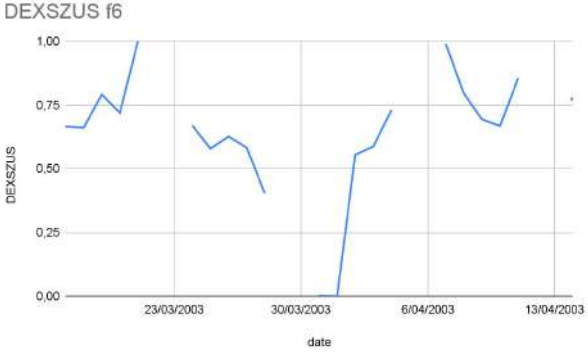




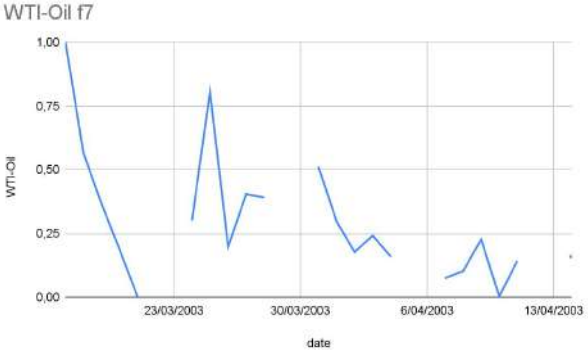


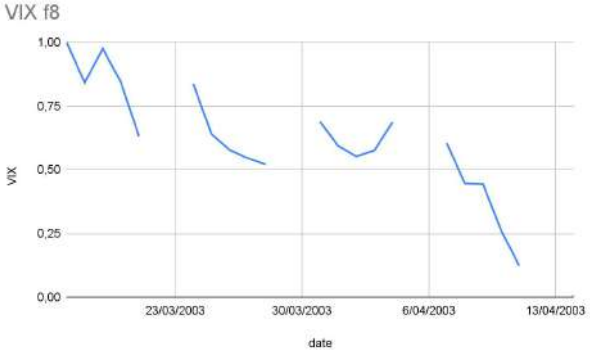


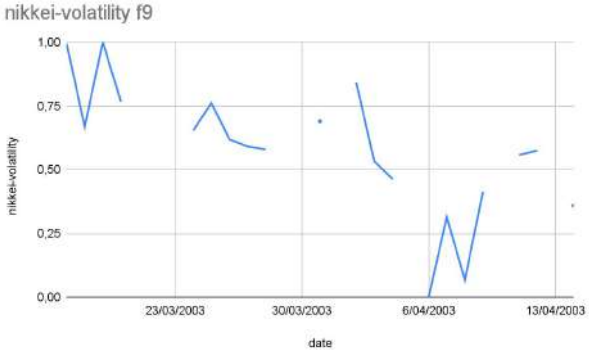


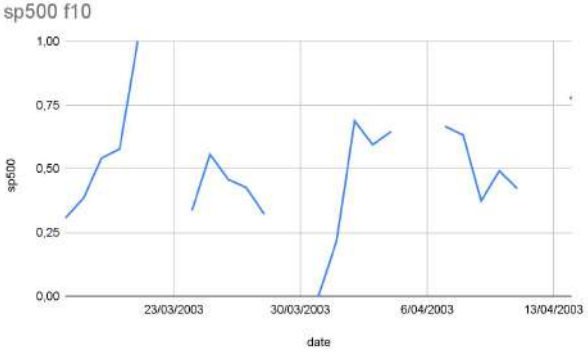


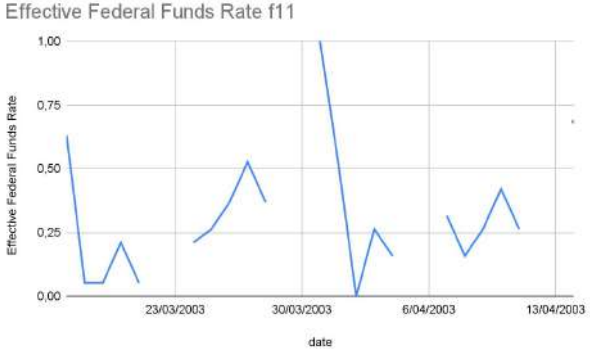
Evento 10 (17/03/2003 - 14/04/2003)

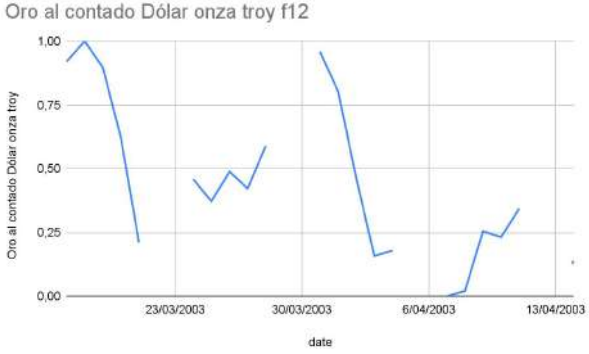




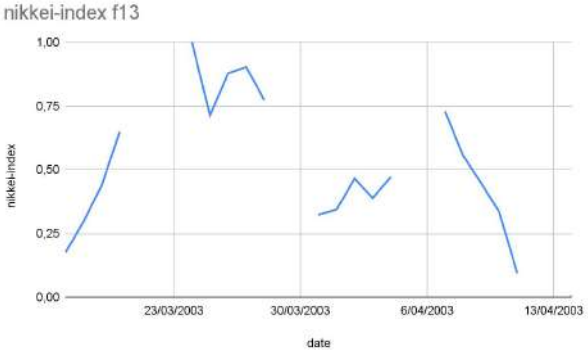


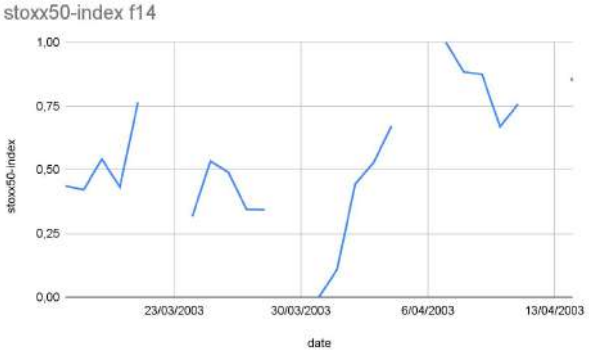




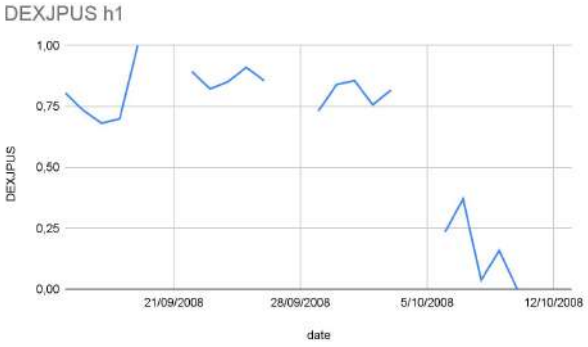


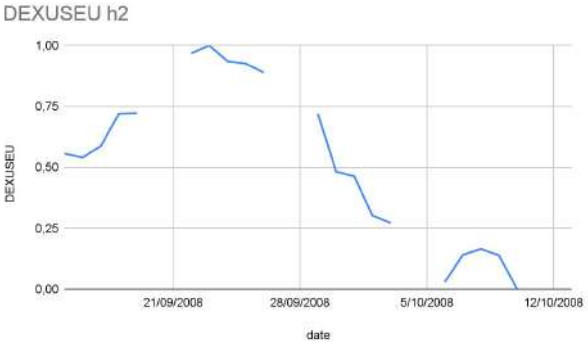
Evento 10 (17/03/2003 - 14/04/2003)

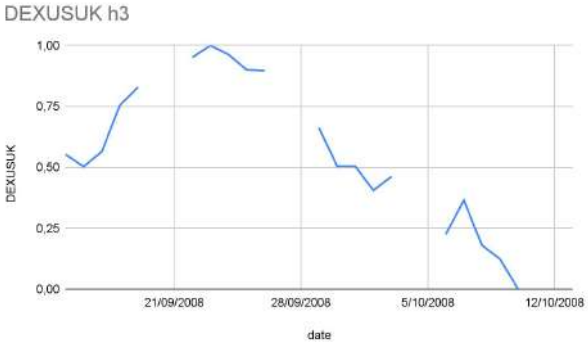


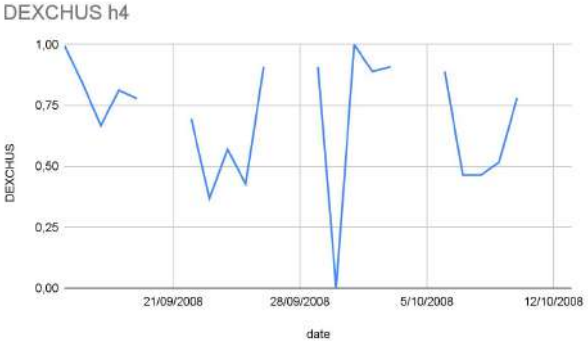


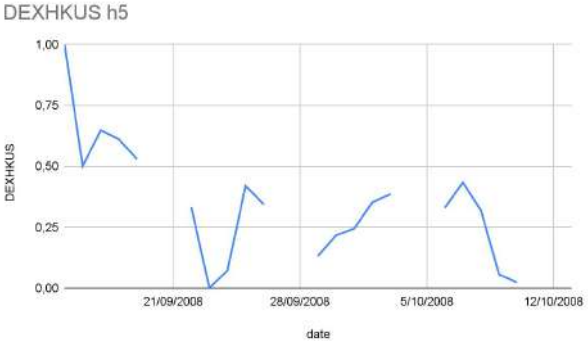
Evento 11 (15/09/2008 - 13/10/2008)

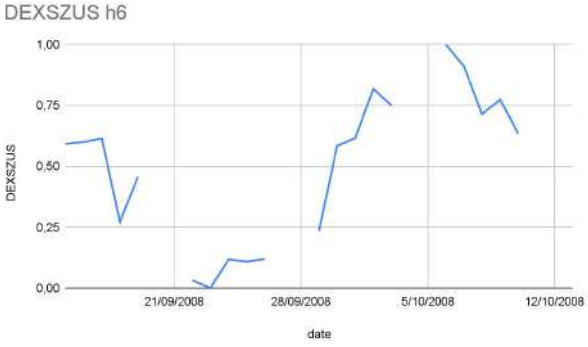




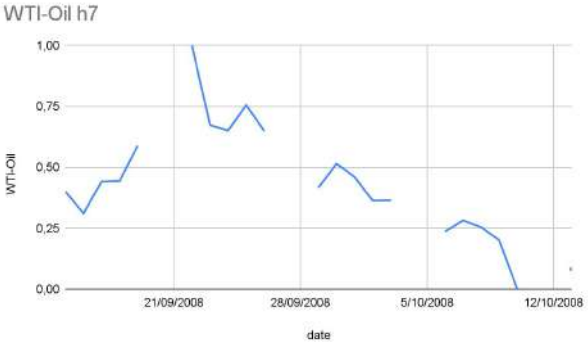


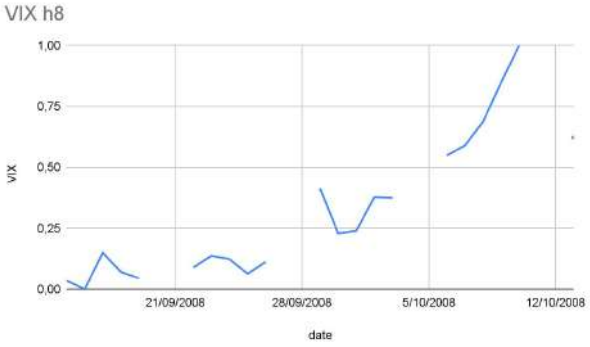


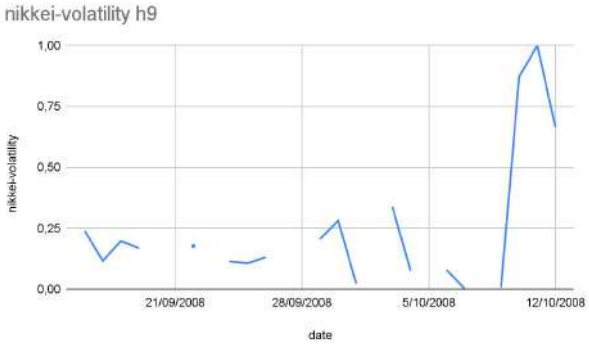


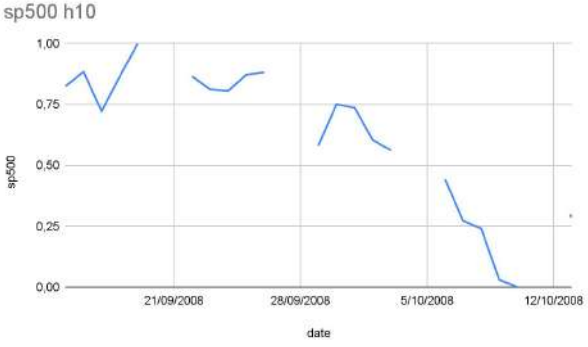


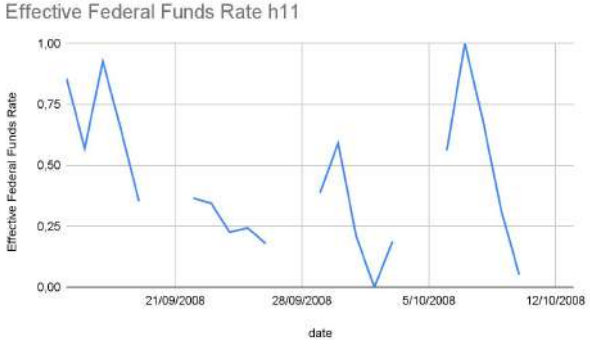
Evento 11 (15/09/2008 - 13/10/2008)

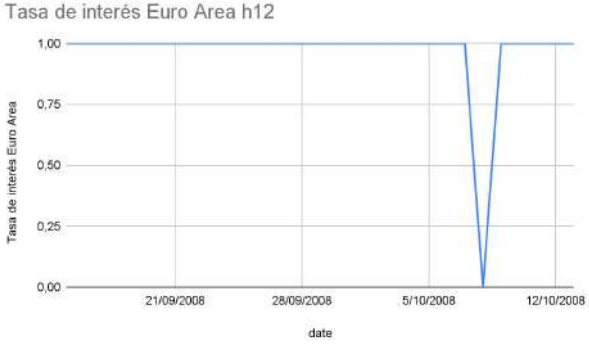




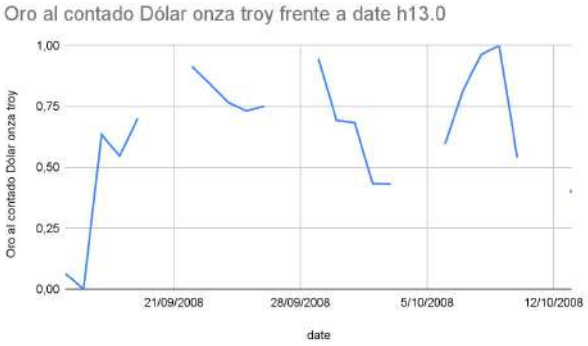


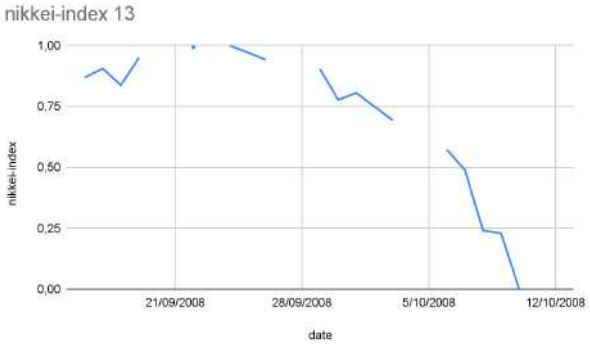


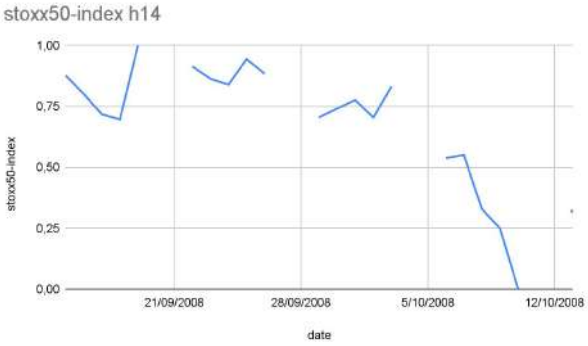




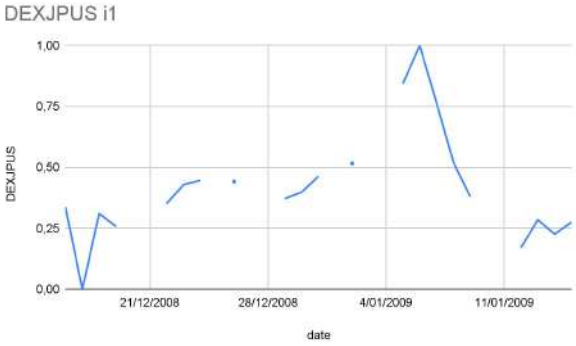
Evento 11 (15/09/2008 - 13/10/2008)

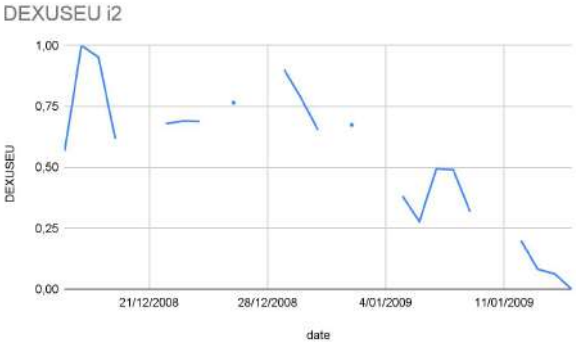


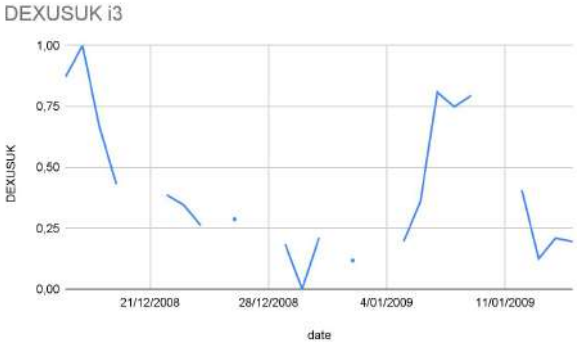


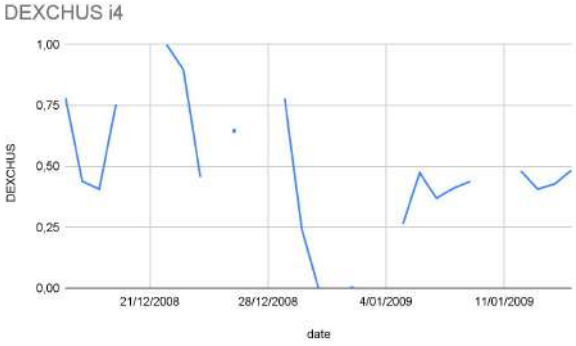


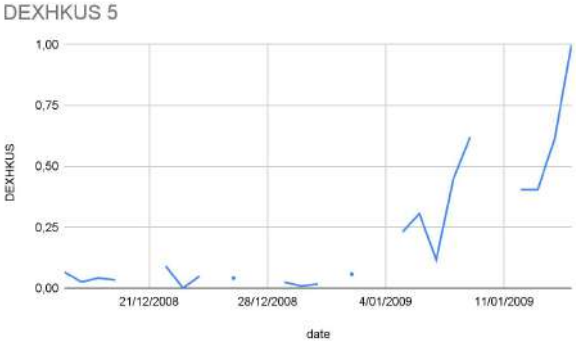
Evento 12 (16/12/2008 - 15/01/2009)

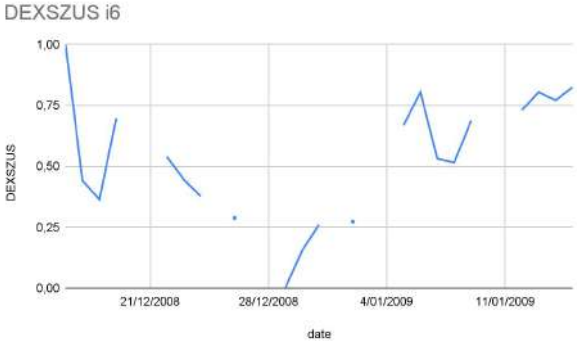




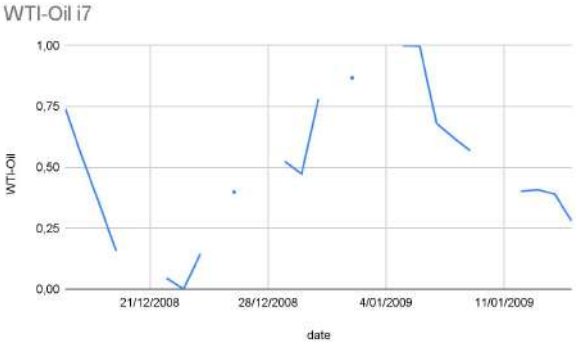


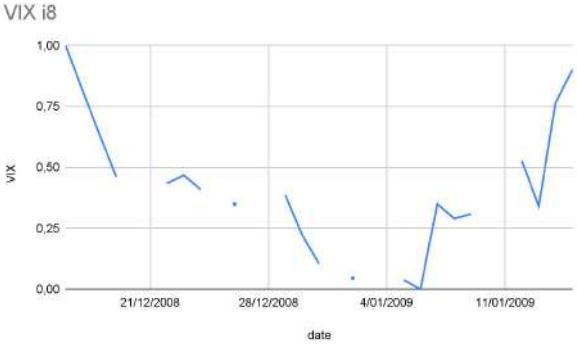


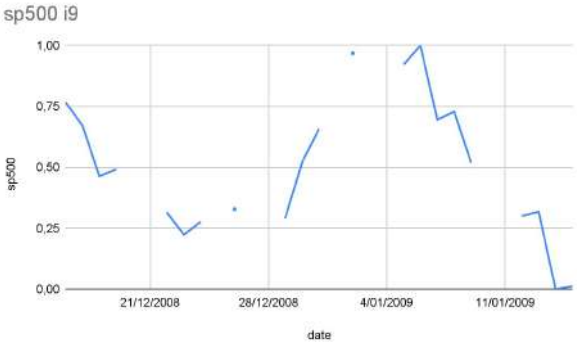


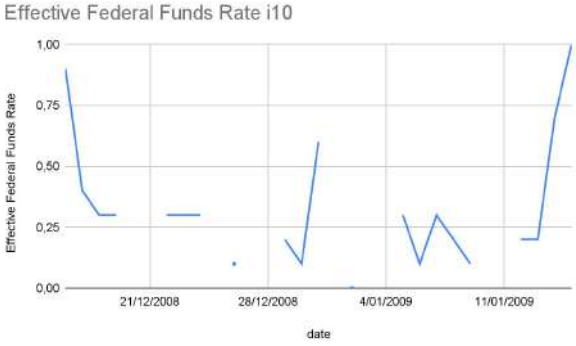


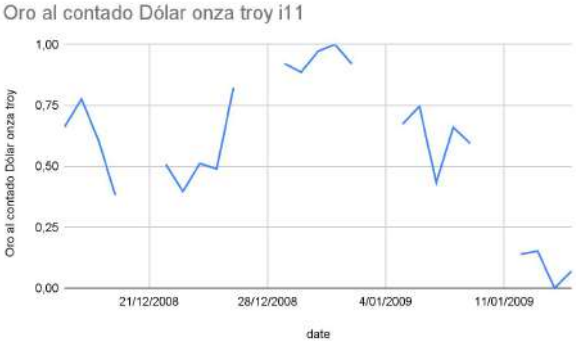
Evento 12 (16/12/2008 - 15/01/2009)

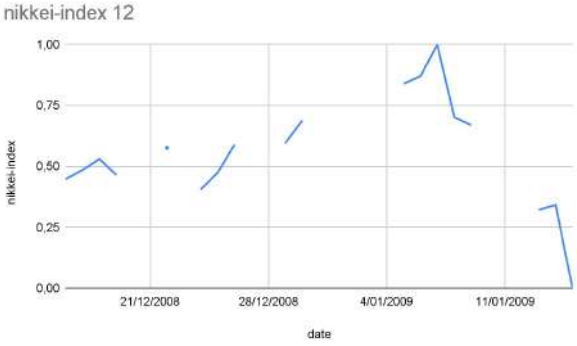




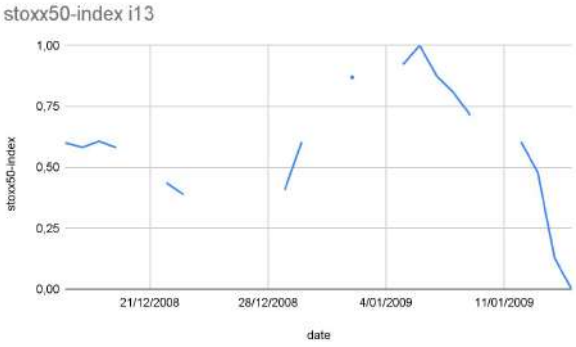




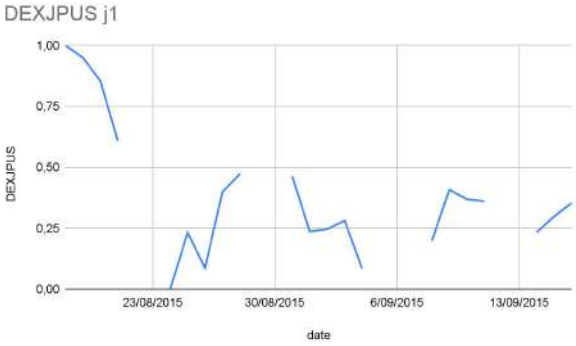


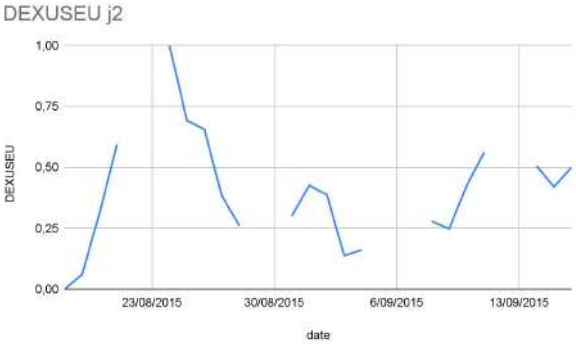


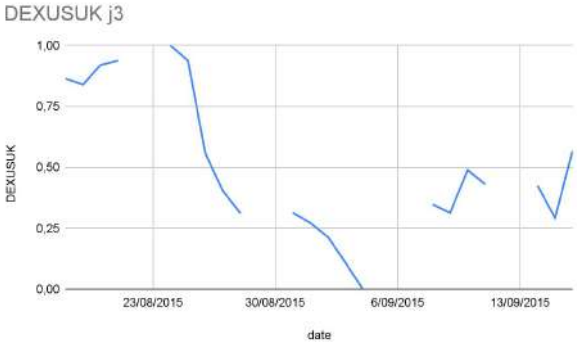
Evento 12 (16/12/2008 - 15/01/2009)

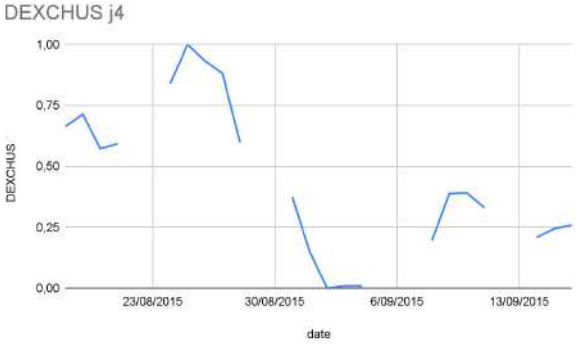


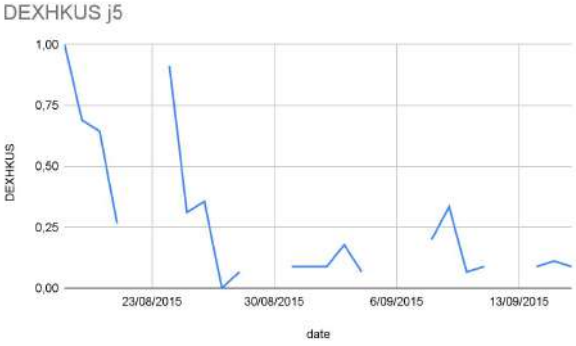
Evento 13 (18/08/2015 - 16/09/2015)

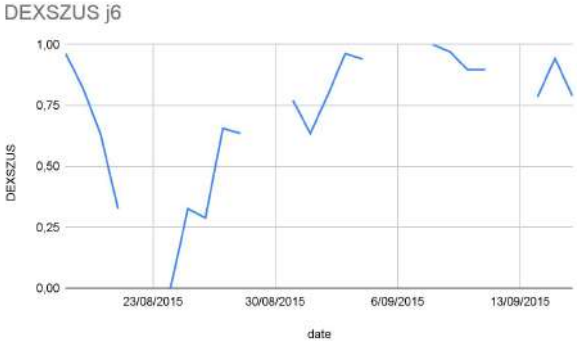




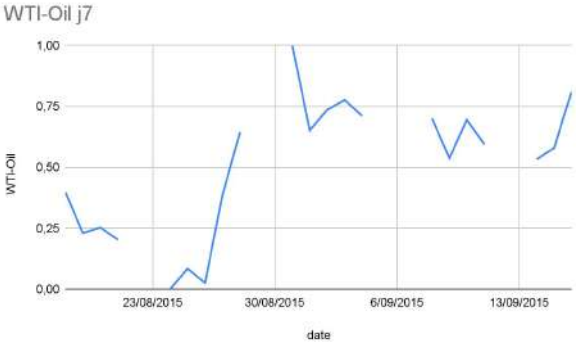


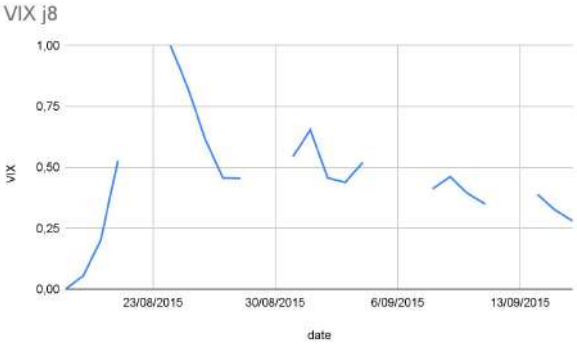


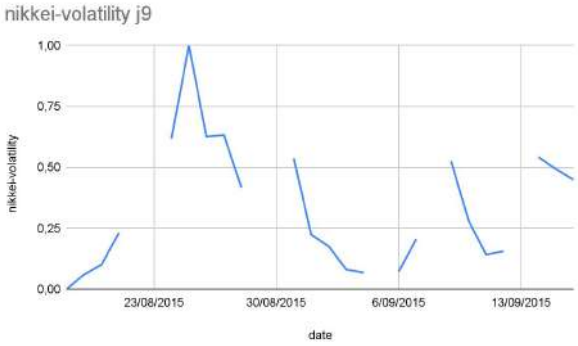


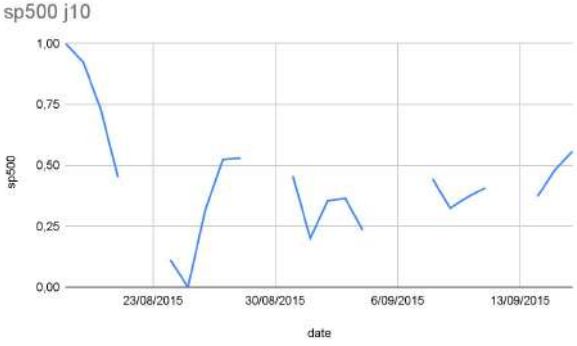


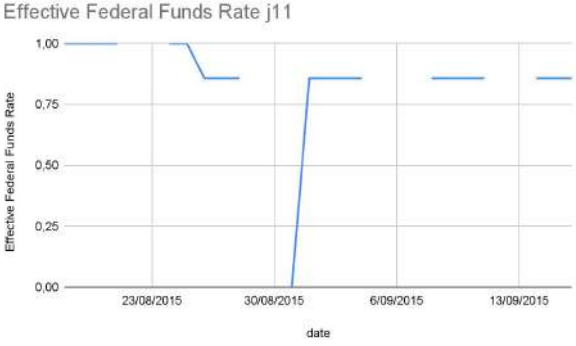
Evento 13 (18/08/2015 - 16/09/2015)



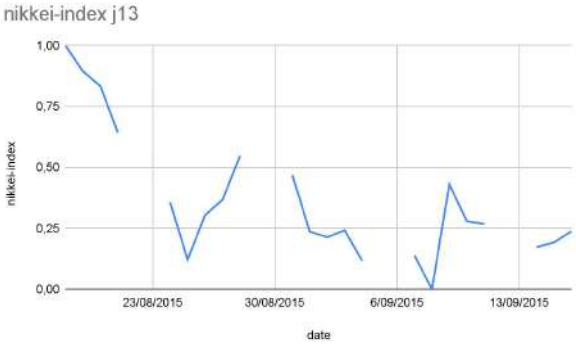


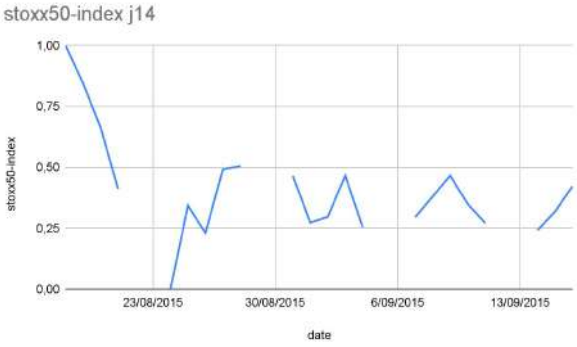


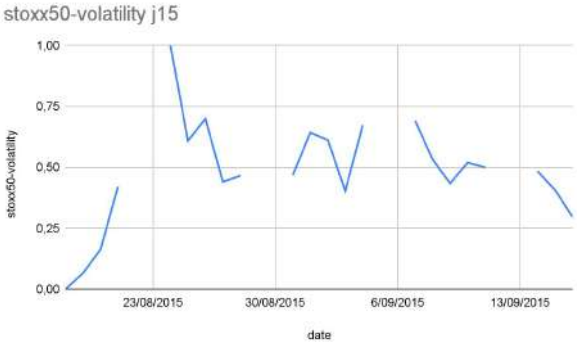




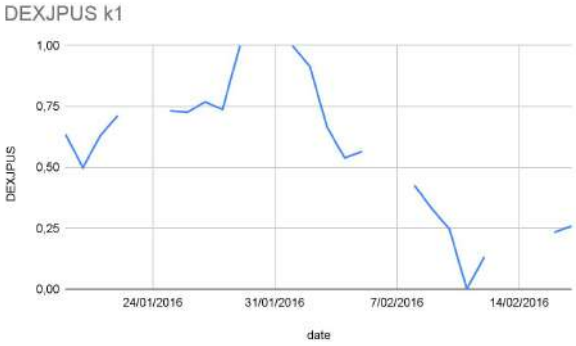
Evento 13 (18/08/2015 - 16/09/2015)

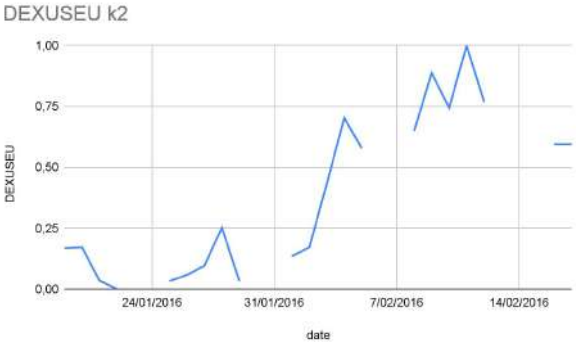


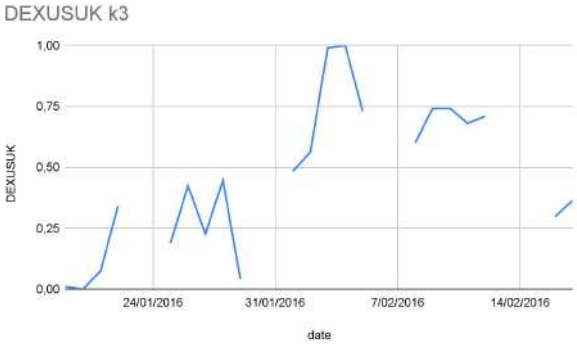


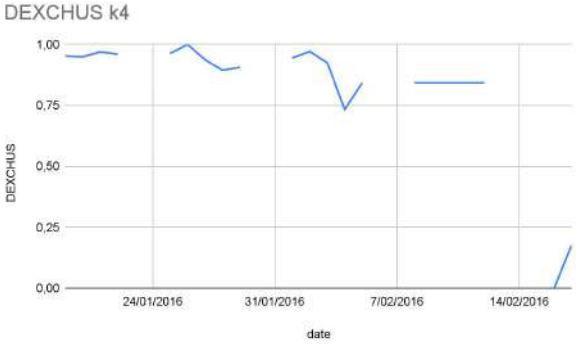


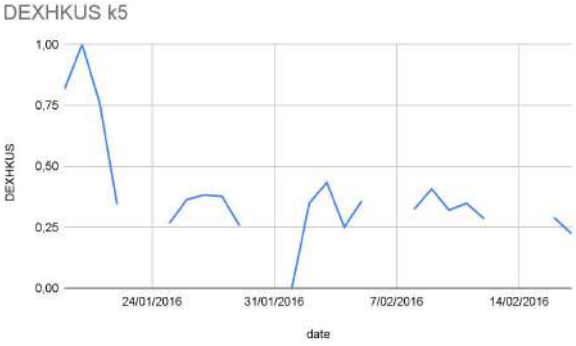
Evento 14 (19/01/2016 - 17/02/2016)

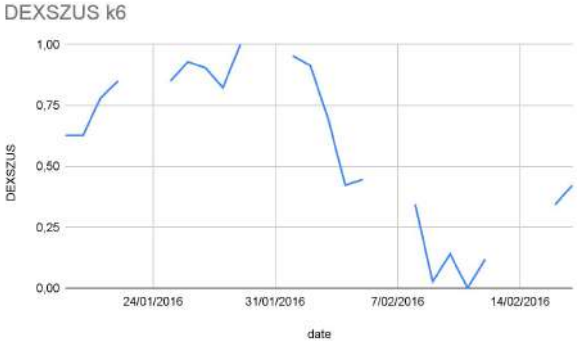




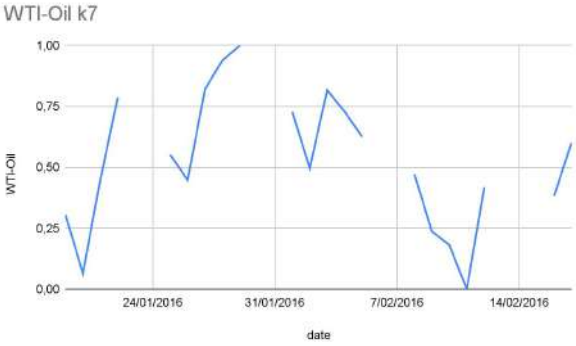


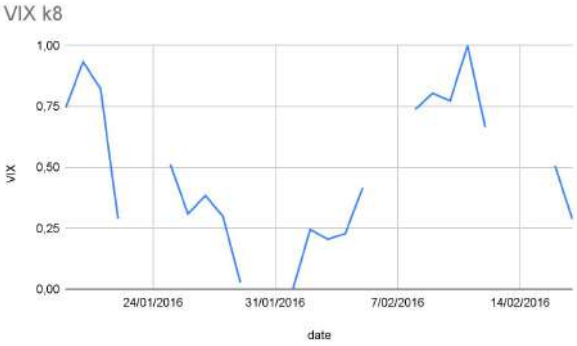


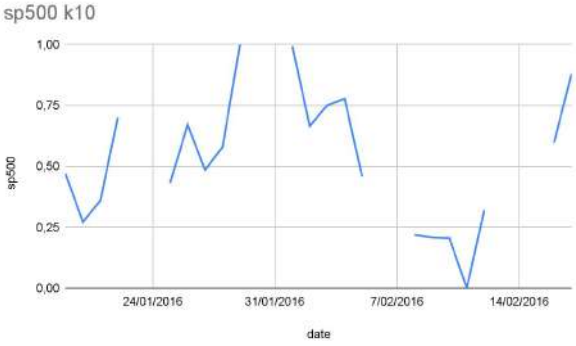


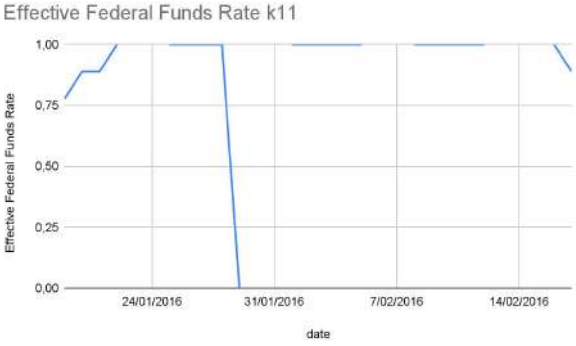


Evento 14 (19/01/2016 - 17/02/2016)

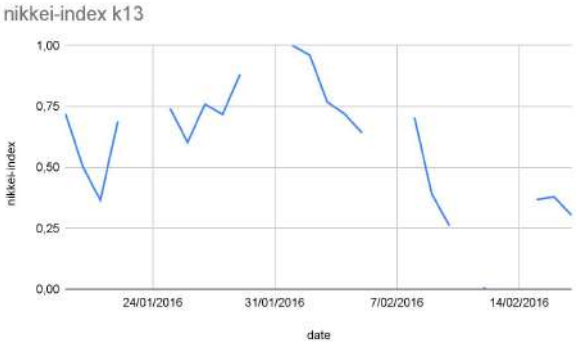


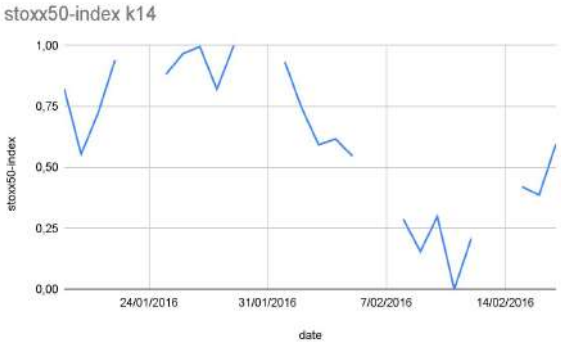




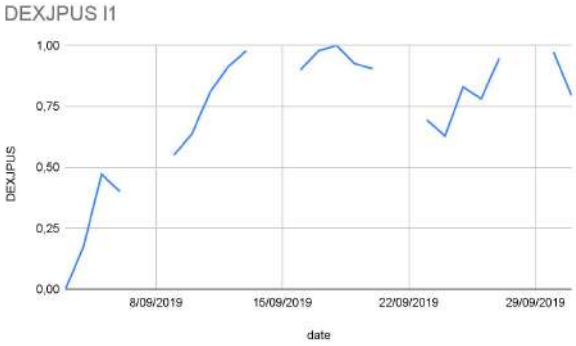


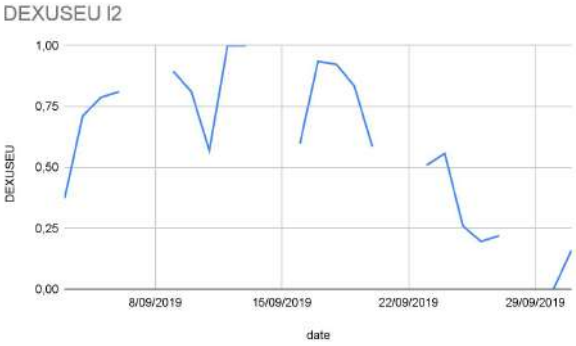
Evento 14 (19/01/2016 - 17/02/2016)

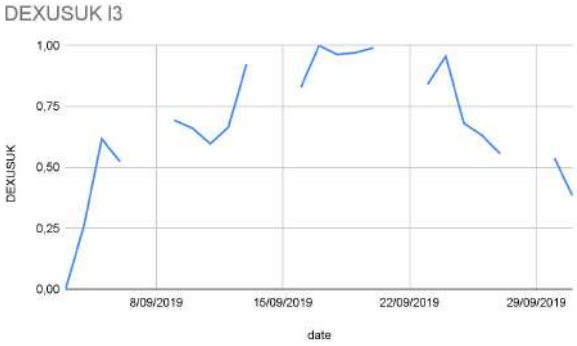


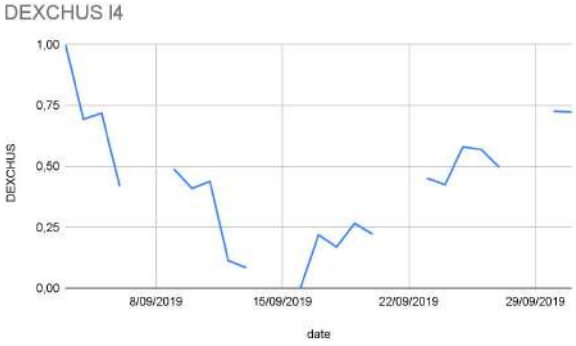


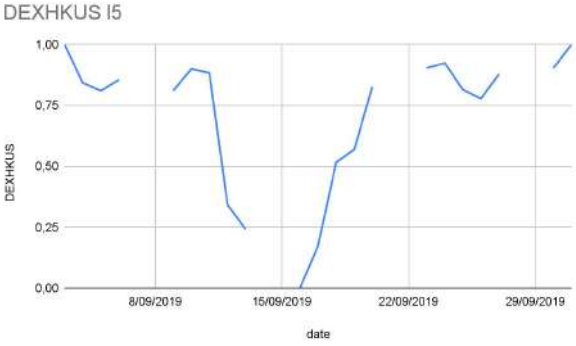
Evento 15 (3/09/2019 - 1/10/2019)

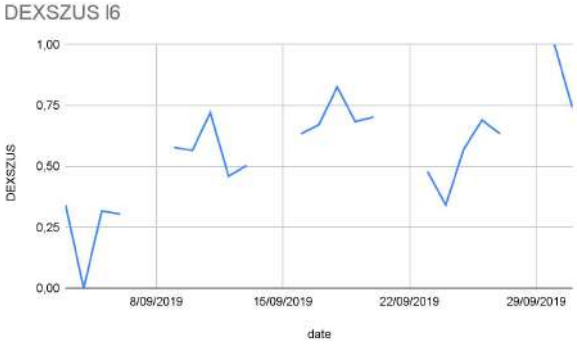




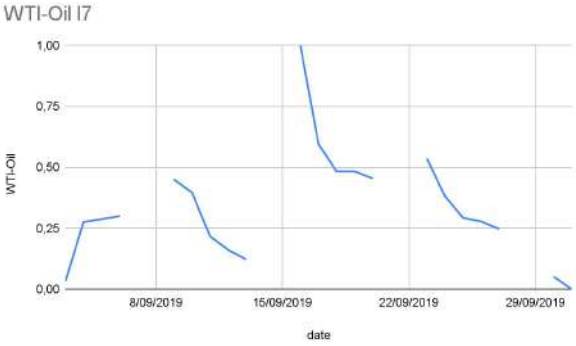


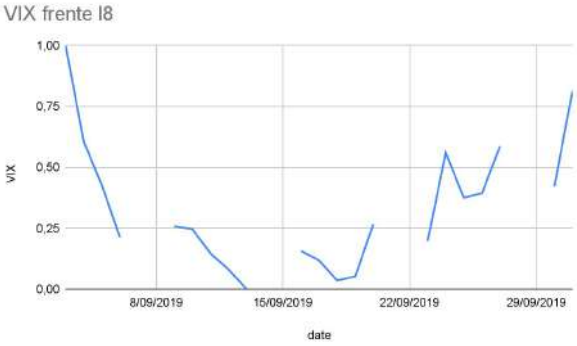


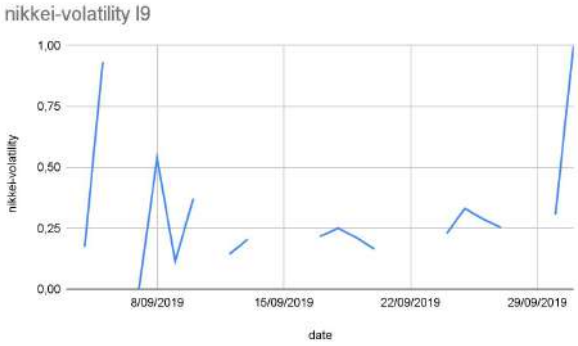


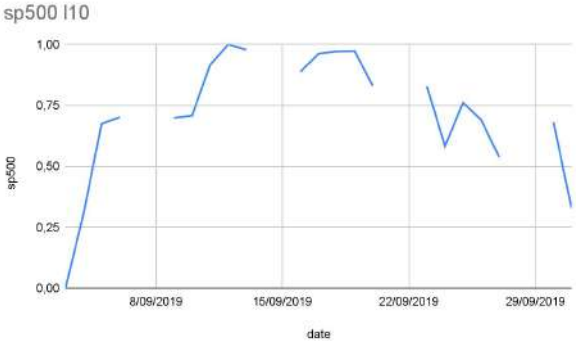


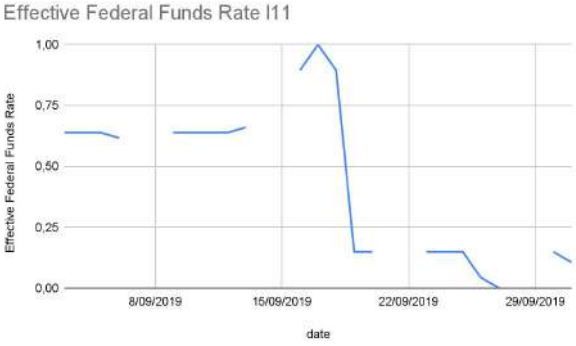
Evento 15 (3/09/2019 - 1/10/2019)

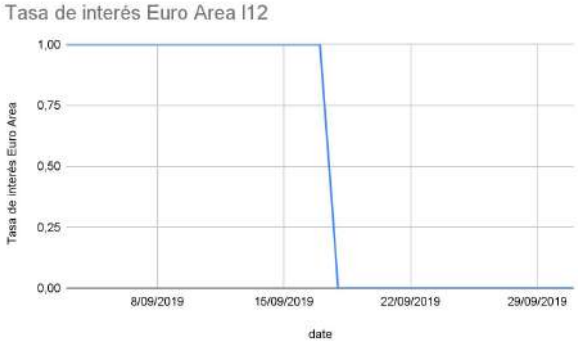




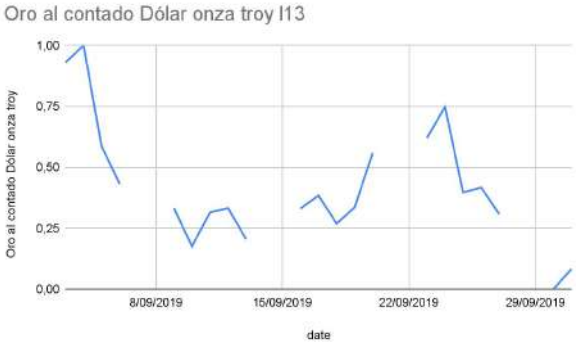


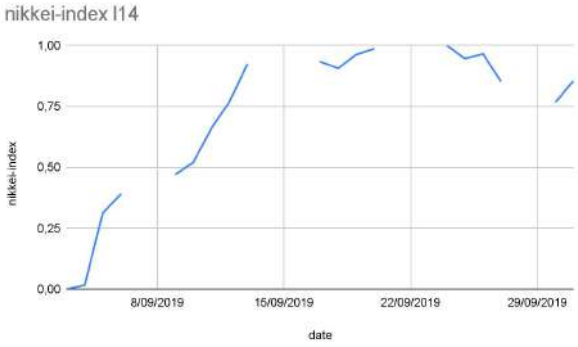


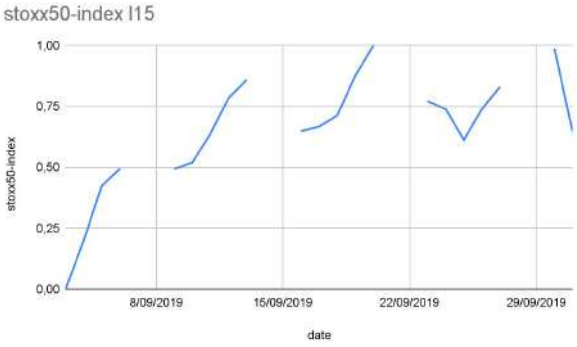


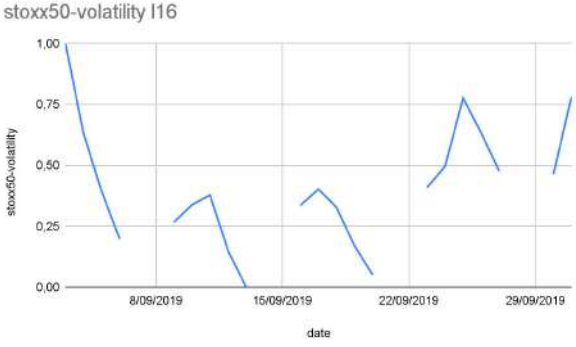


Evento 15 (3/09/2019 - 1/10/2019)

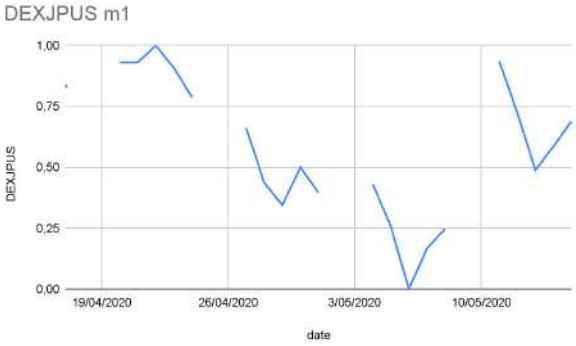


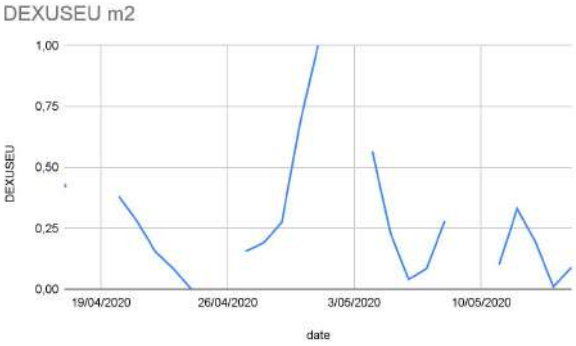


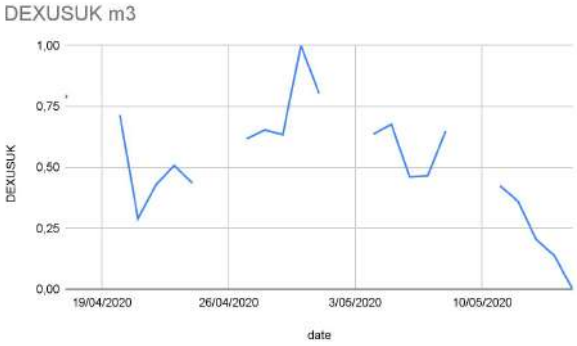


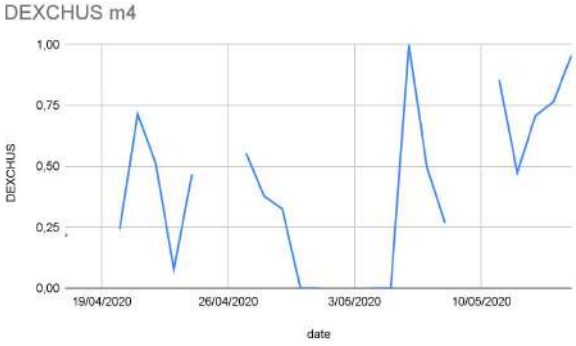


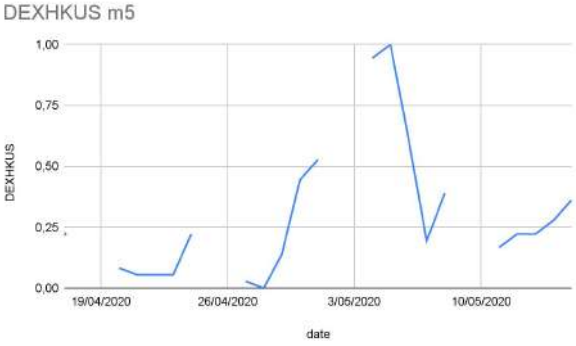
Evento 16 (17/04/2020 - 15/05/2020)

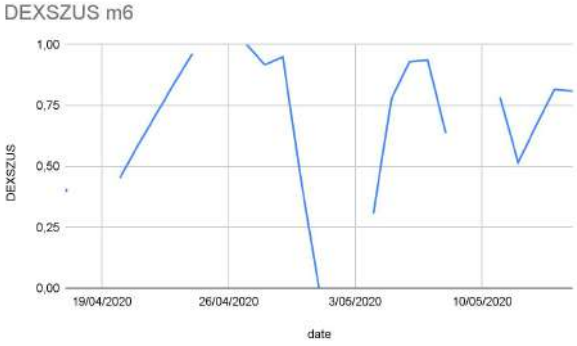




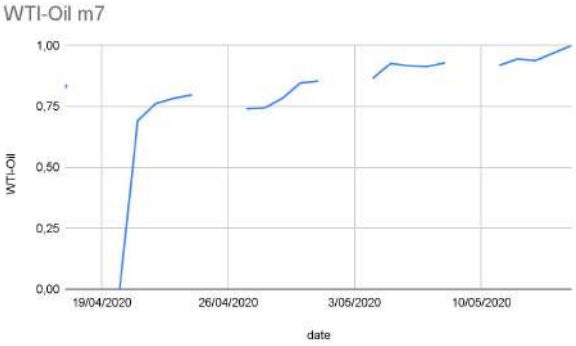


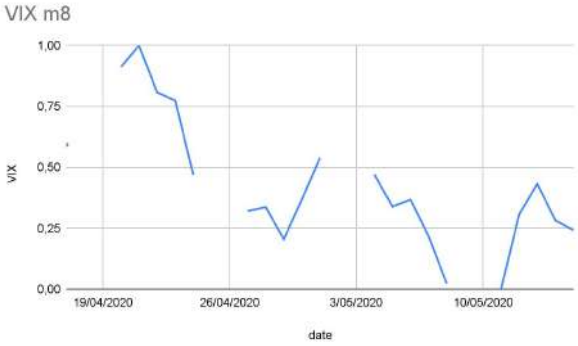


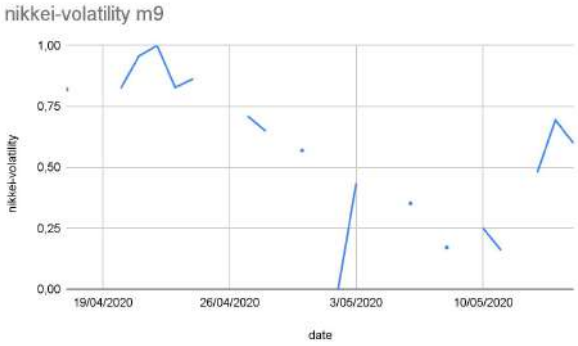


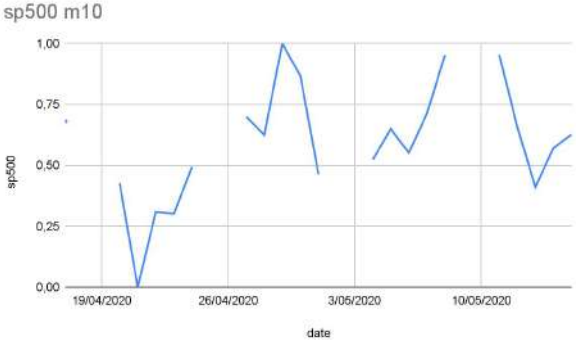


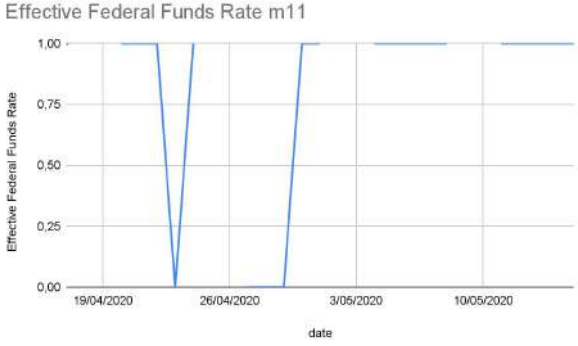
Evento 16 (17/04/2020 - 15/05/2020)

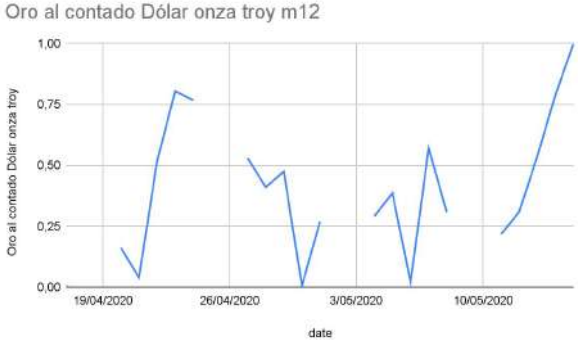




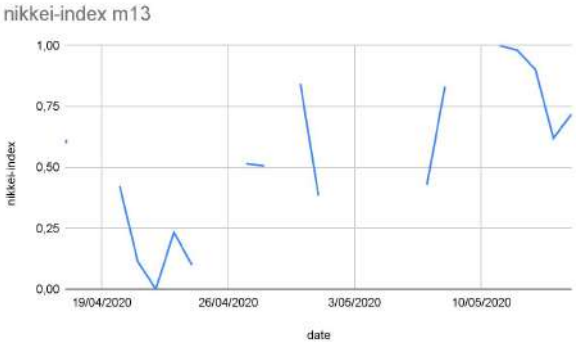


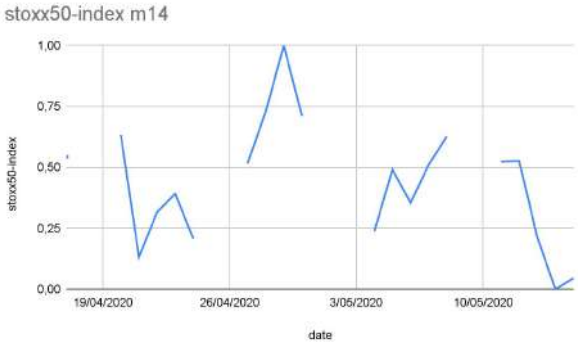


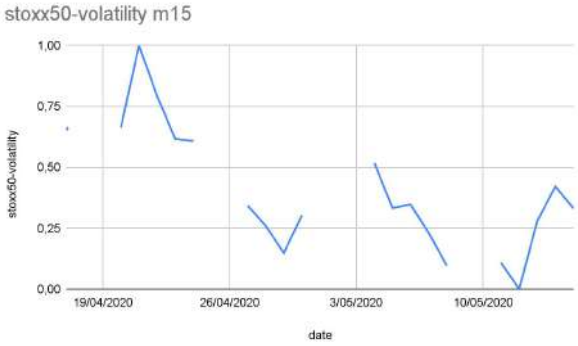




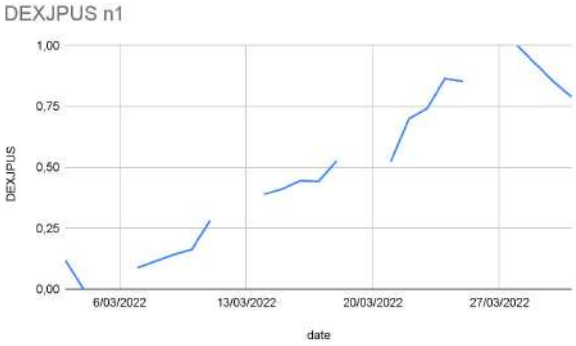
Evento 16 (17/04/2020 - 15/05/2020)

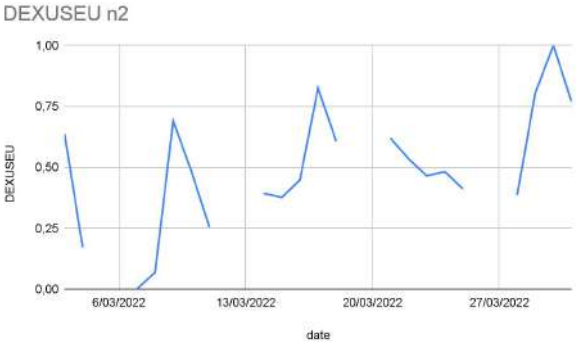


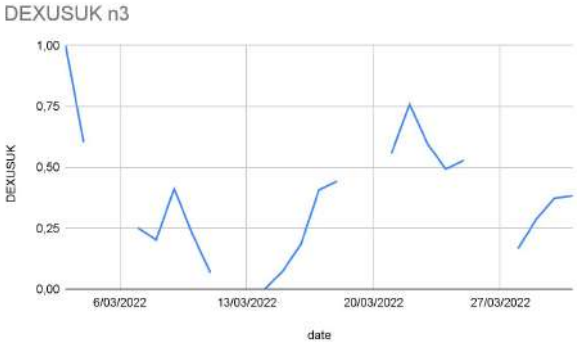


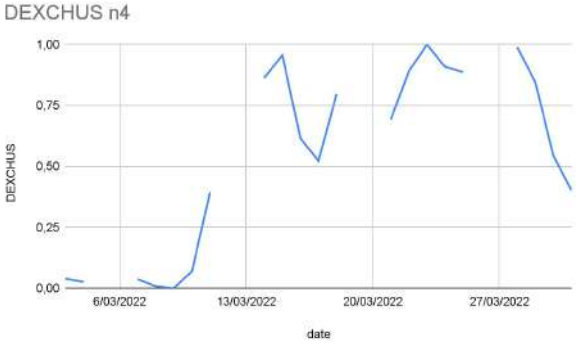


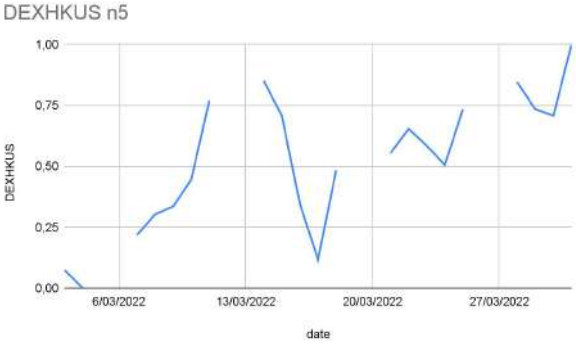
Evento 17 (3/03/2022 - 31/03/2022)

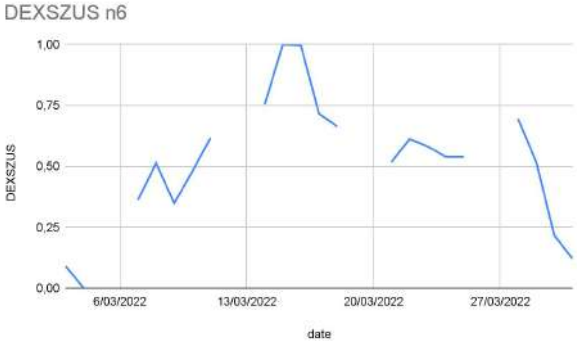




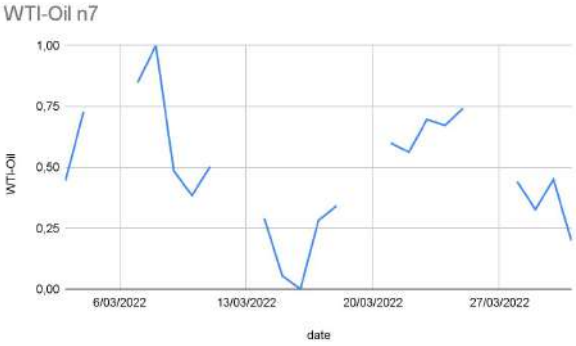


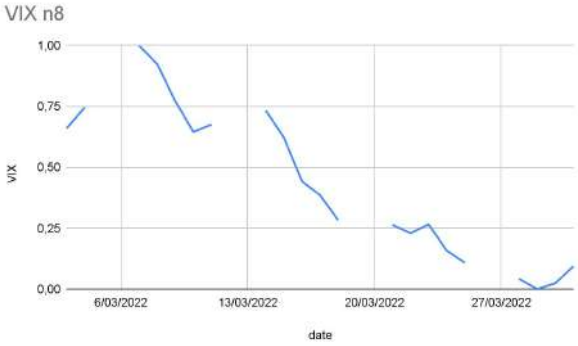


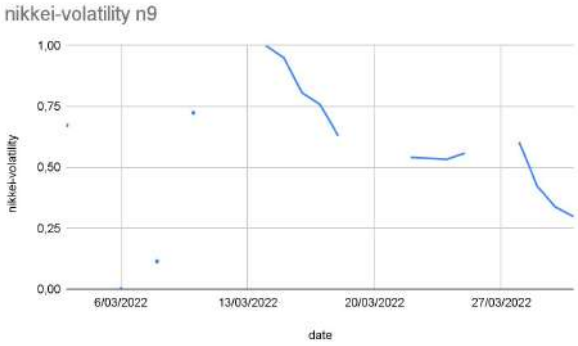


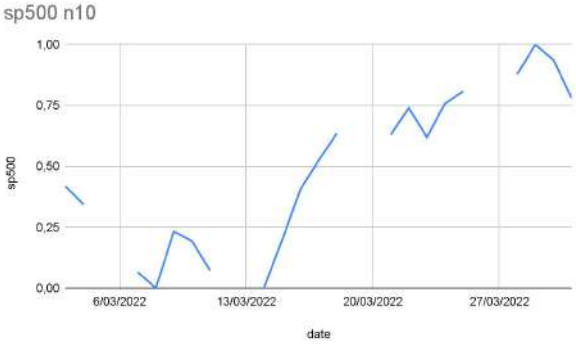


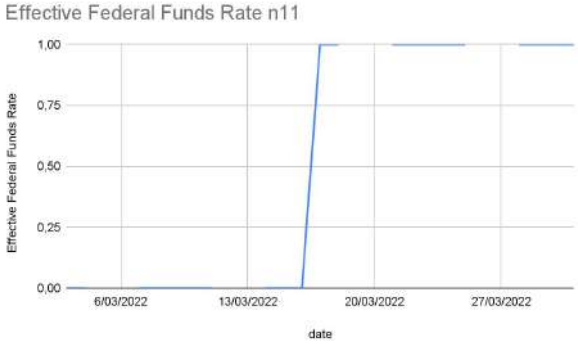
Evento 17 (3/03/2022 - 31/03/2022)



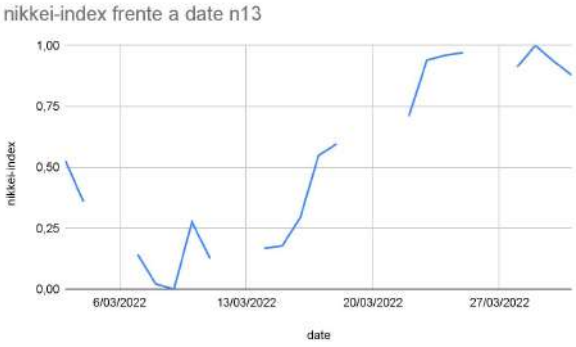


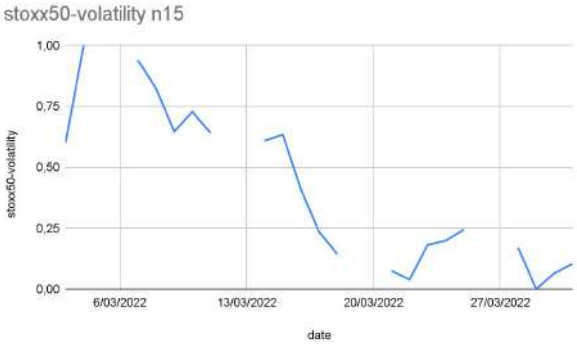




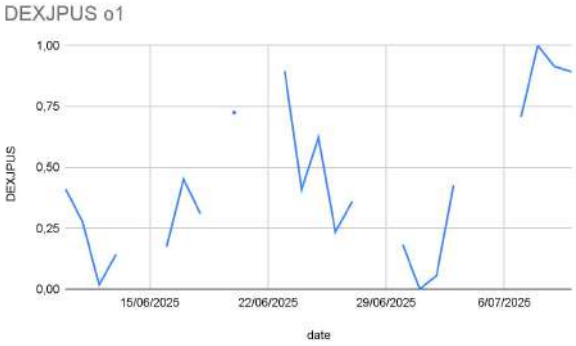


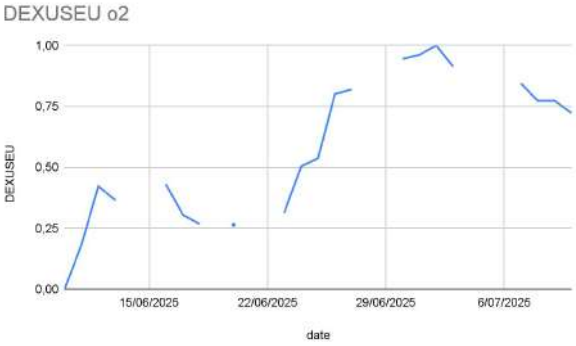
Evento 17 (3/03/2022 - 31/03/2022)

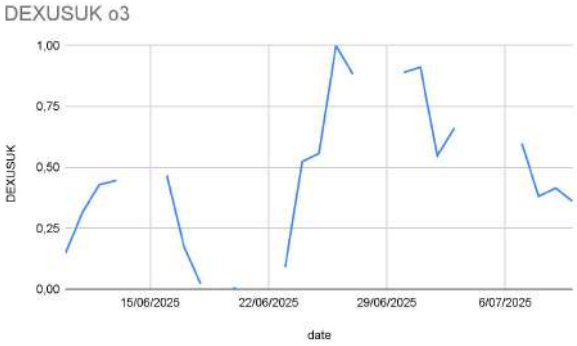


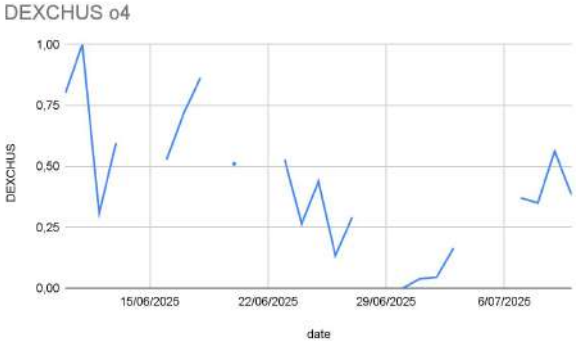


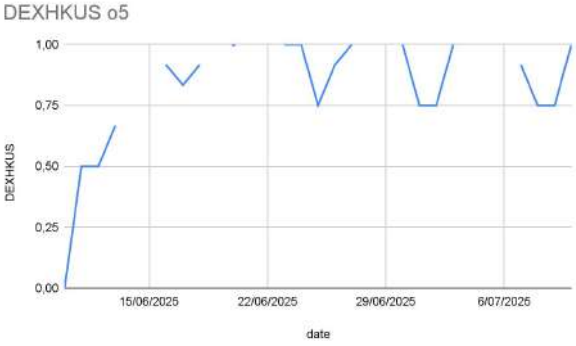
Evento 18 (10/06/2025 - 10/07/2025)

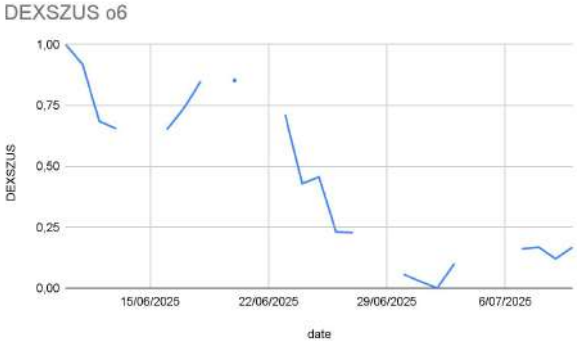




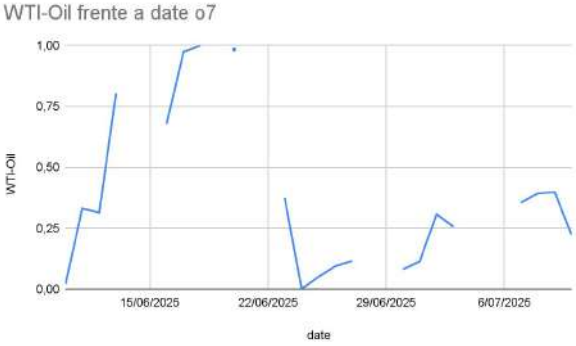


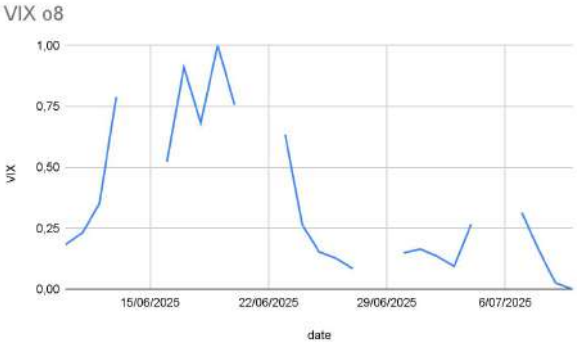


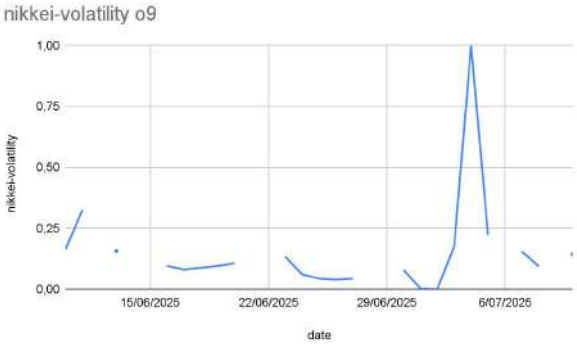


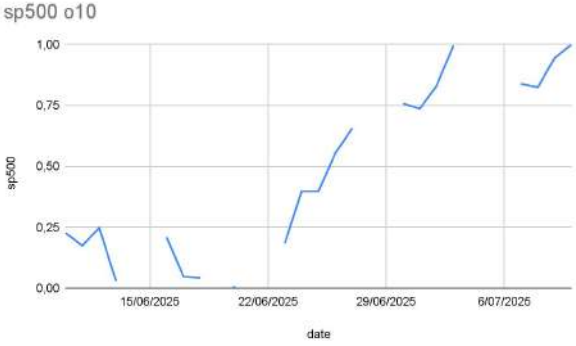


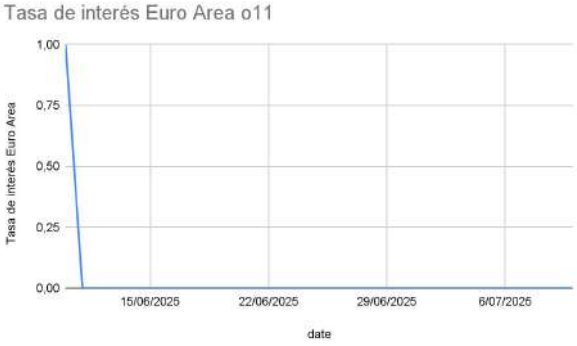
Evento 18 (10/06/2025 - 10/07/2025)



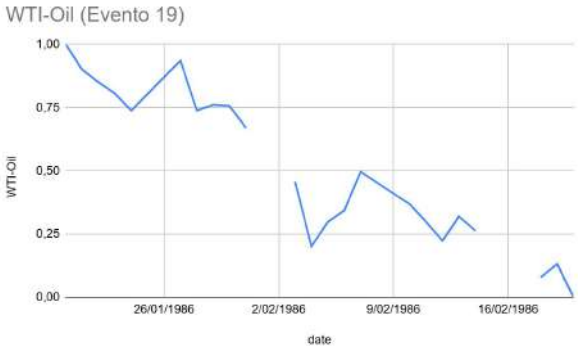


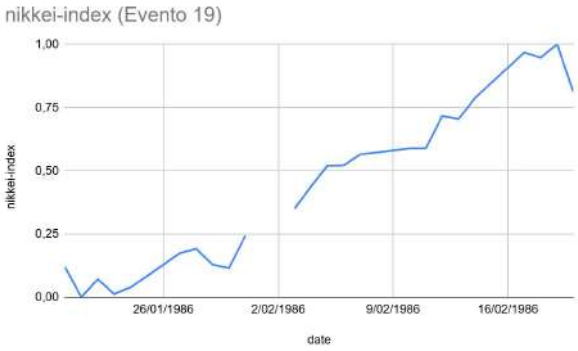


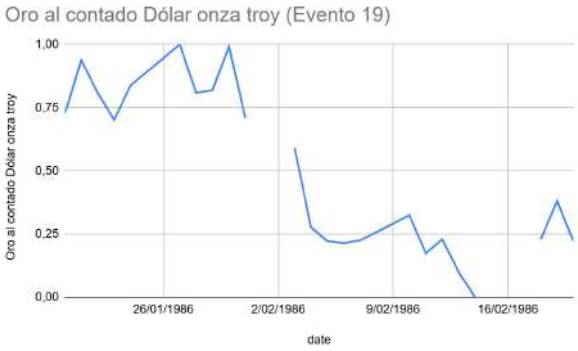




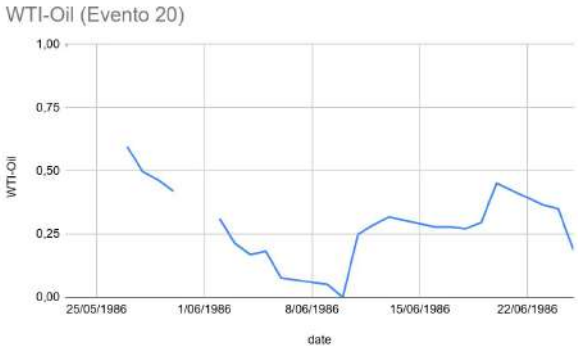
Evento 19 (20/01/1986 - 20/02/1986)

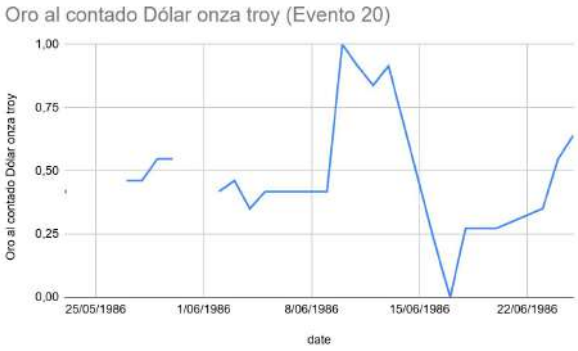


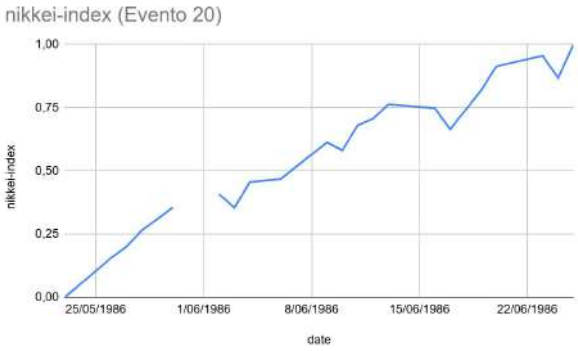




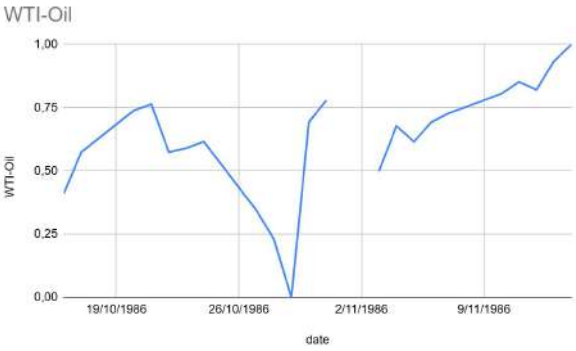
Evento 20 (23/05/1986 - 25/06/1986)

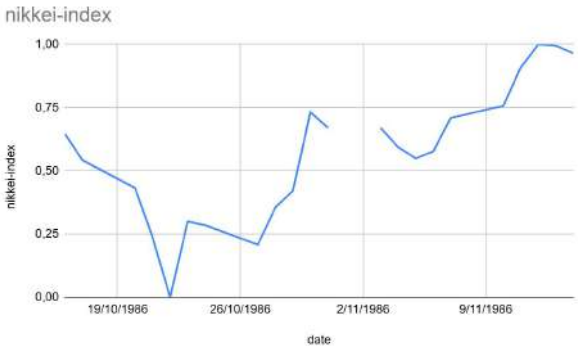




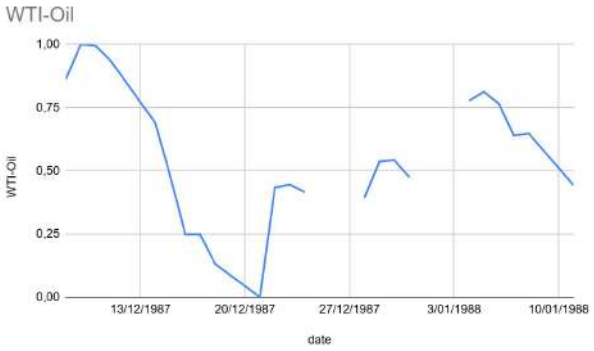


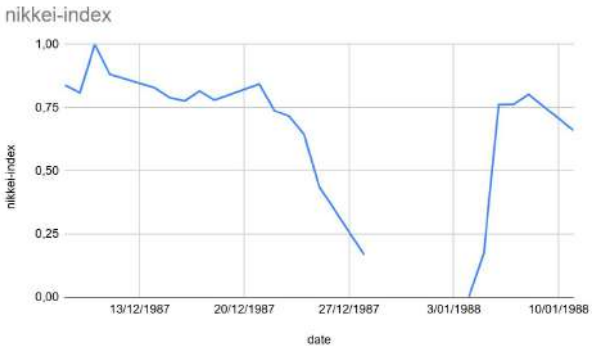
Evento 21 (16/10/1986 - 14/11/1986)

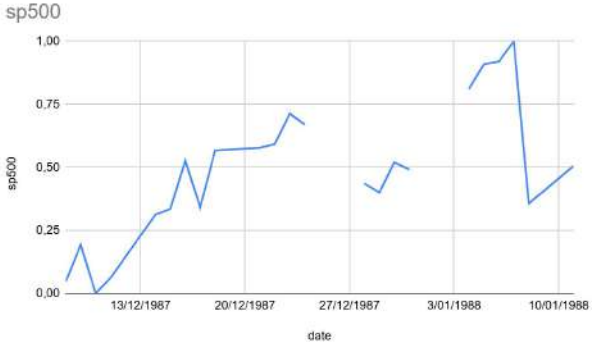




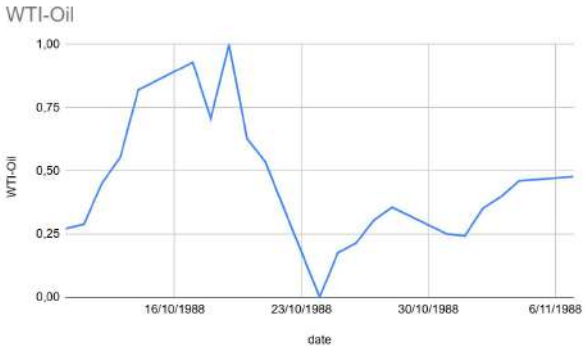
Evento 22 (8/12/1987 - 11/01/1988)

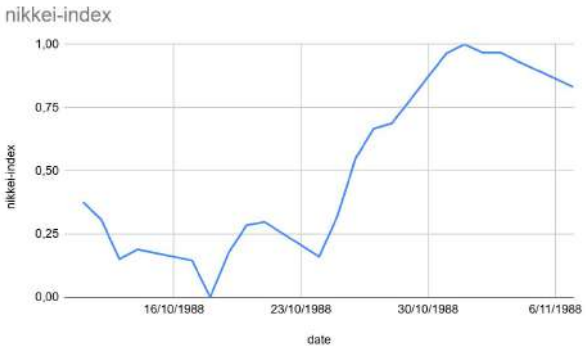


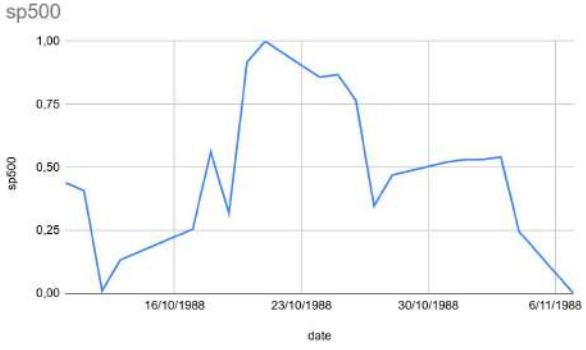




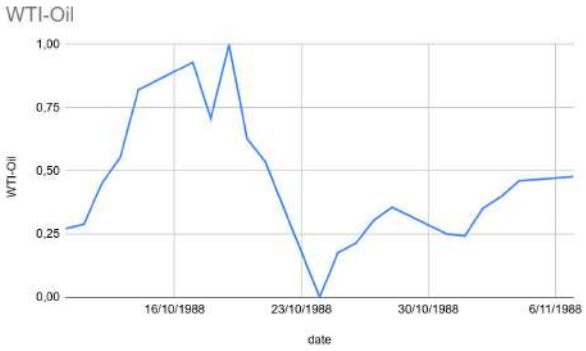
Evento 23 (10/10/1988 - 7/11/1988)

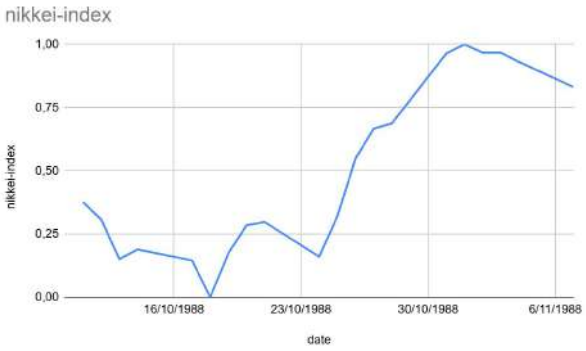


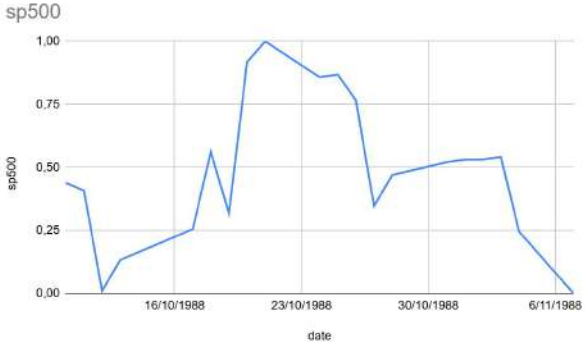




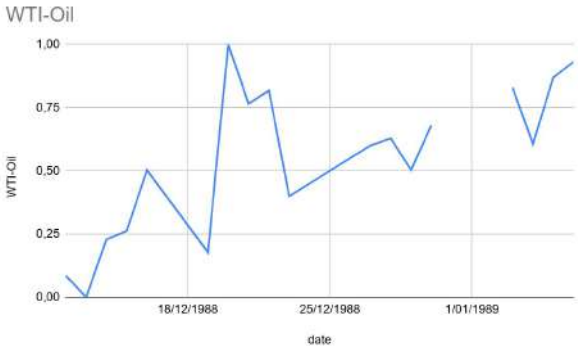
Evento 24 (10/10/1988 - 7/11/1988)

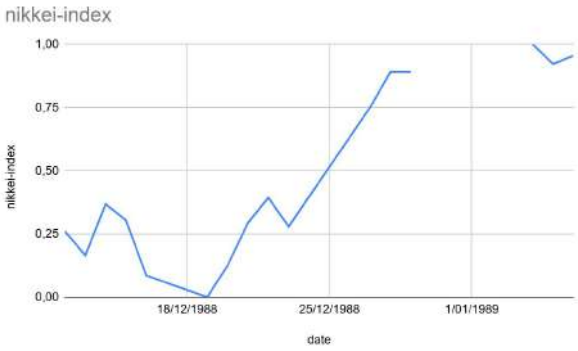


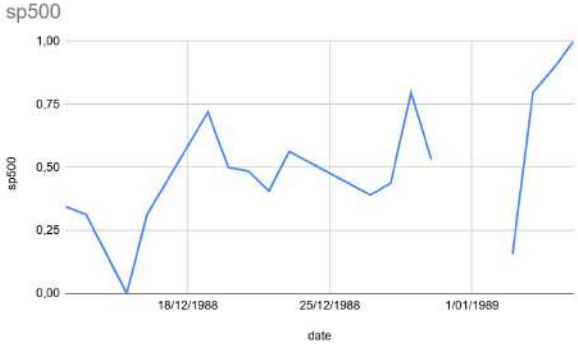




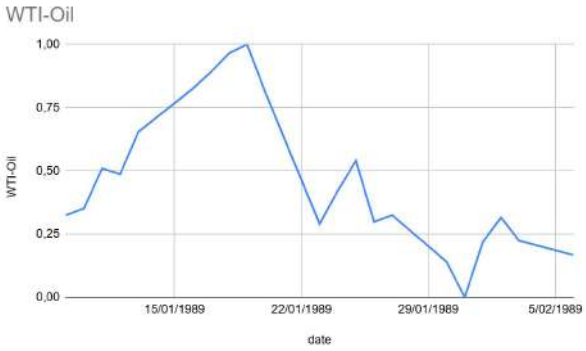
Evento 25 (12/12/1988 - 6/01/1989)

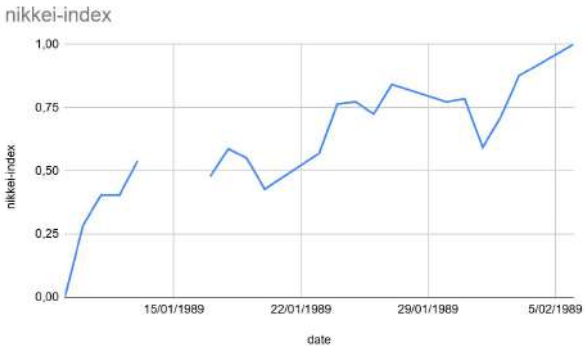


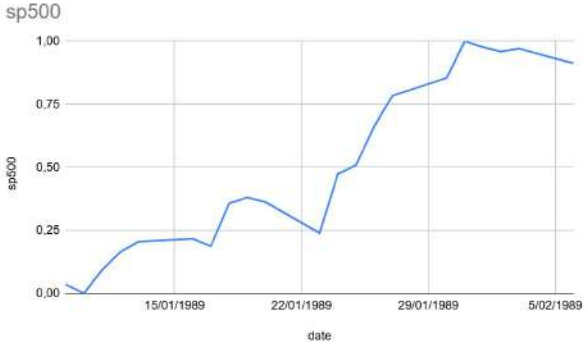




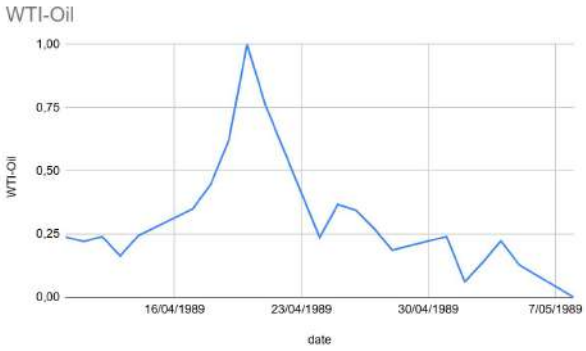
Evento 26 (9/01/1989 - 6/02/1989)

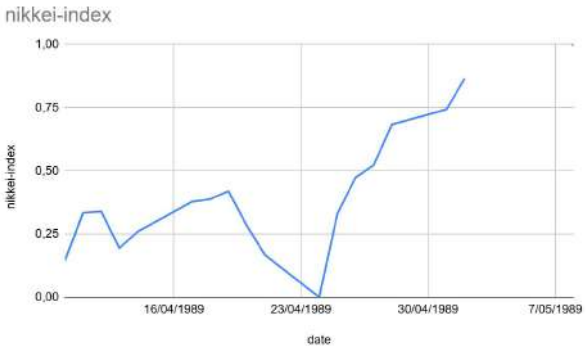


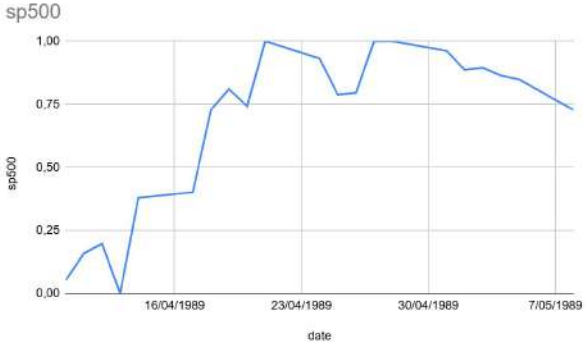




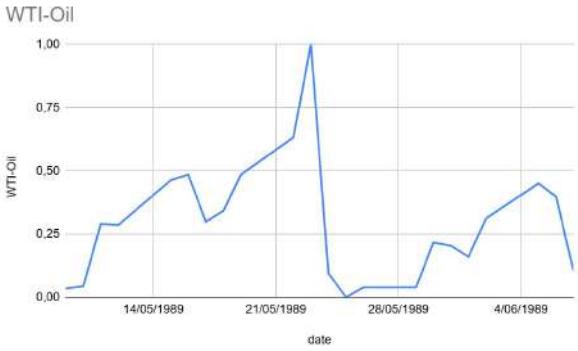
Evento 27 (10/04/1989 - 8/05/1989)

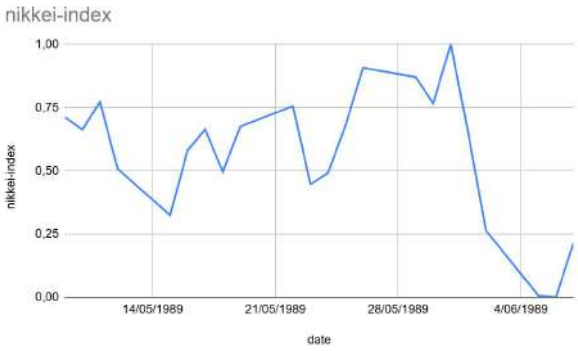


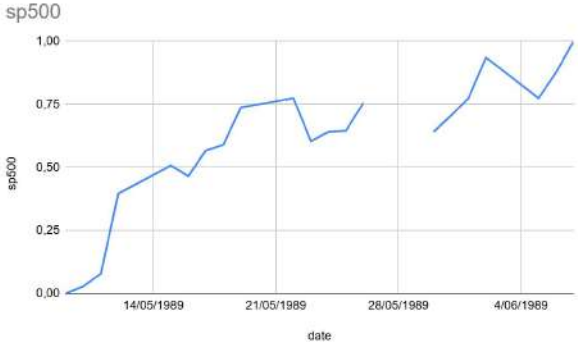




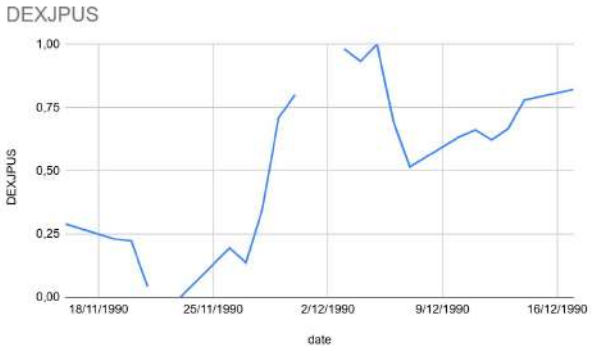
Evento 28 (9/05/1989 - 7/06/1989)

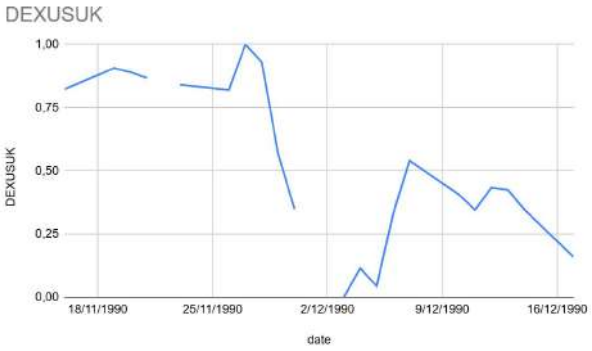


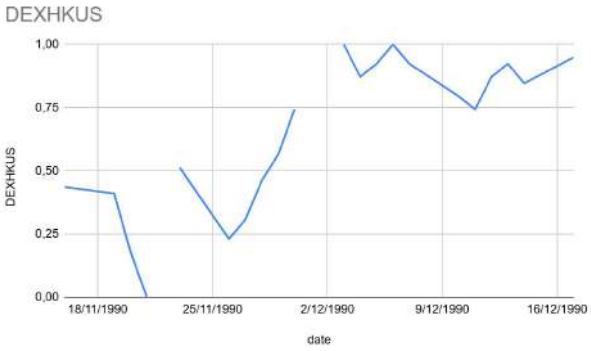


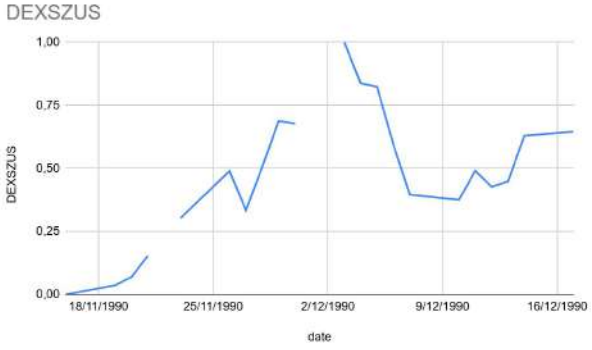


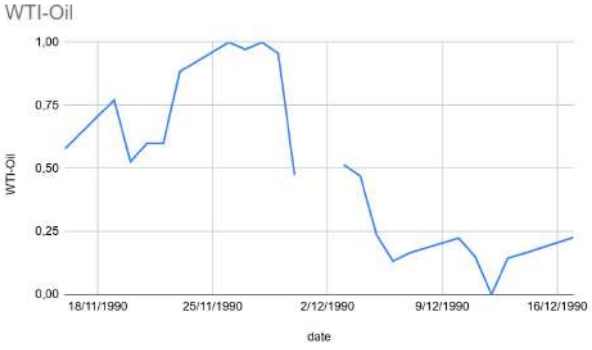
Evento 29 (16/11/1990- 17/12/1990)

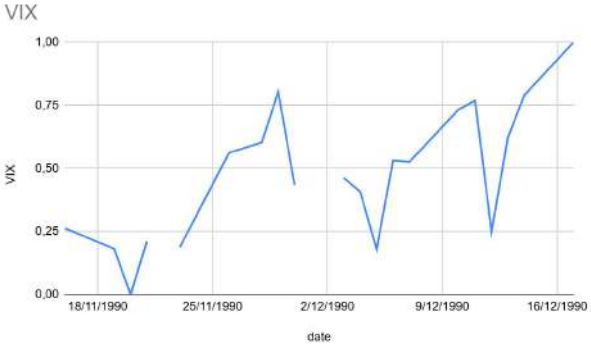




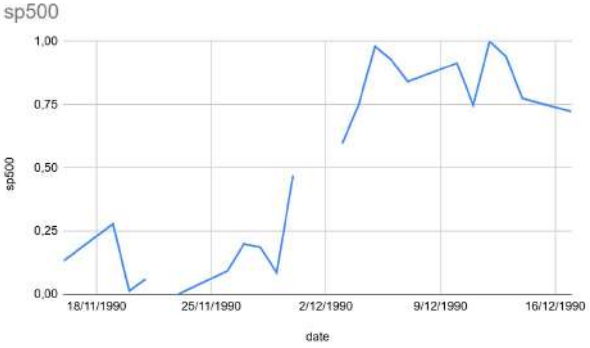


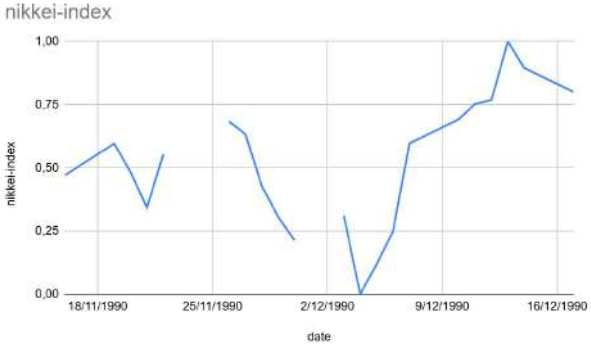




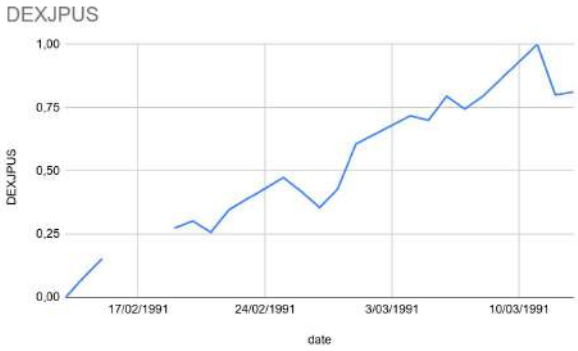


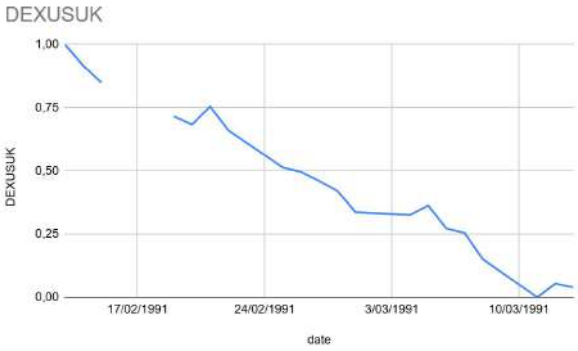
Evento 29 (16/11/1990- 17/12/1990)

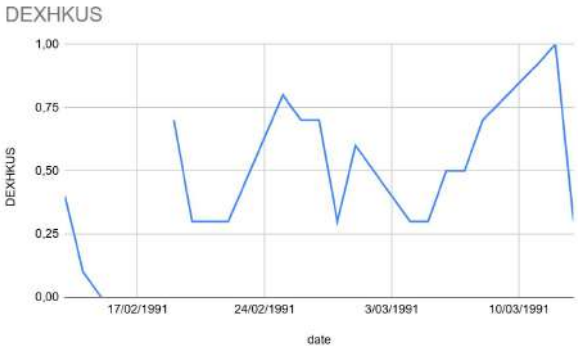


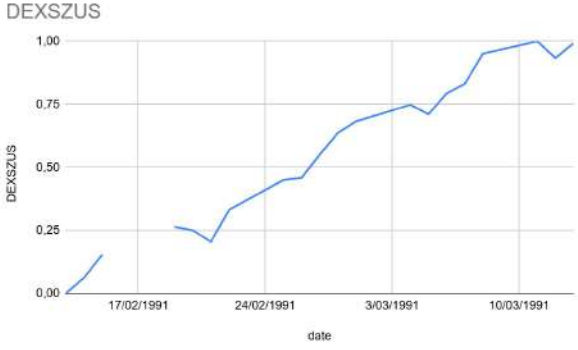


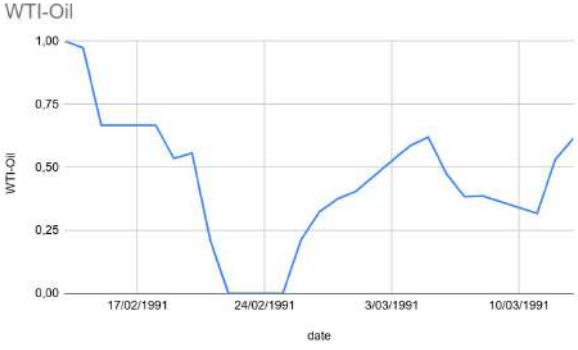
Evento 30 (13/02/1991- 13/03/1991)

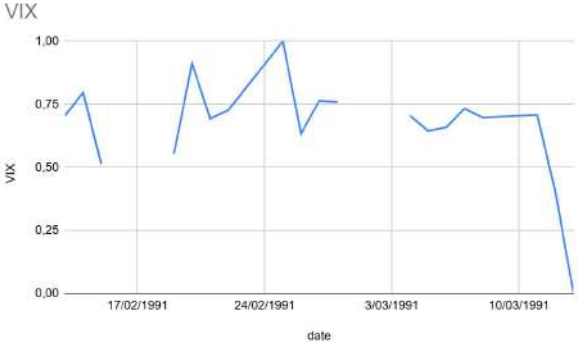




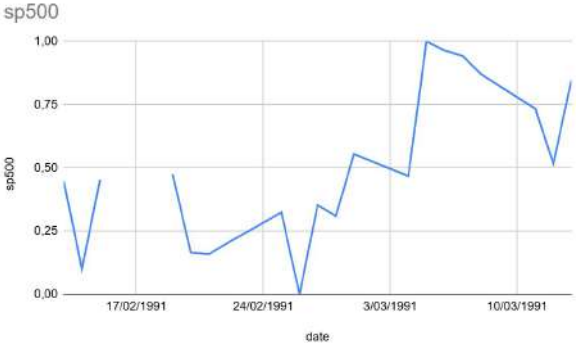


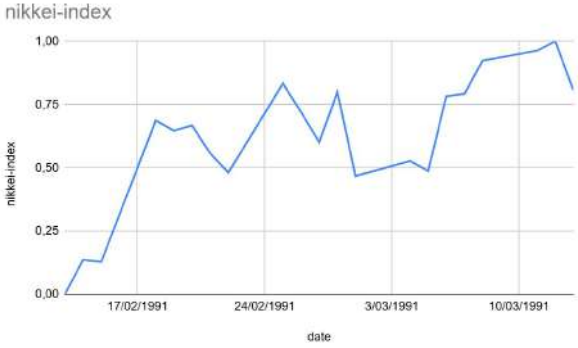




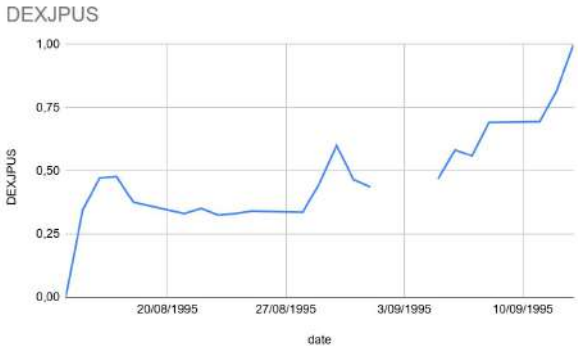


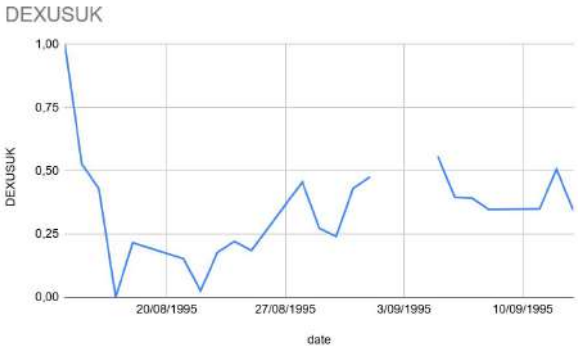
Evento 30 (13/02/1991- 13/03/1991)

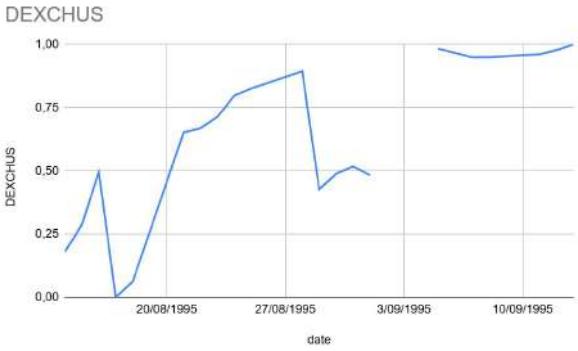


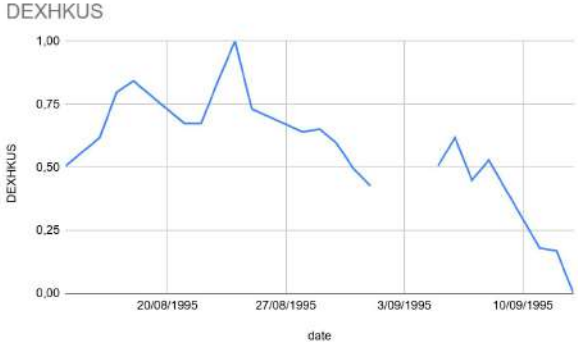


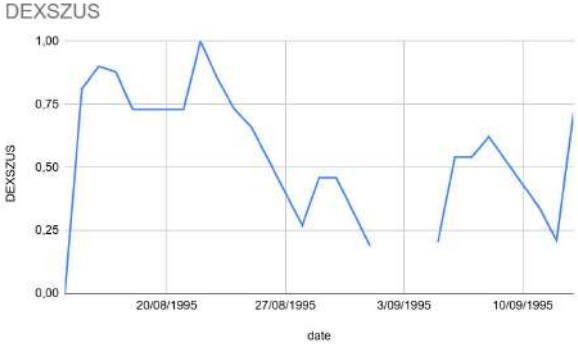
Evento 31 (14/08/1995 - 13/09/1995)

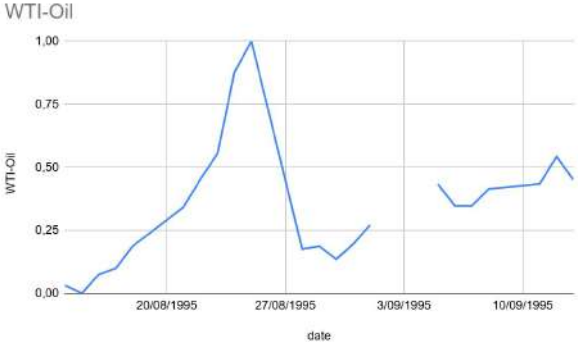




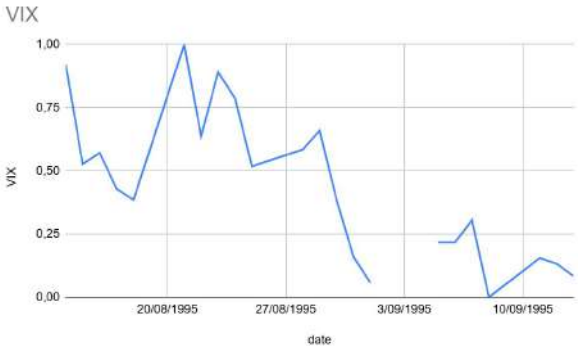


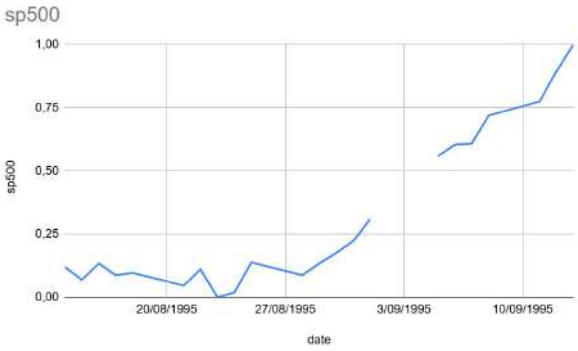


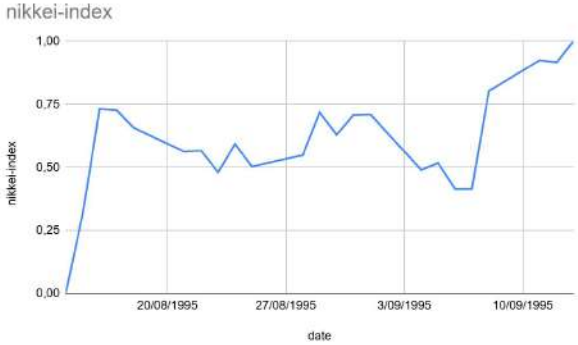




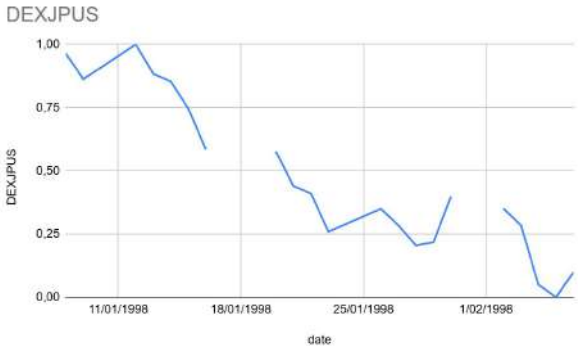
Evento 31 (14/08/1995 - 13/09/1995)

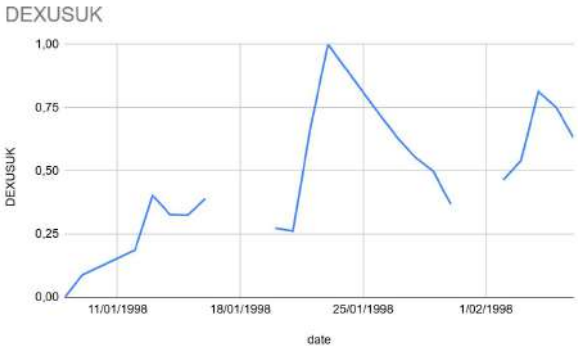


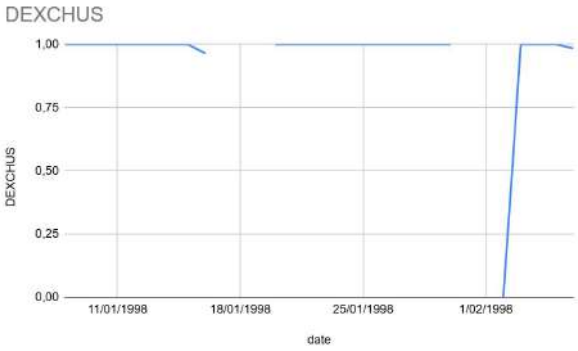


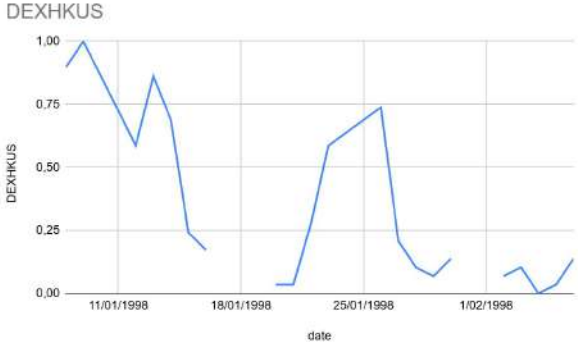


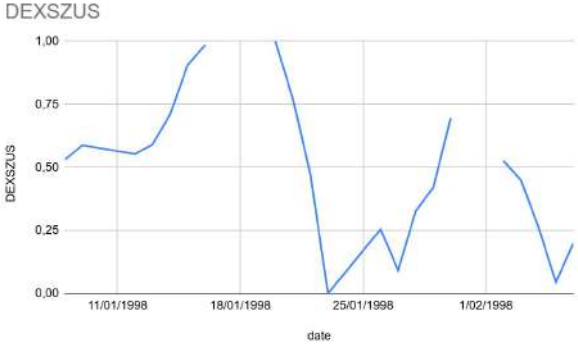
Evento 32 (8/01/1998 - 10/02/1998)

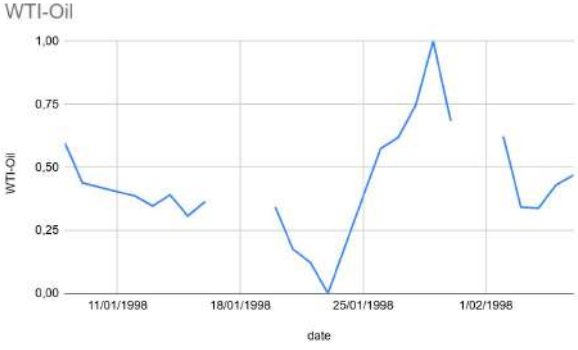




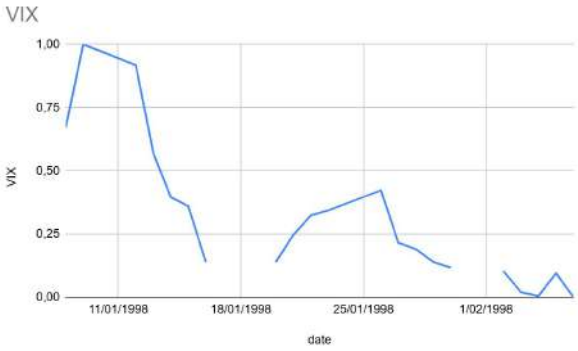


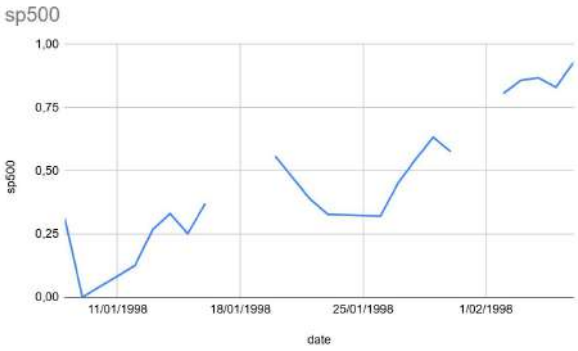


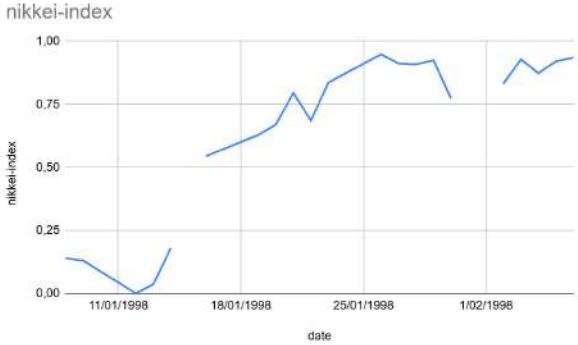




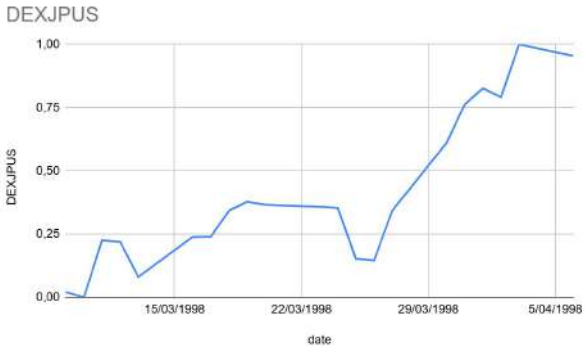
Evento 32 (8/01/1998 - 10/02/1998)

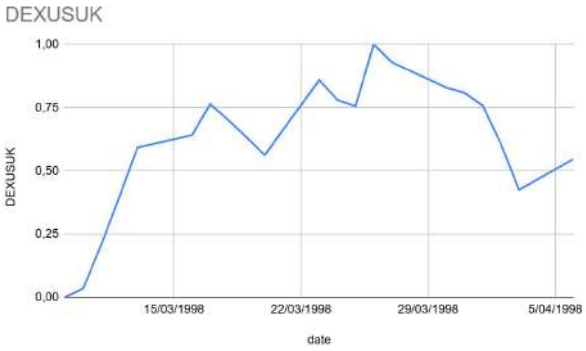


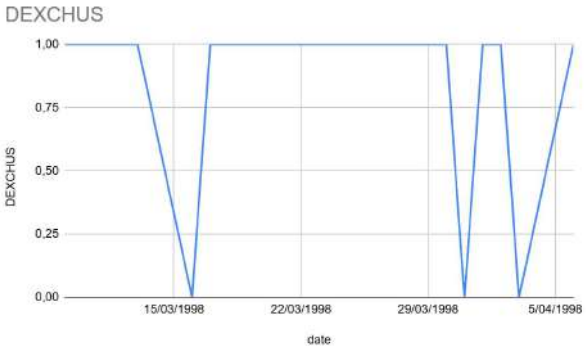


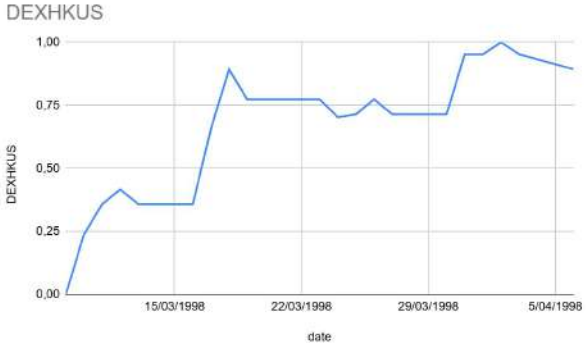


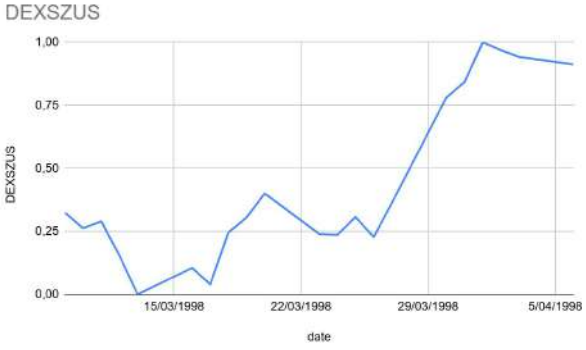
Evento 33 (9/03/1998 - 6/04/1998)

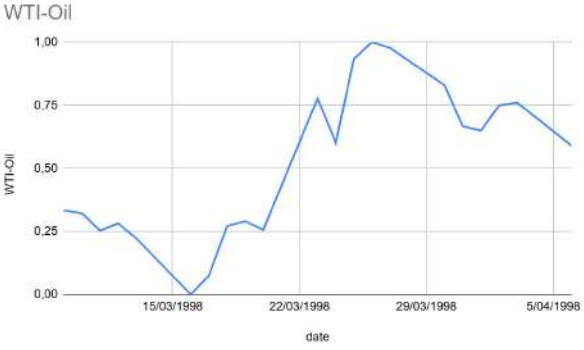




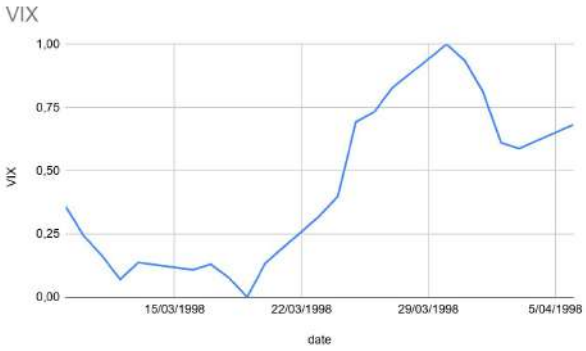


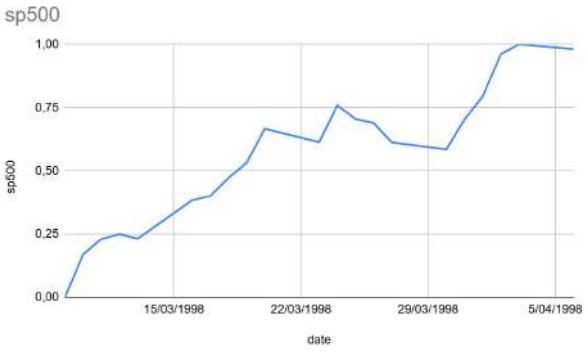


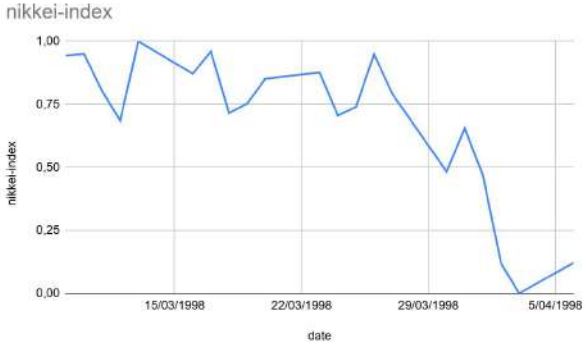




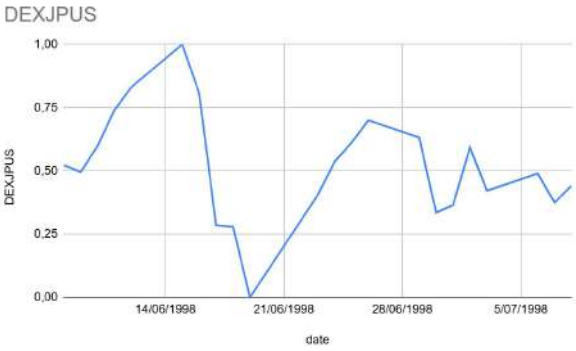
Evento 33 (9/03/1998 - 6/04/1998)

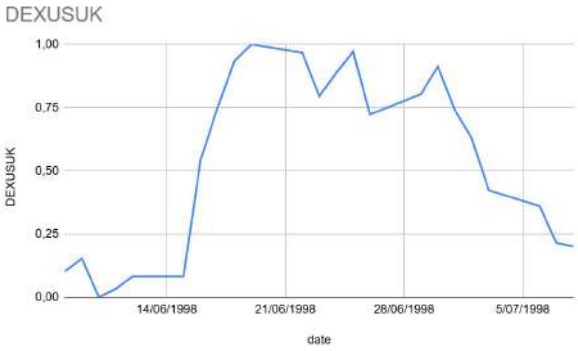


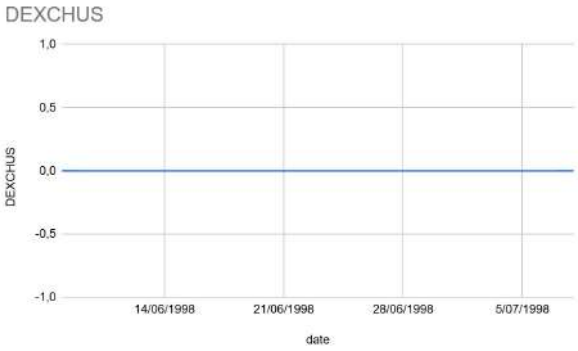


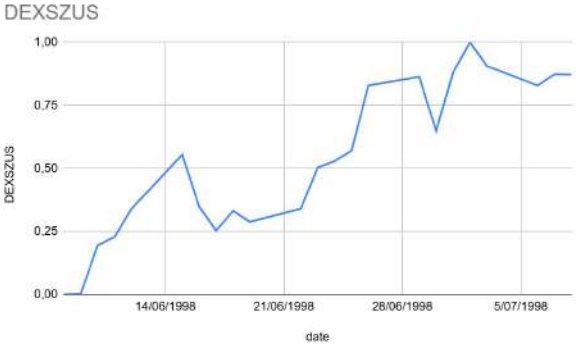


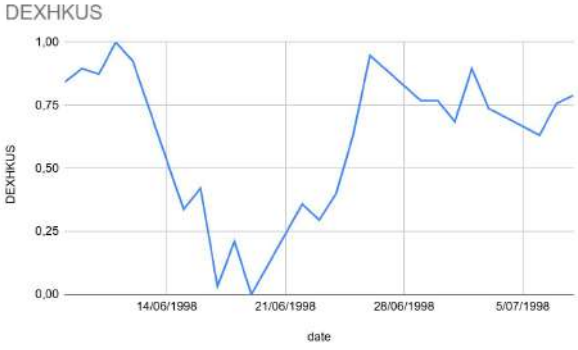
Evento 34 (8/06/1998 - 8/07/1998)

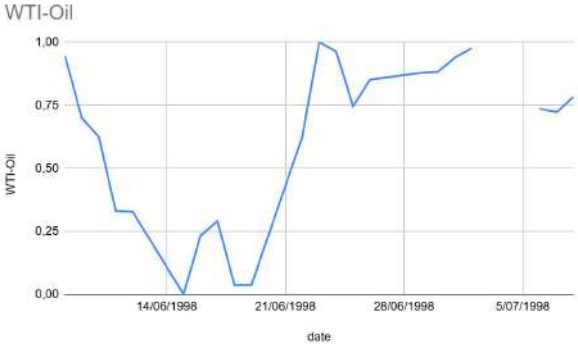




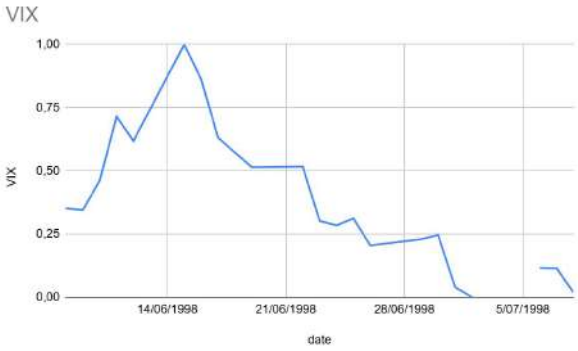


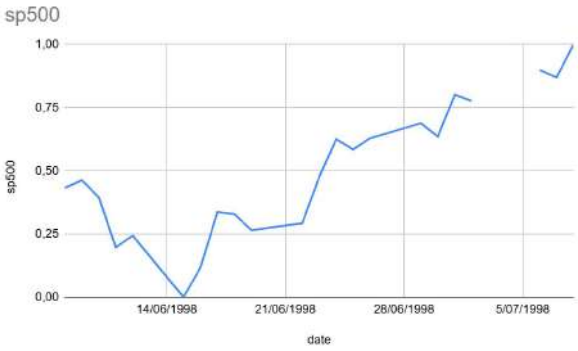


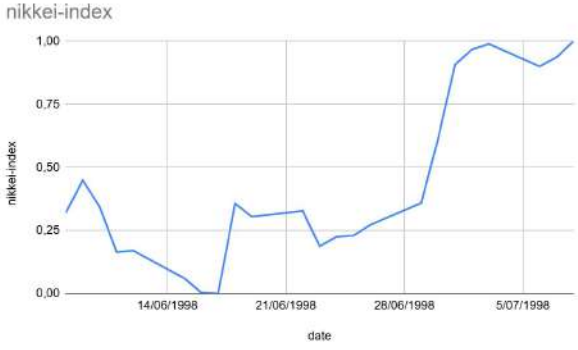




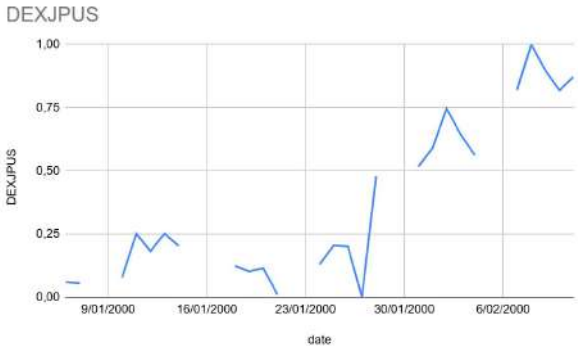
Evento 34 (8/06/1998 - 8/07/1998)

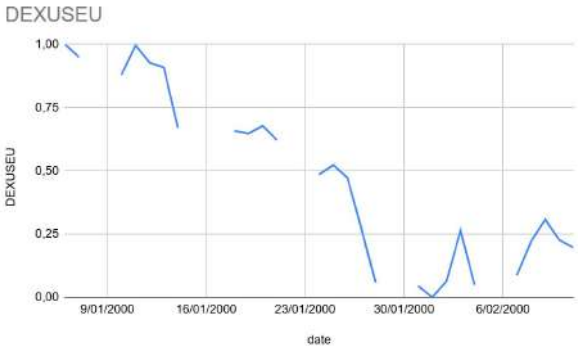


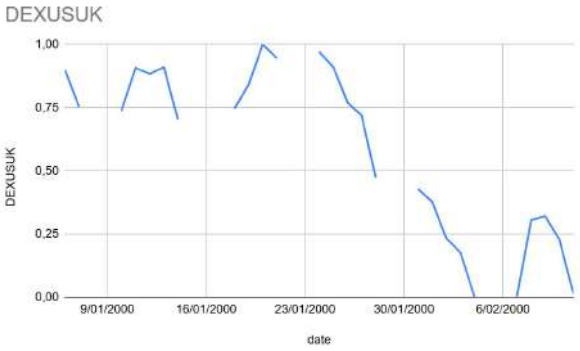


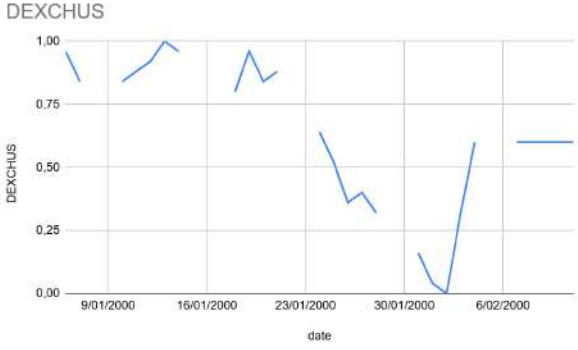


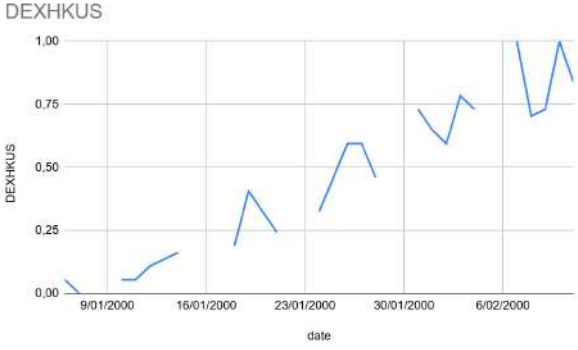
Evento 35 (6/01/2000 - 11/02/2000)

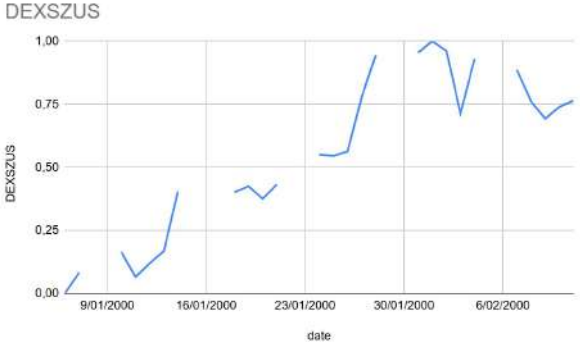




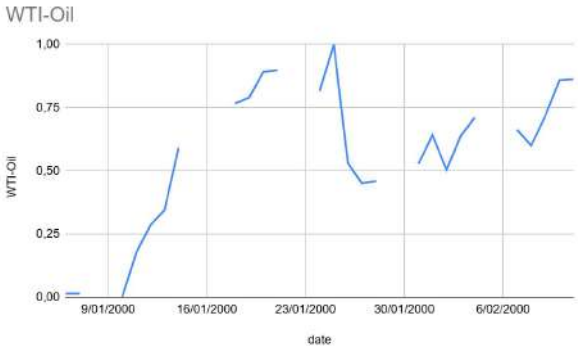


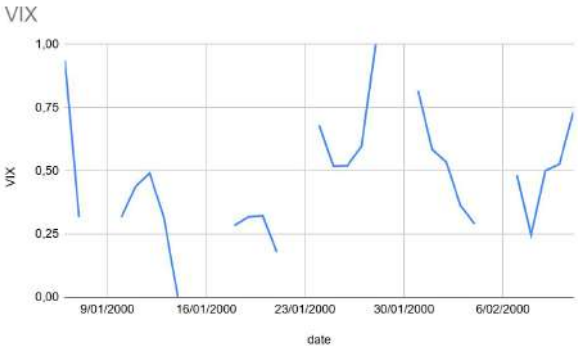


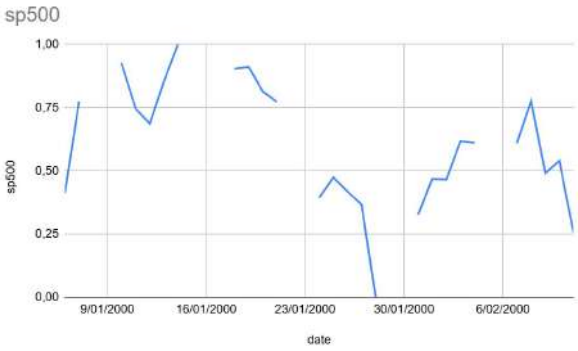


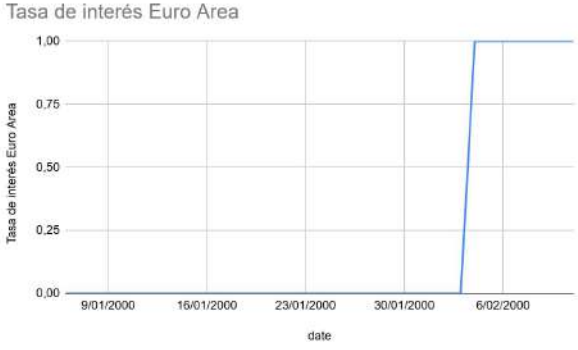


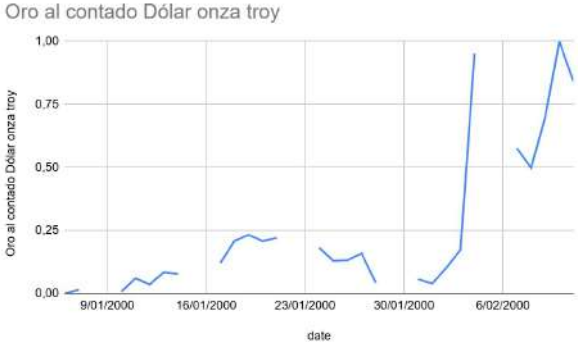
Evento 35 (6/01/2000 - 11/02/2000)

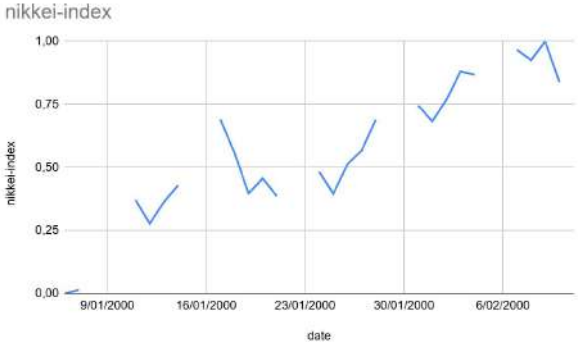




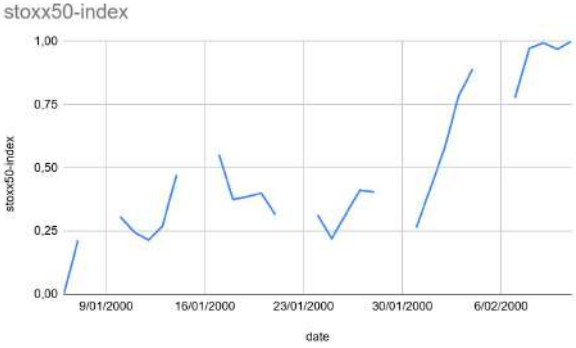




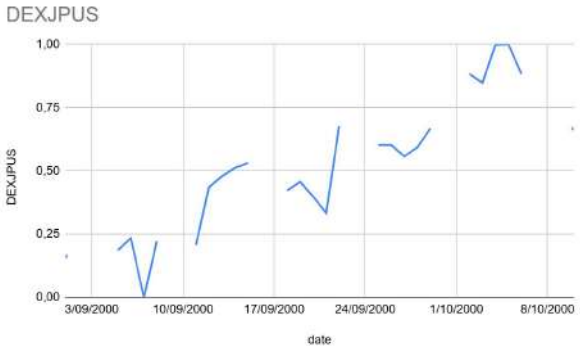


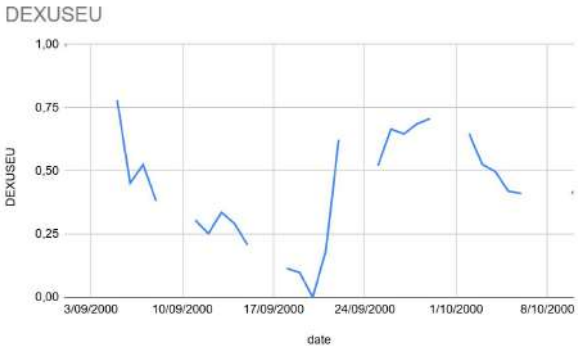


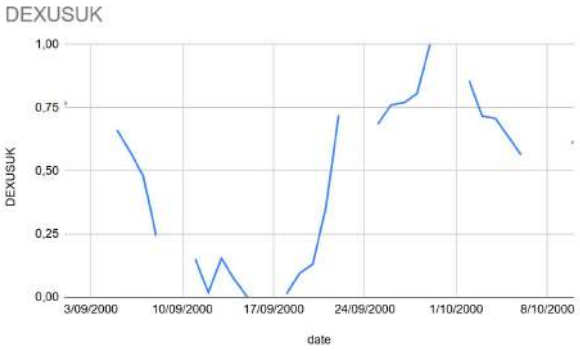
Evento 35 (6/01/2000 - 11/02/2000)

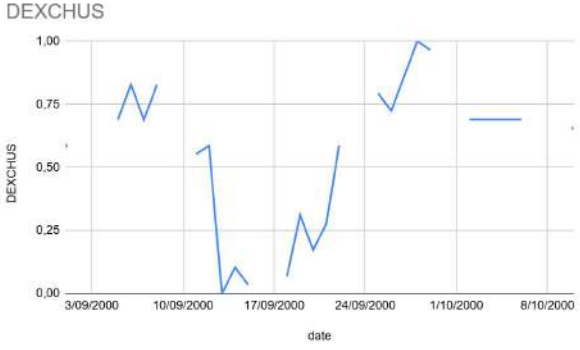


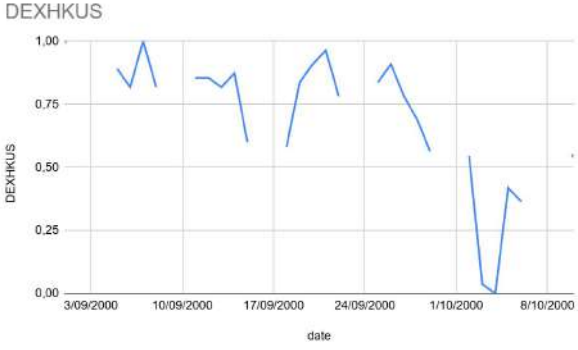
Evento 36 (1/09/2000 - 10/10/2000)

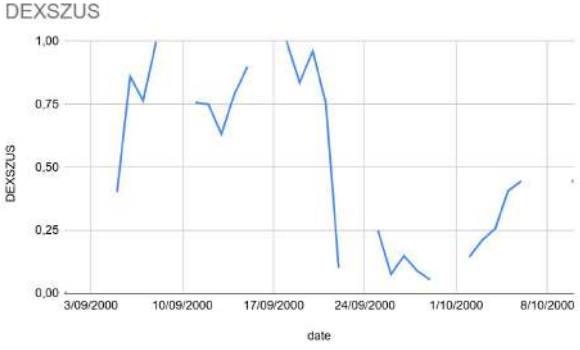




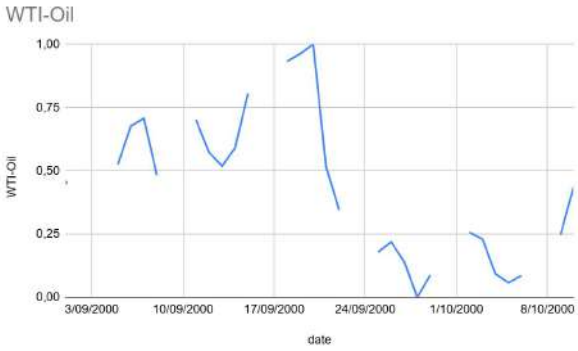


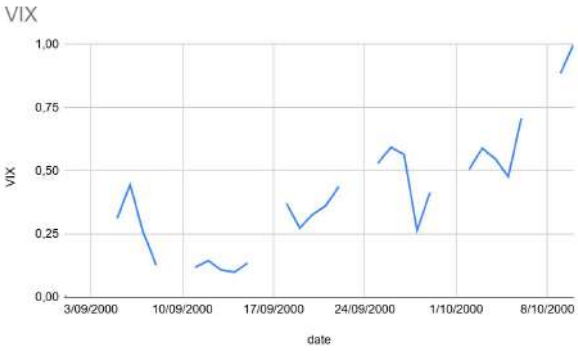


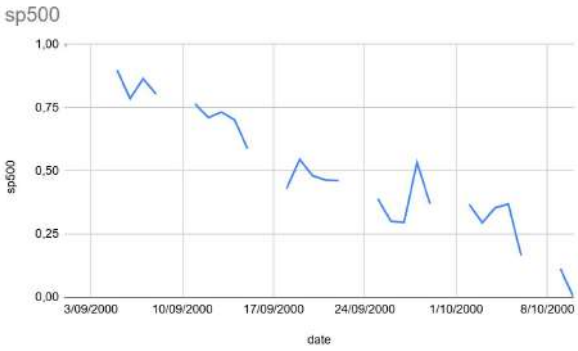


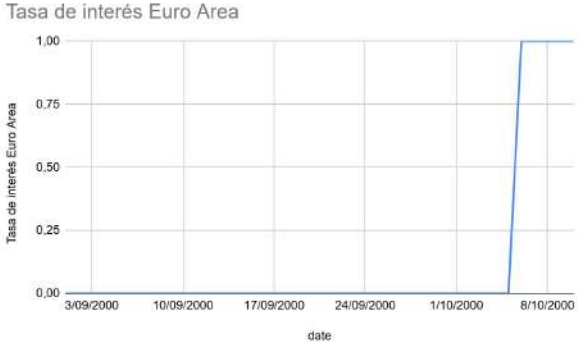


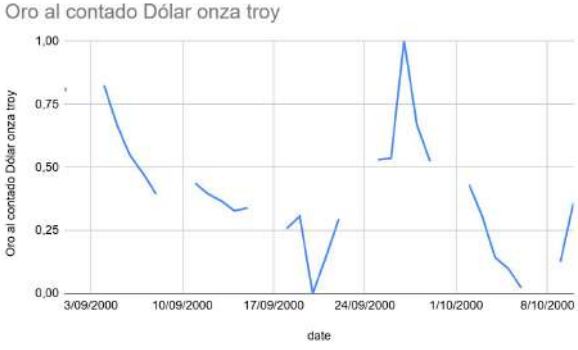
Evento 36 (1/09/2000 - 10/10/2000)

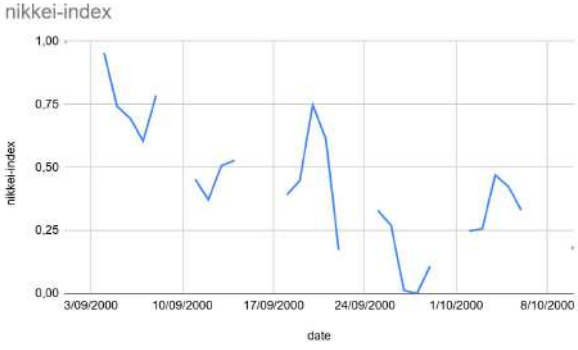




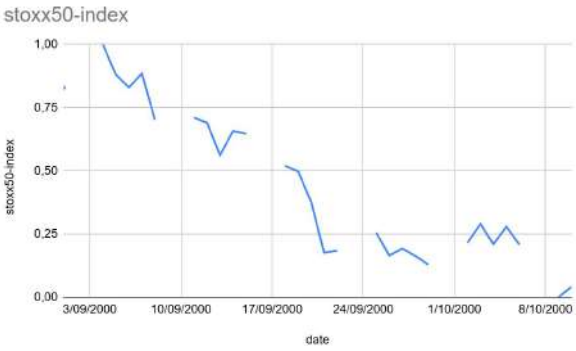


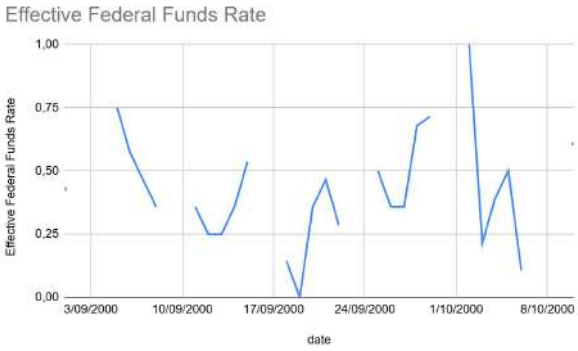




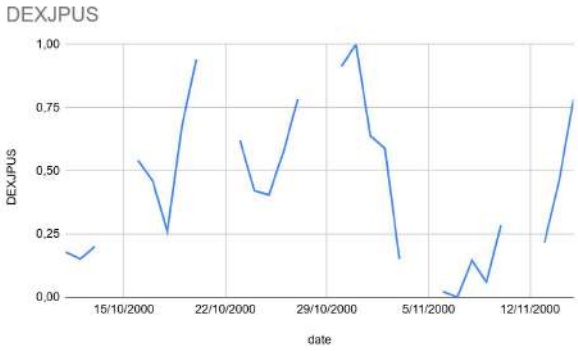


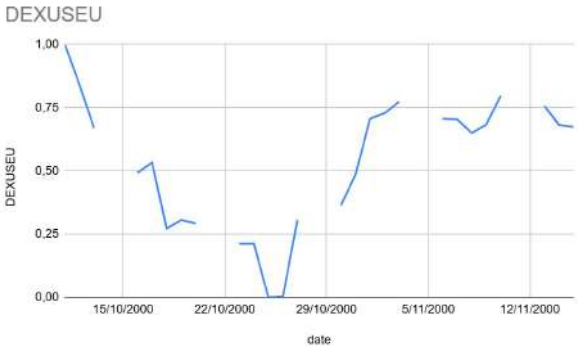
Evento 36 (1/09/2000 - 10/10/2000)

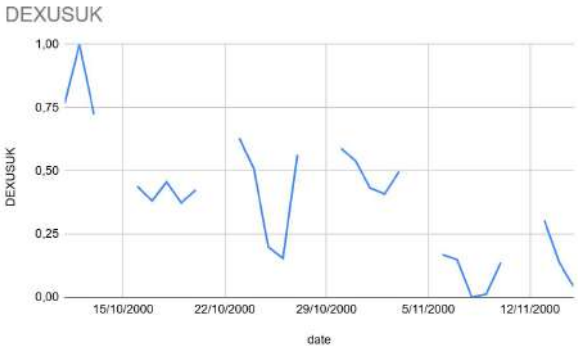


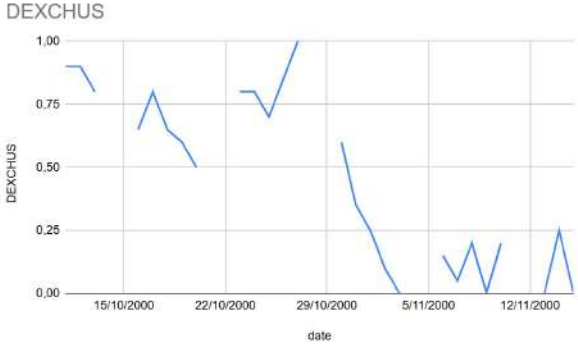


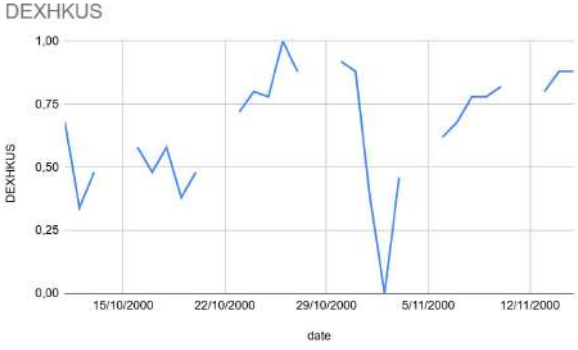
Evento 37 (11/10/2000 - 15/11/2000)

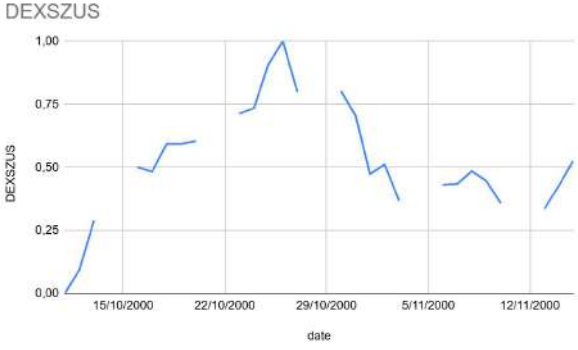




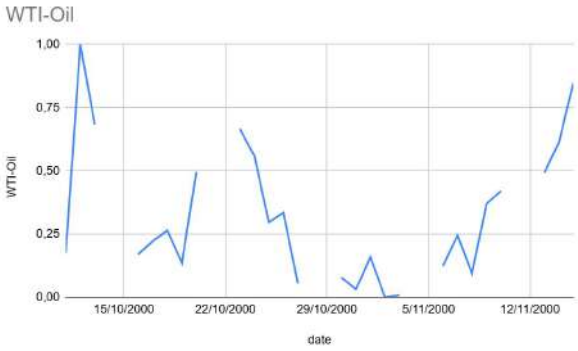


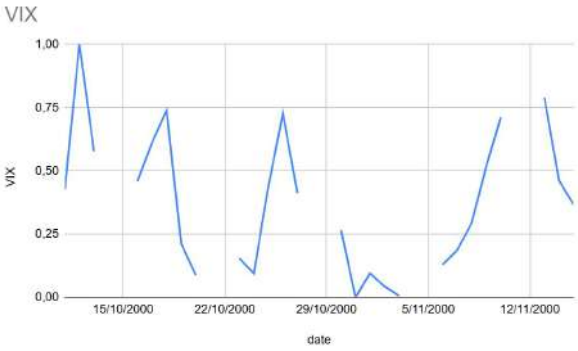


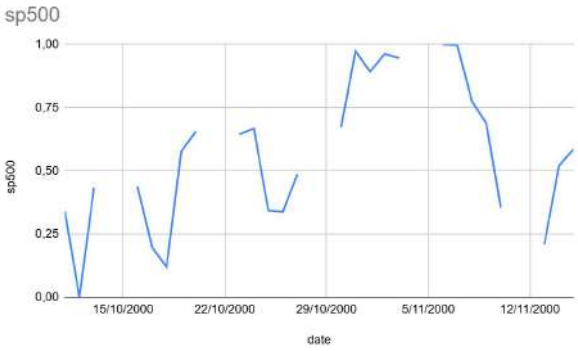


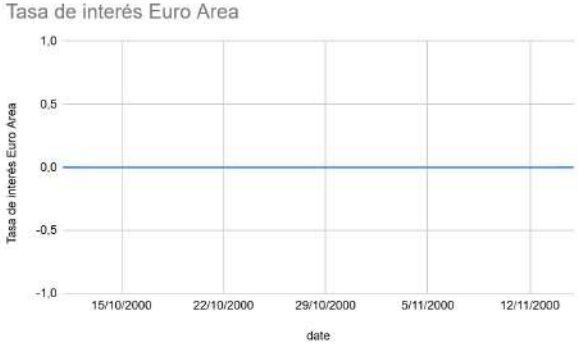


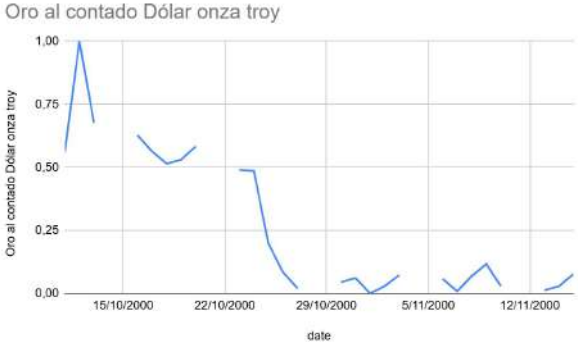
Evento 37 (11/10/2000 - 15/11/2000)

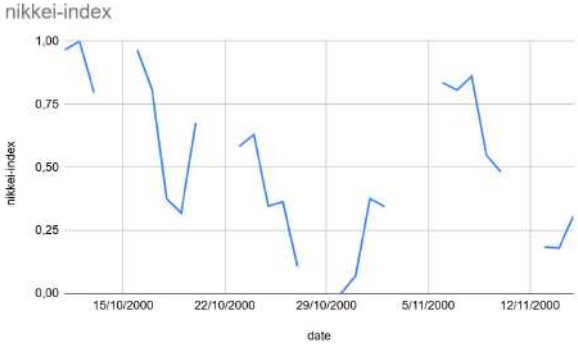




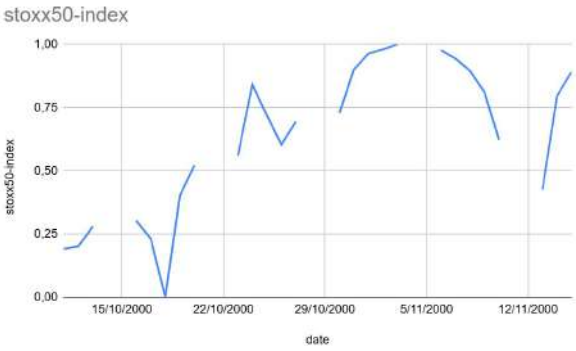


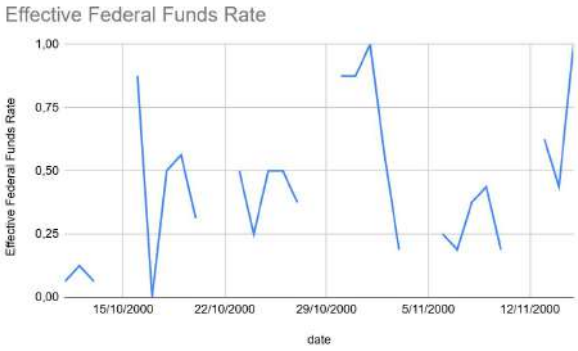




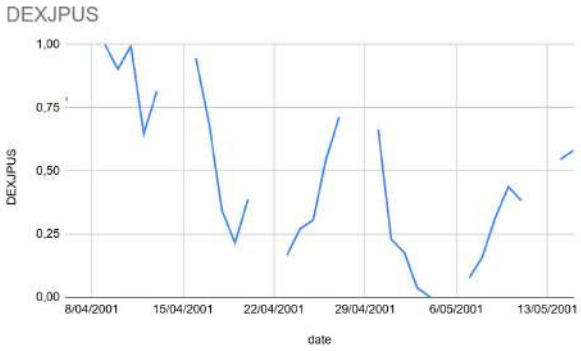


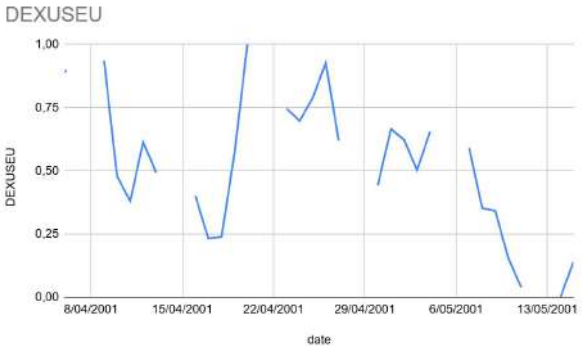
Evento 37 (11/10/2000 - 15/11/2000)

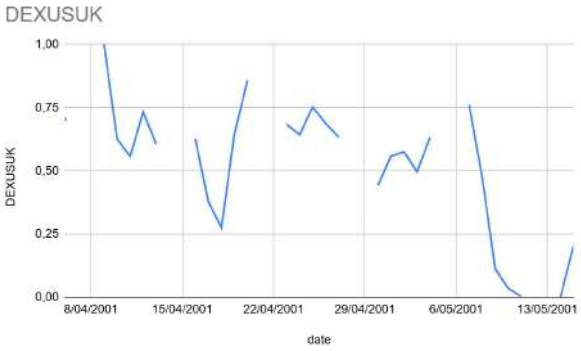


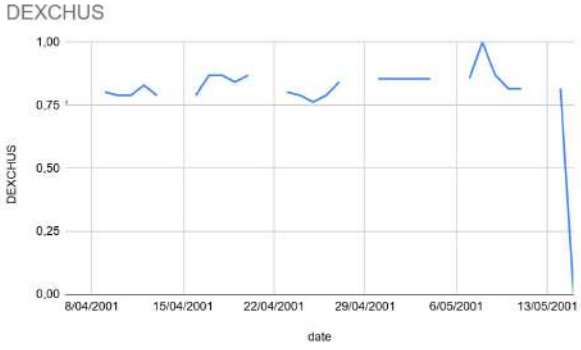


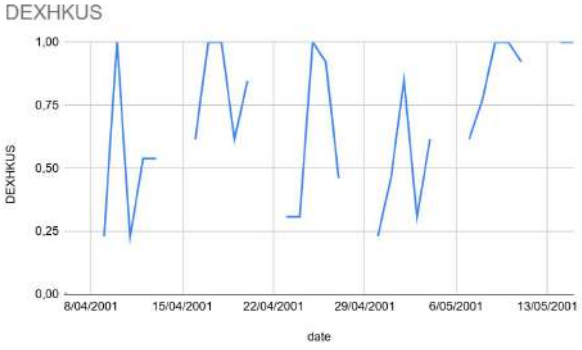
Evento 38 (6/04/2001 - 15/05/2001)

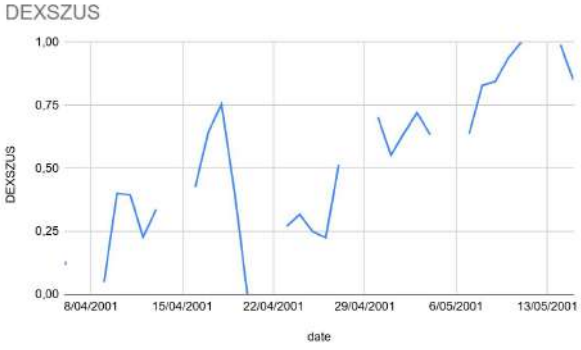




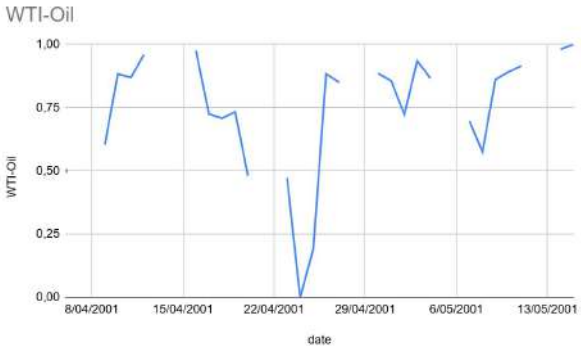


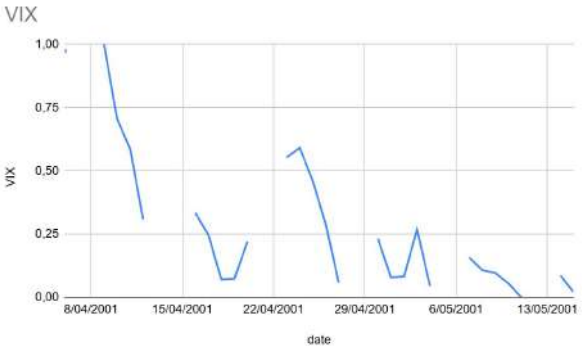


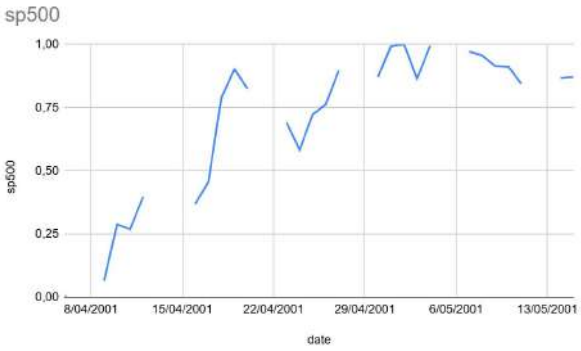


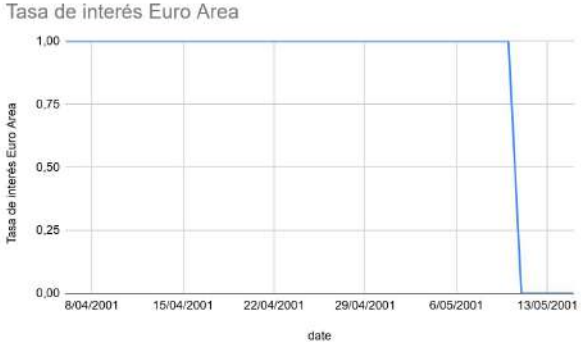


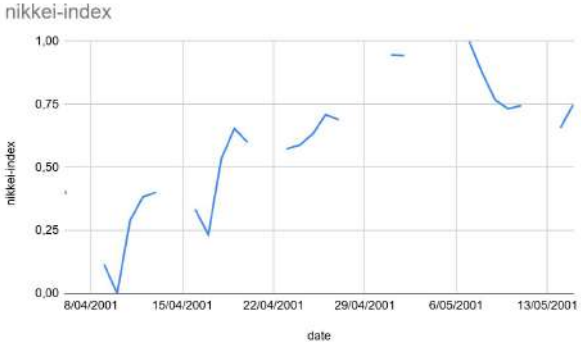
Evento 38 (6/04/2001 - 15/05/2001)



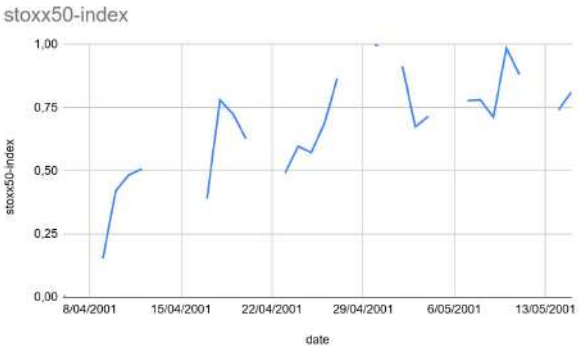


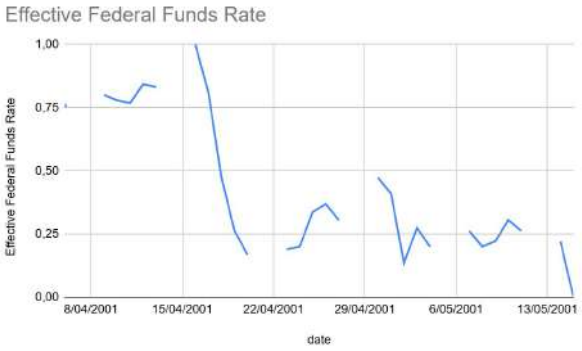


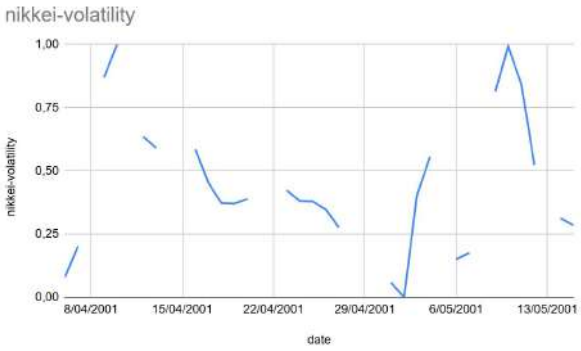




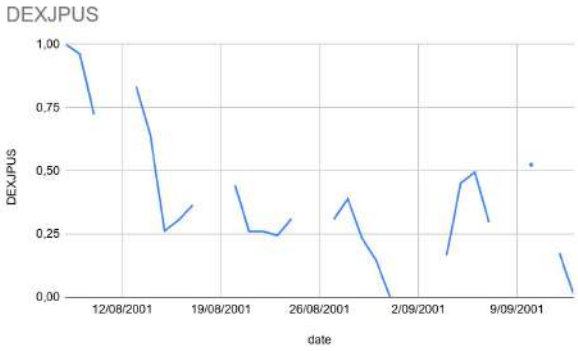
Evento 38 (6/04/2001 - 15/05/2001)

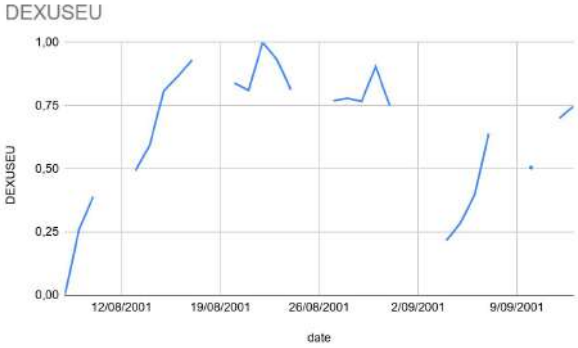


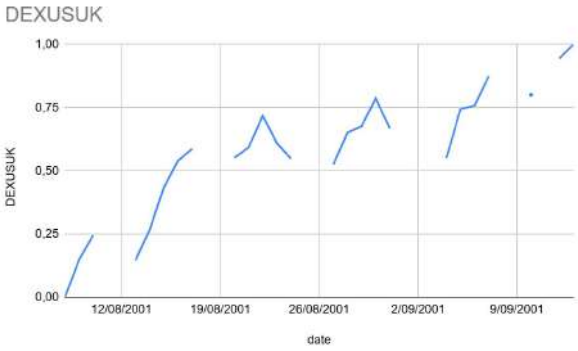


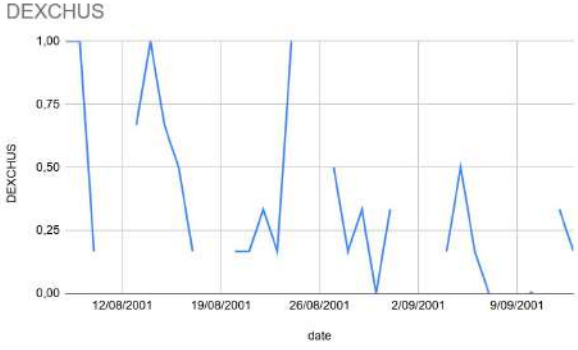


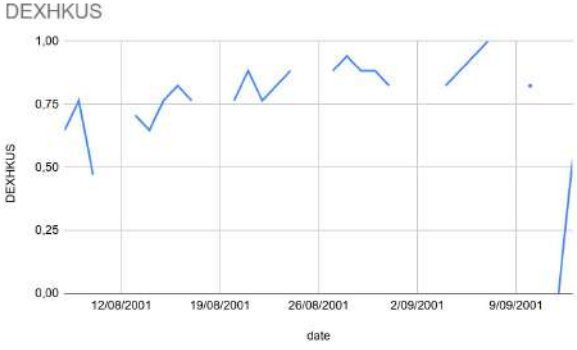
Evento 39 (8/08/2001 - 13/09/2001)

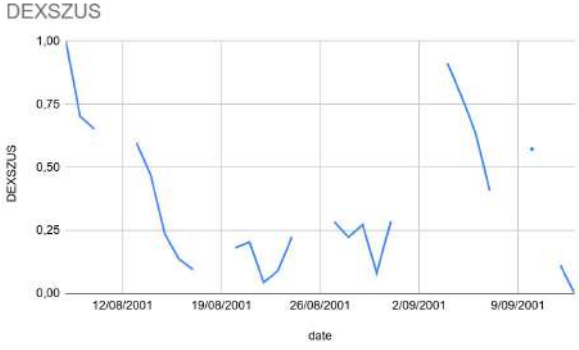




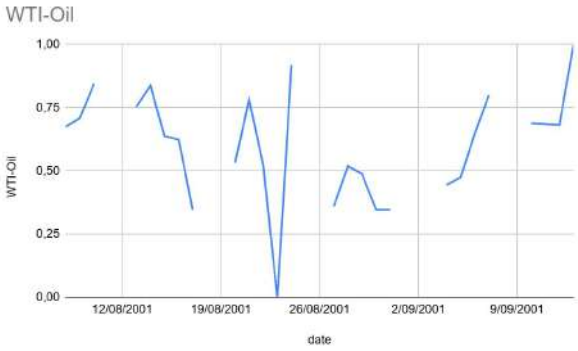


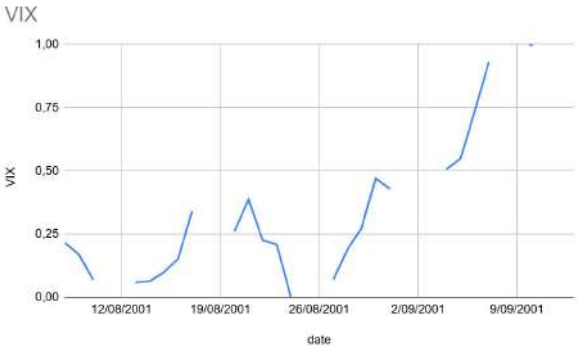


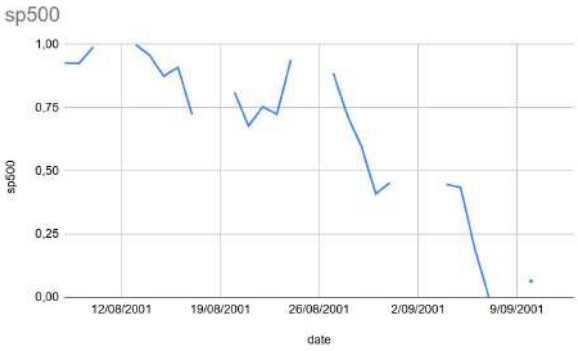


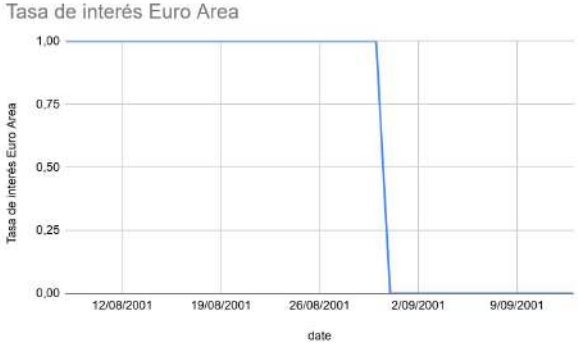


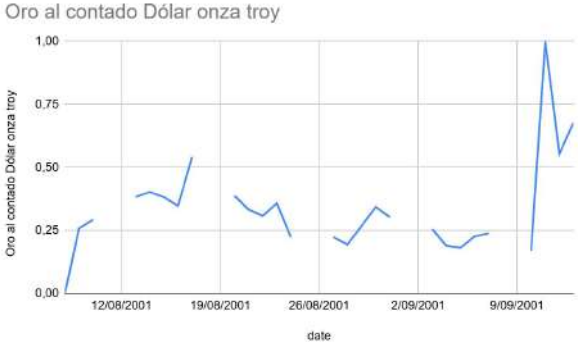
Evento 39 (8/08/2001 - 13/09/2001)

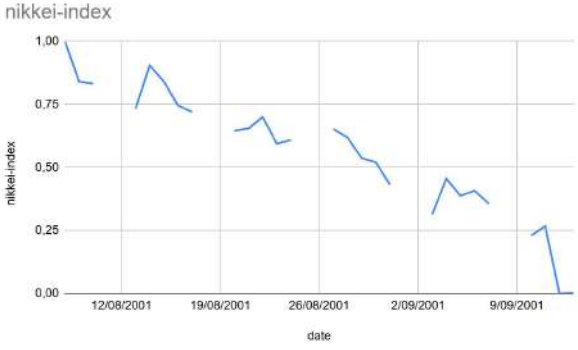




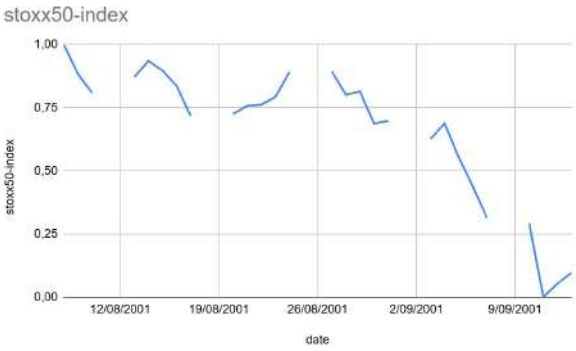


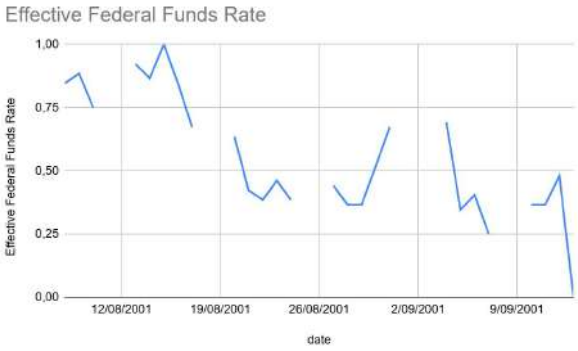


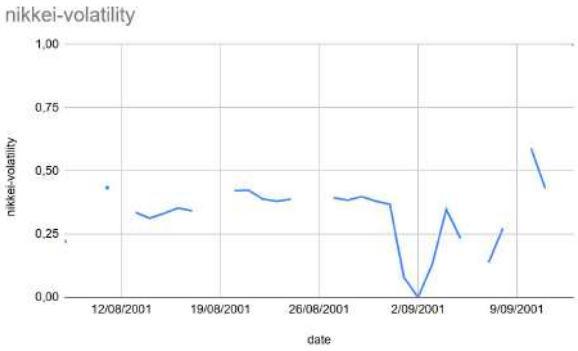




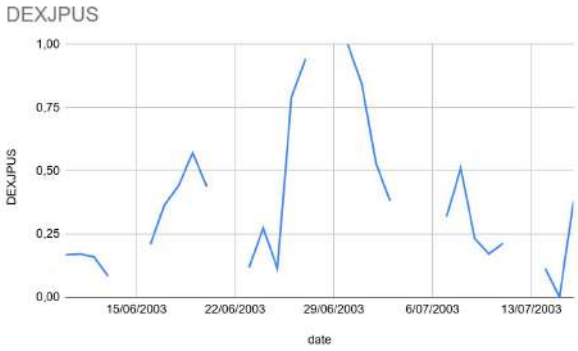
Evento 39 (8/08/2001 - 13/09/2001)

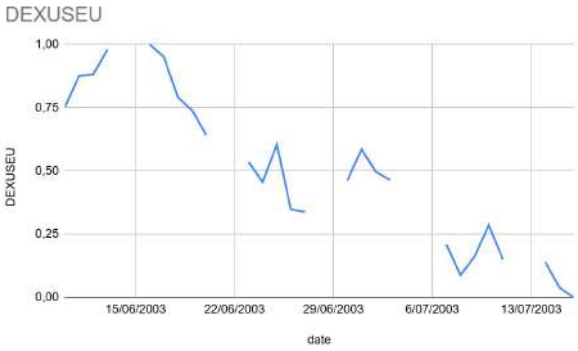


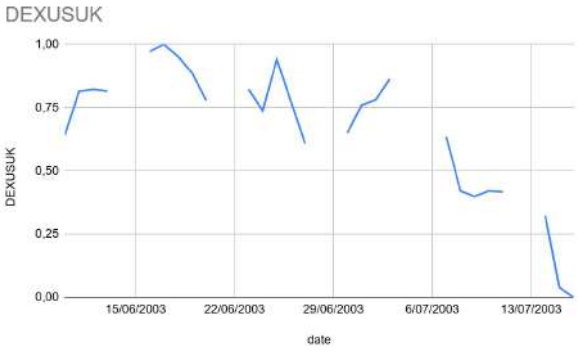


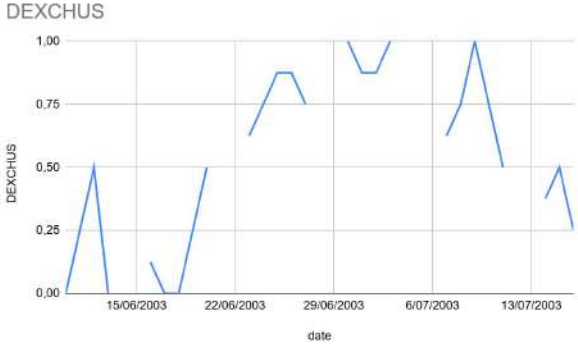


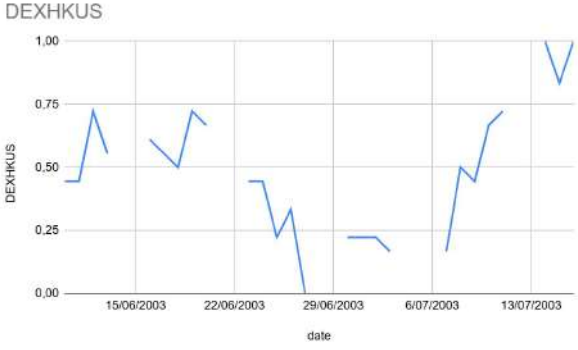
Evento 40 (10/06/2003 - 16/07/2003)

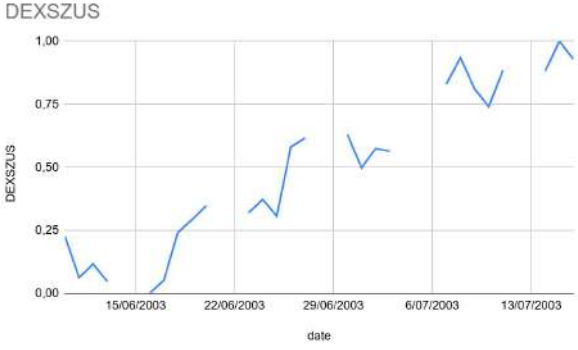




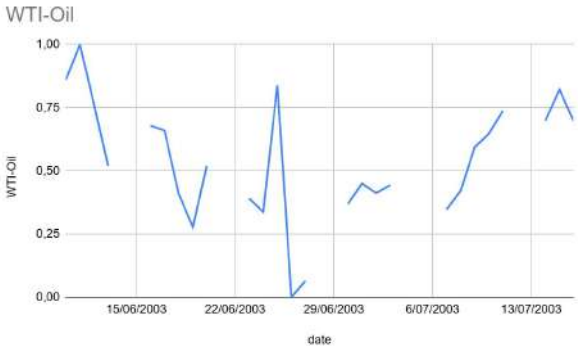


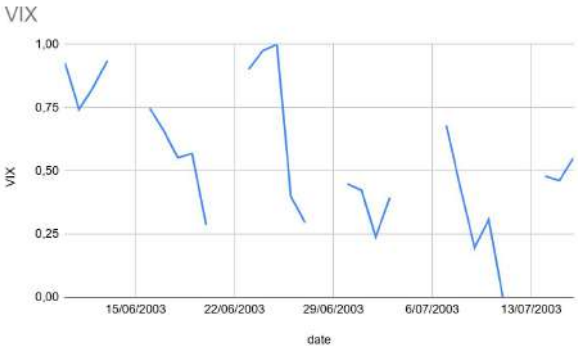


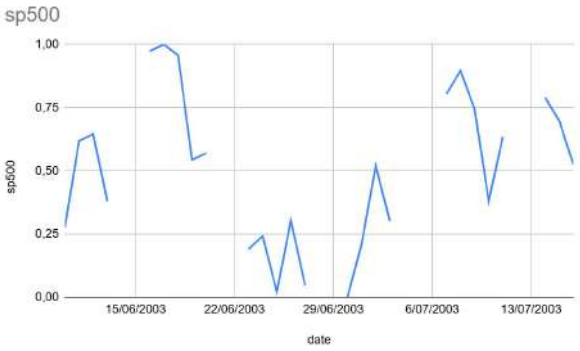


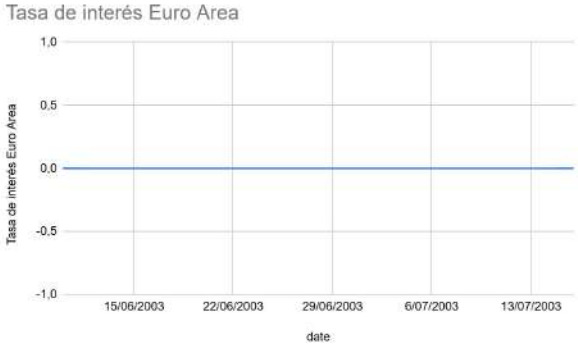


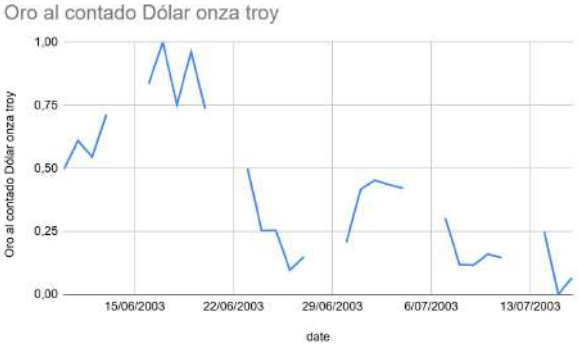
Evento 40 (10/06/2003 - 16/07/2003)

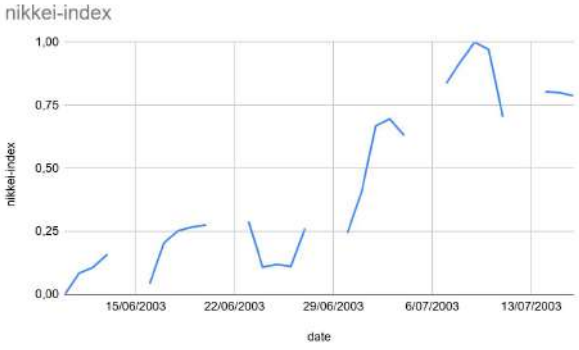




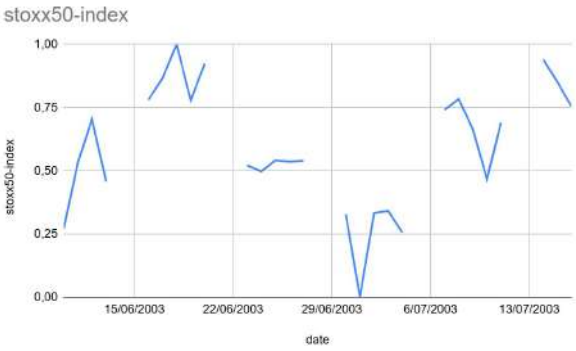


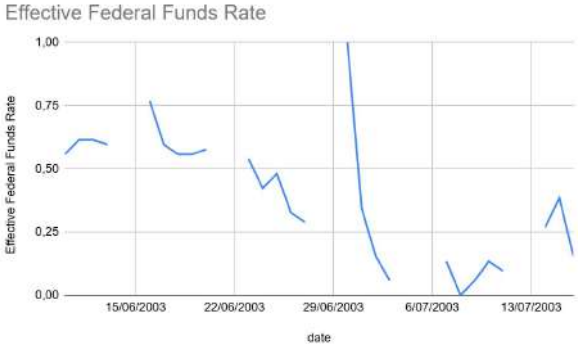


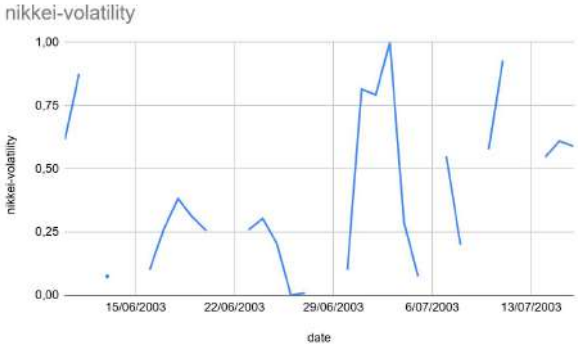




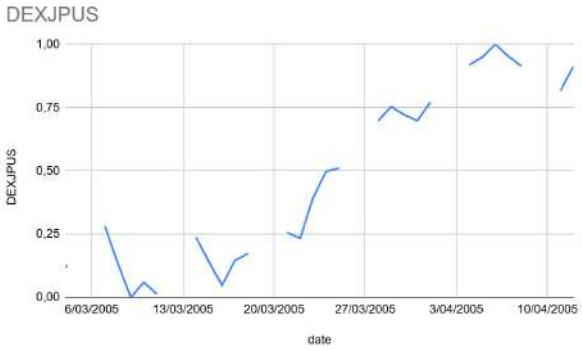
Evento 40 (10/06/2003 - 16/07/2003)

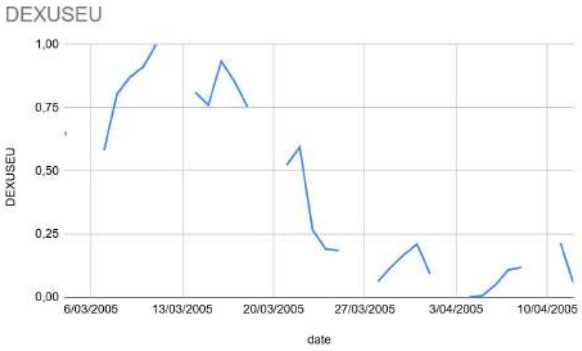


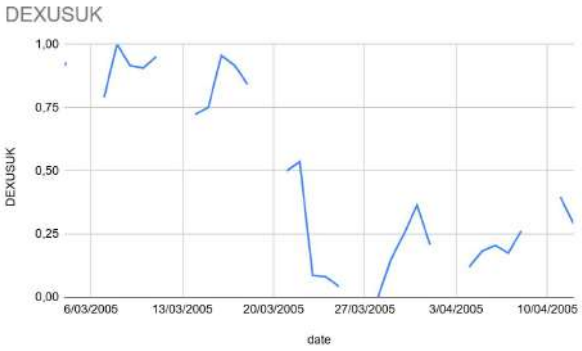


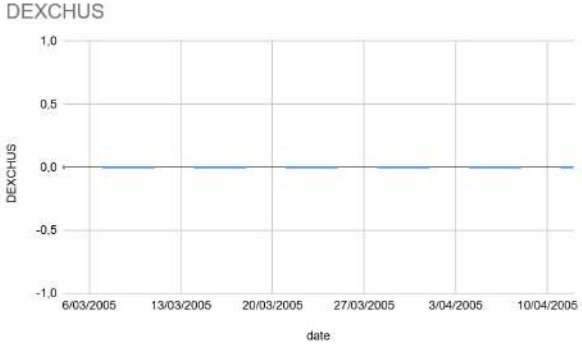


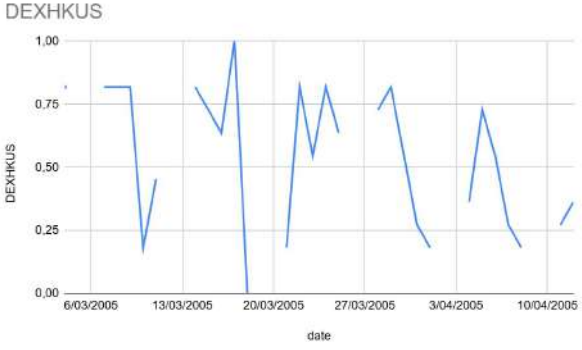
Evento 41 (4/03/2005 - 12/04/2005)

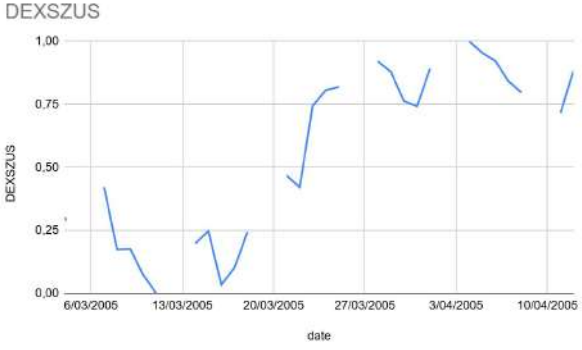




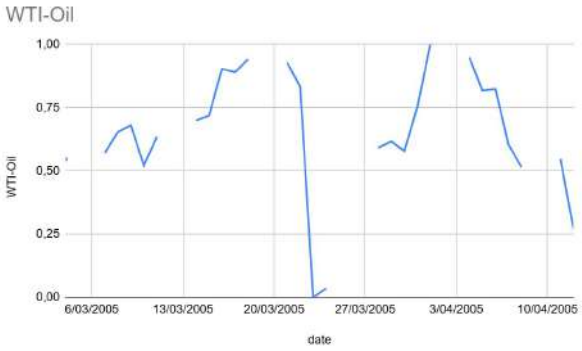


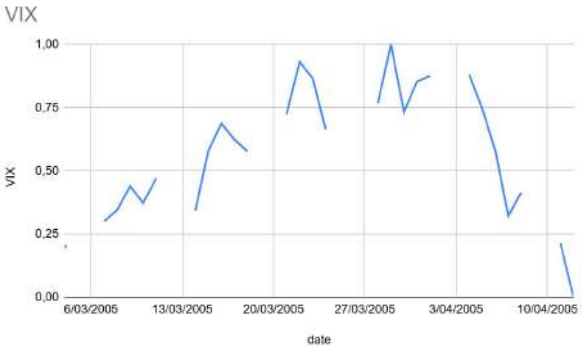


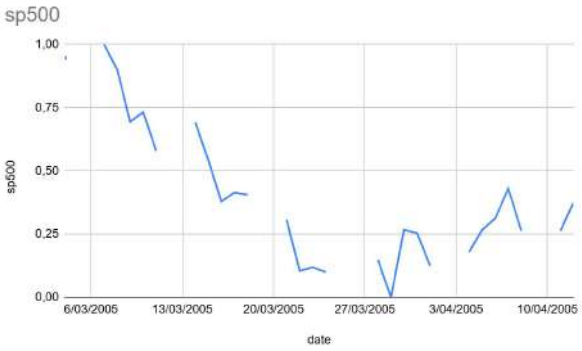


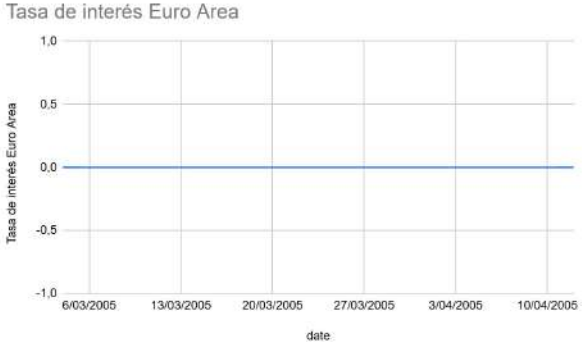


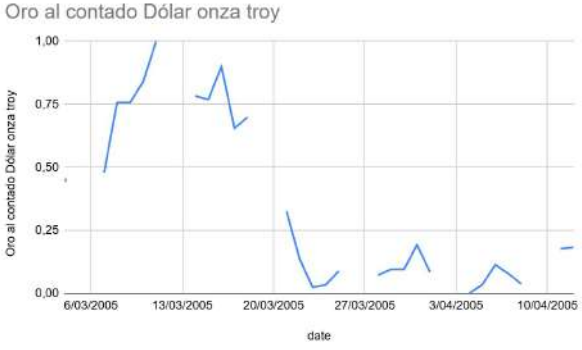
Evento 41 (4/03/2005 - 12/04/2005)

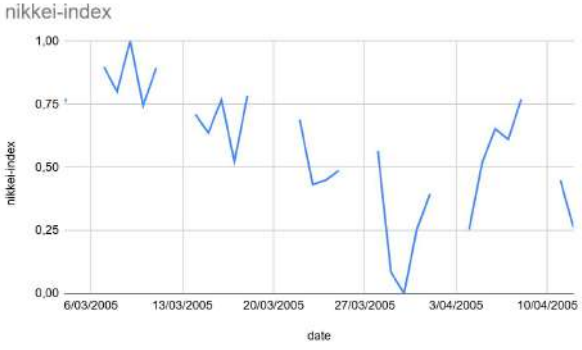




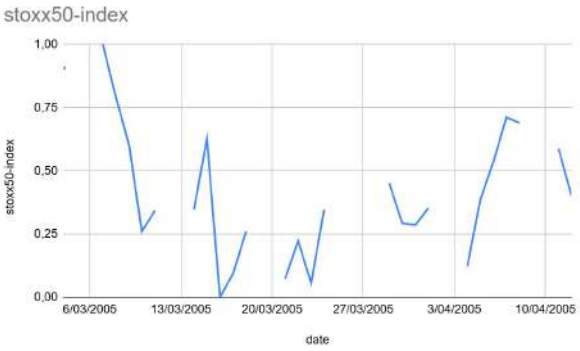


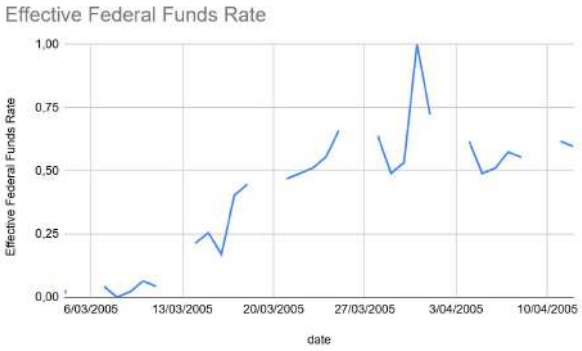


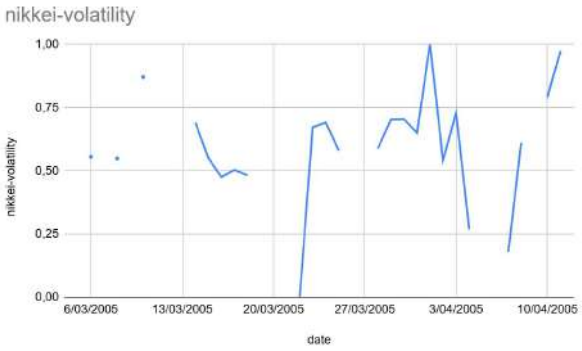




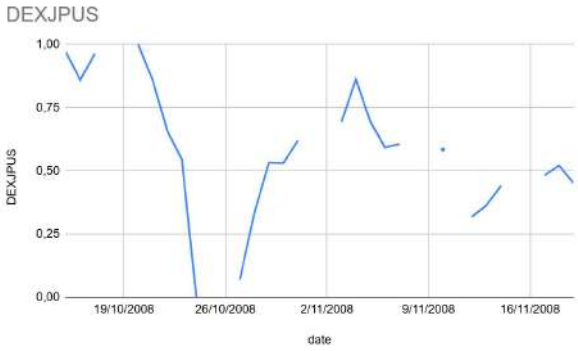
Evento 41 (4/03/2005 - 12/04/2005)

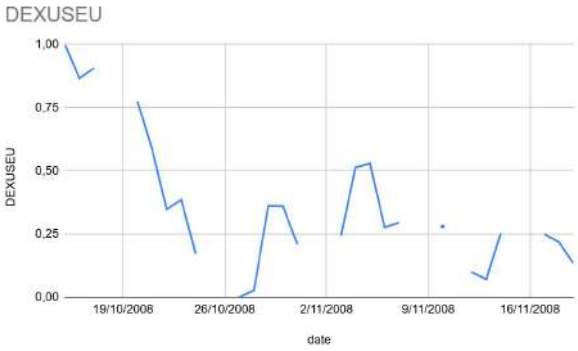


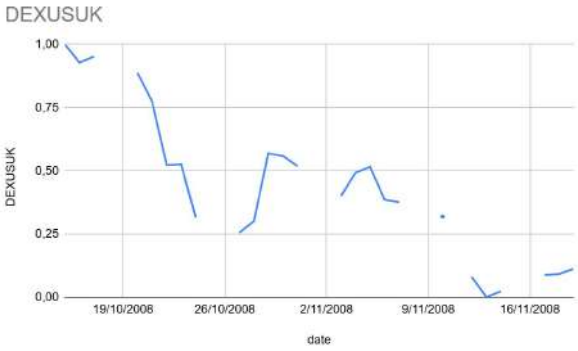


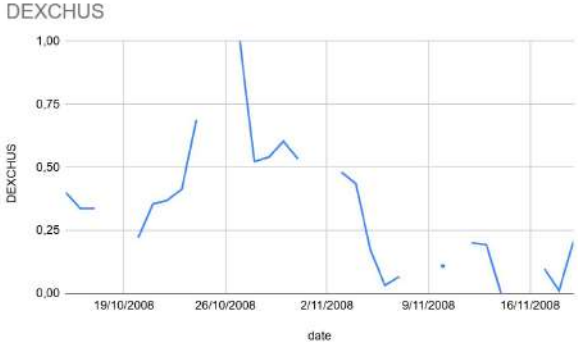


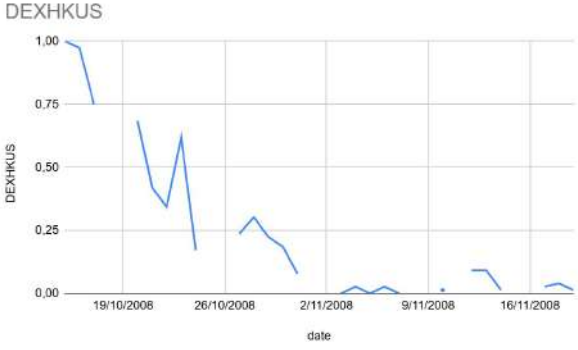
Evento 42 (15/10/2008 - 19/11/2008)

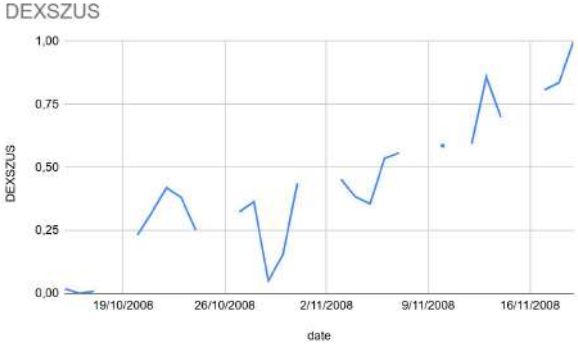




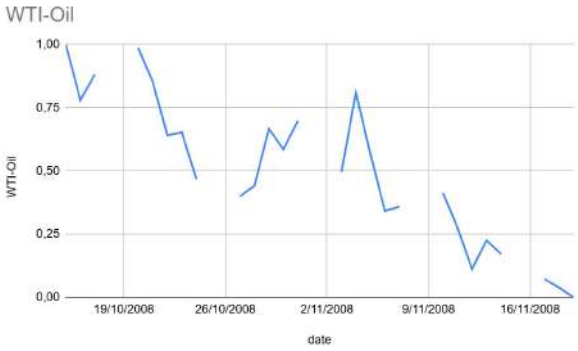


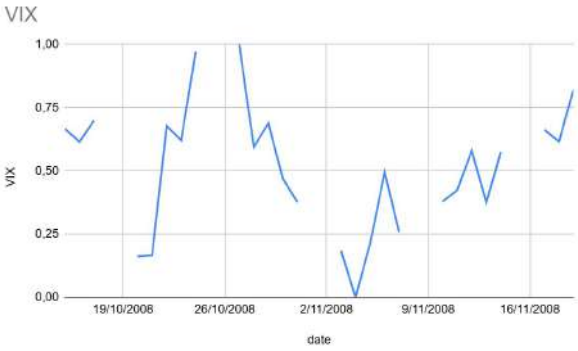


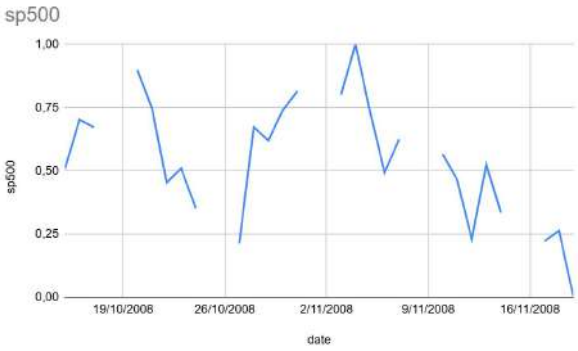


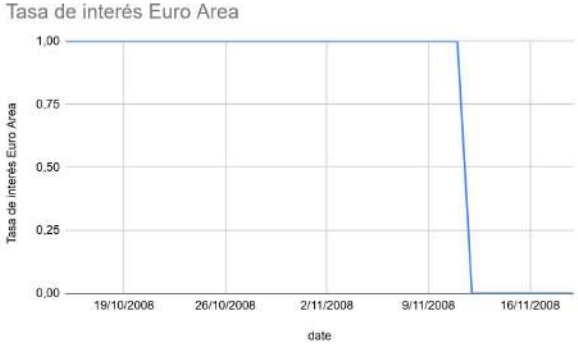


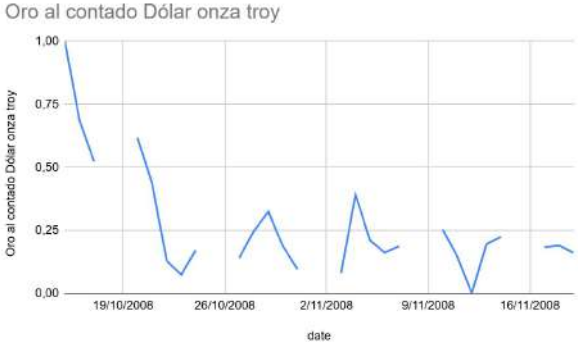
Evento 42 (15/10/2008 - 19/11/2008)

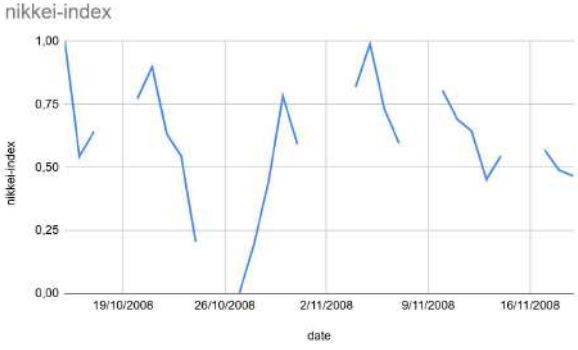




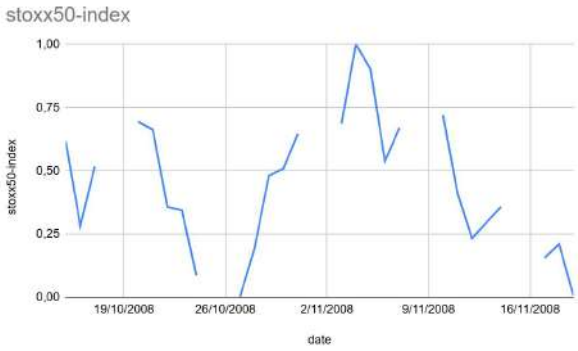


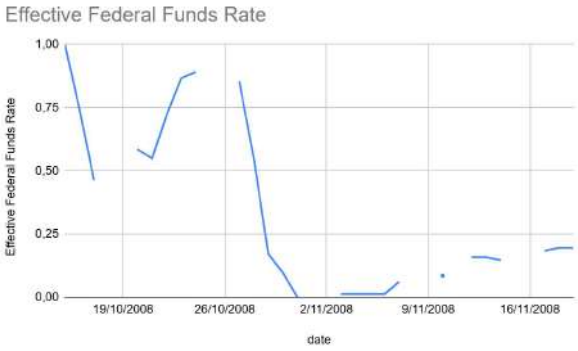


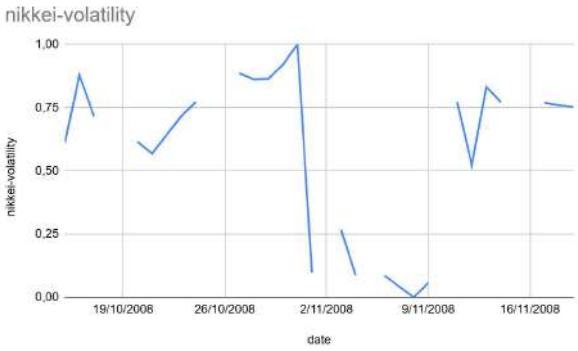




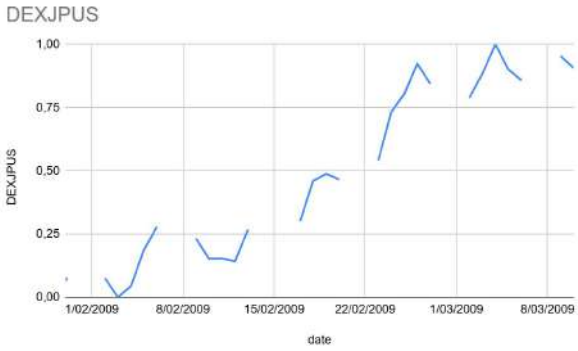
Evento 42 (15/10/2008 - 19/11/2008)

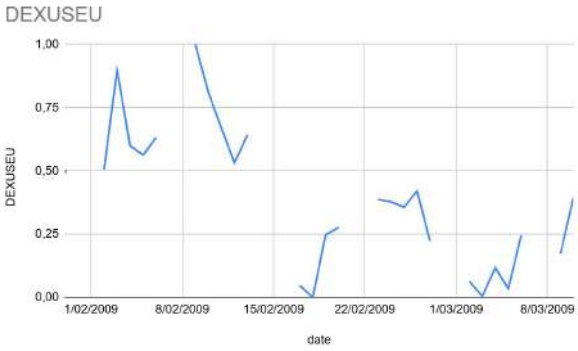


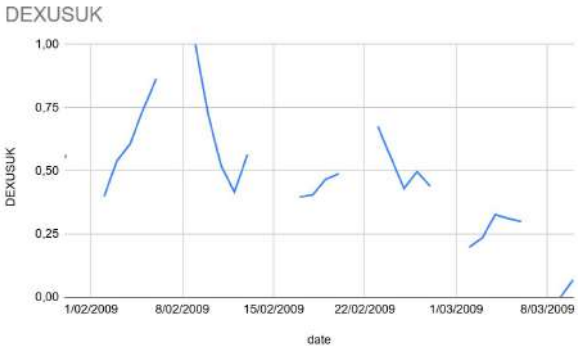


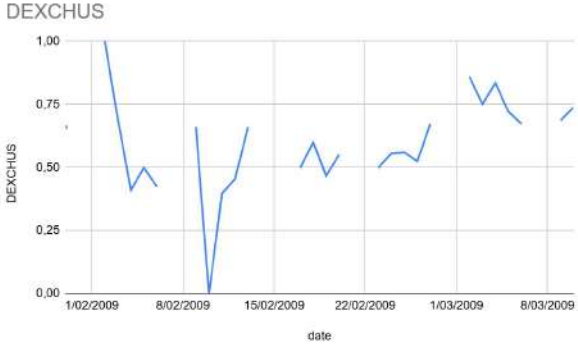


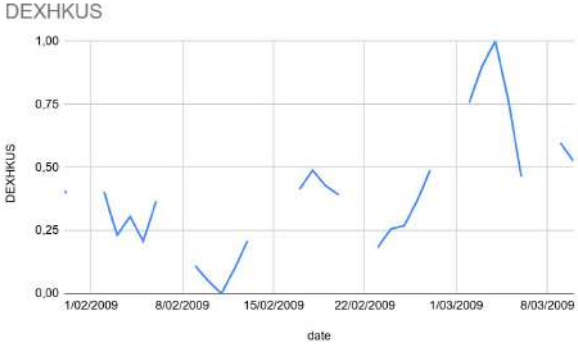
Evento 43 (30/01/2009 - 10/03/2009)

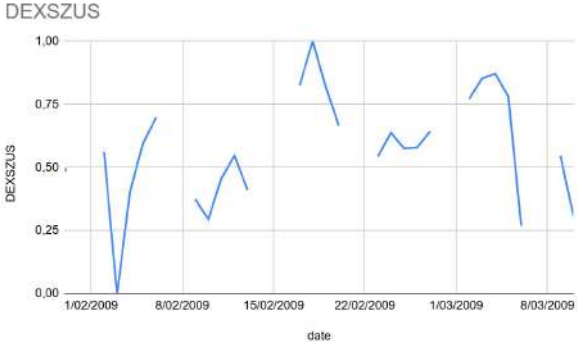




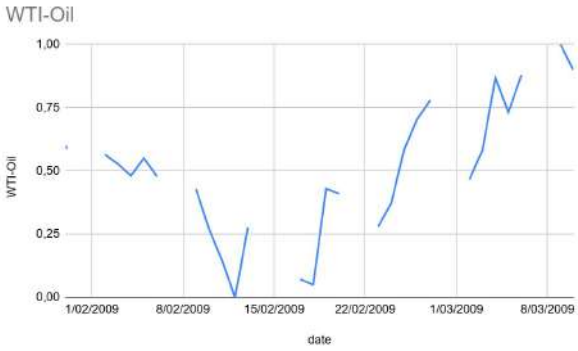


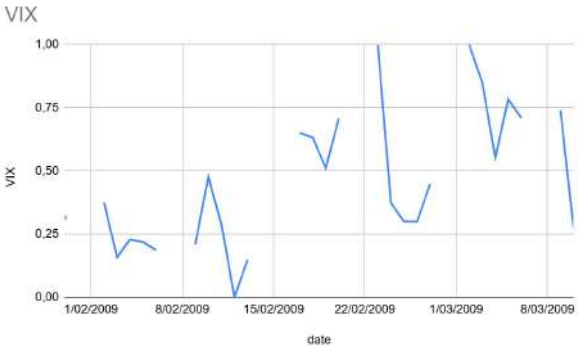


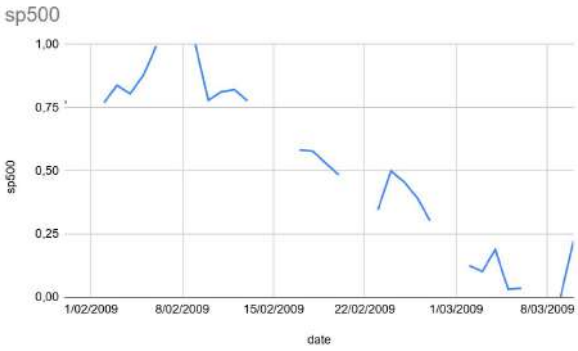


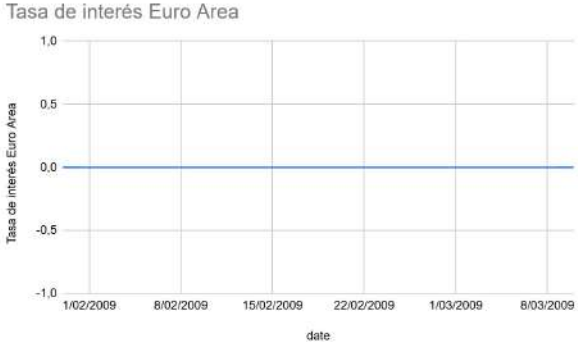


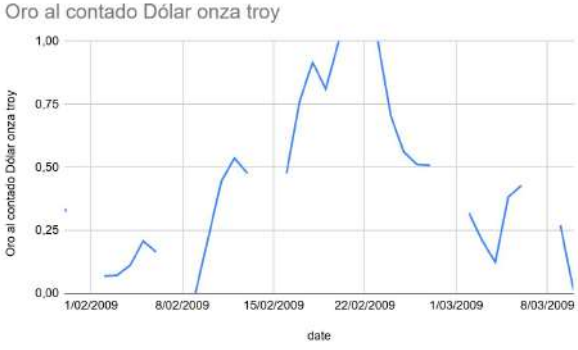
Evento 43 (30/01/2009 - 10/03/2009)

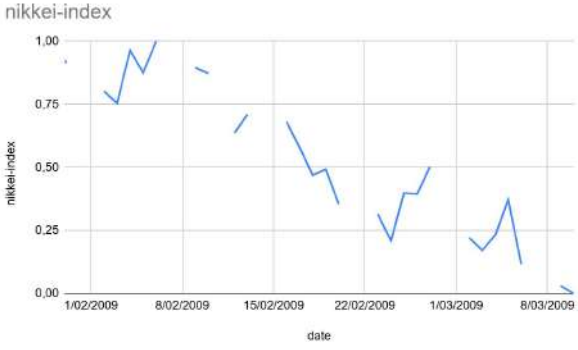




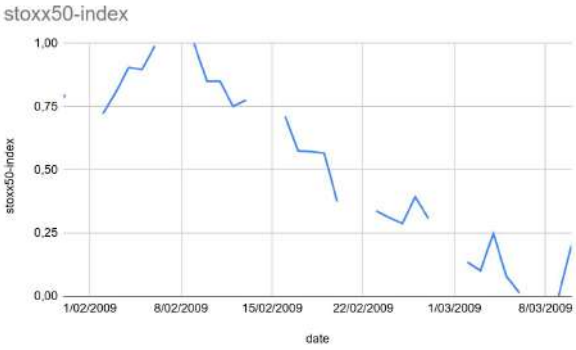


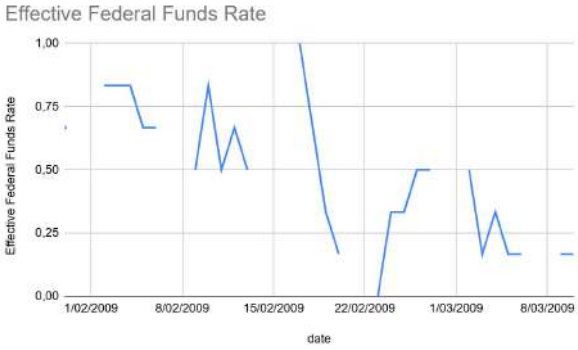


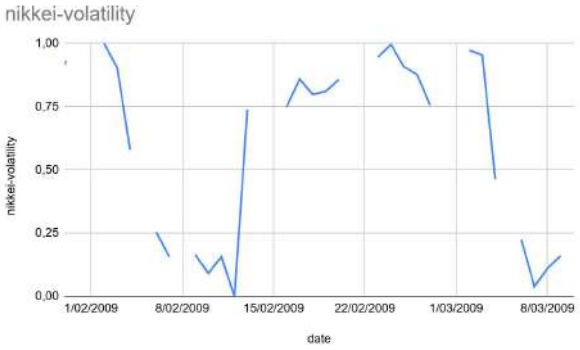




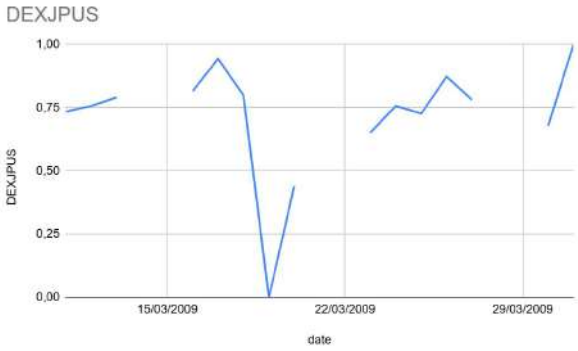
Evento 43 (30/01/2009 - 10/03/2009)

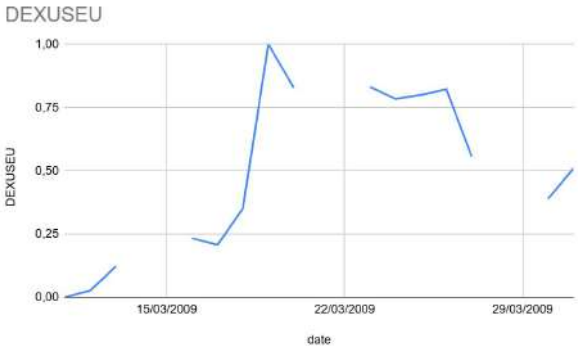


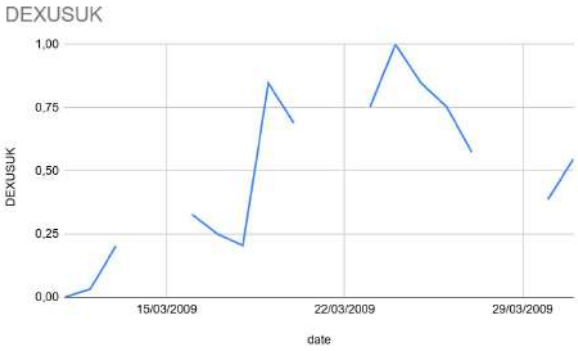


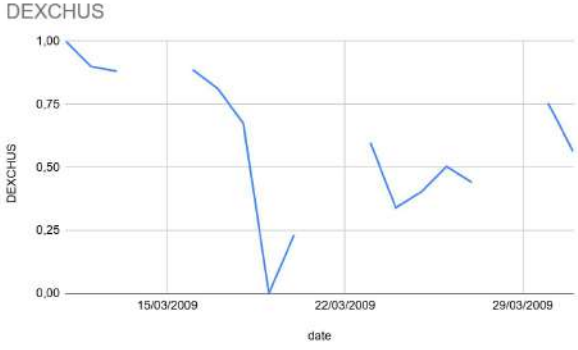


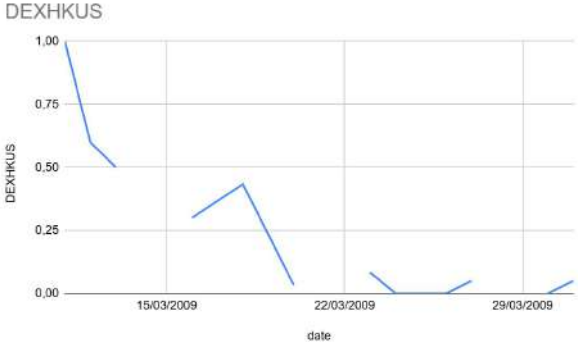
Evento 44 (11/03/2009 - 31/03/2009)

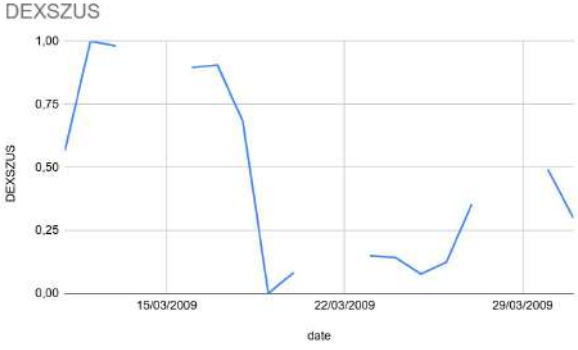




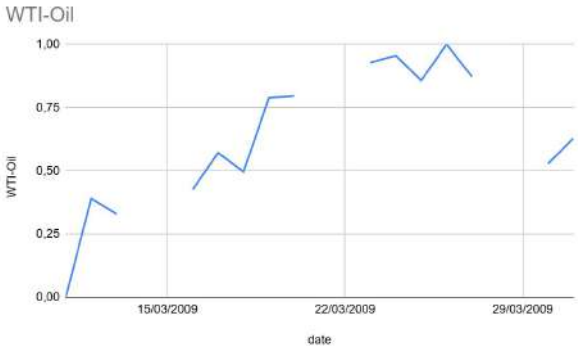


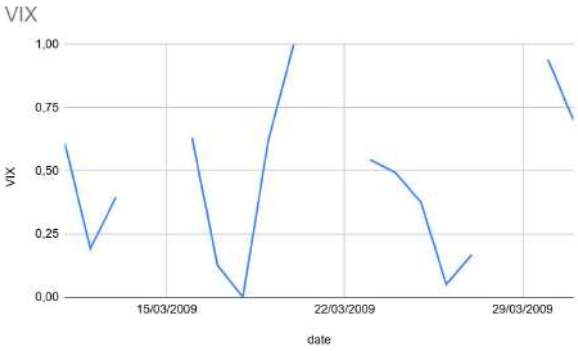


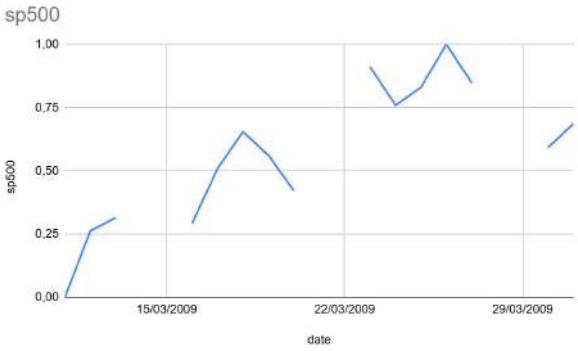


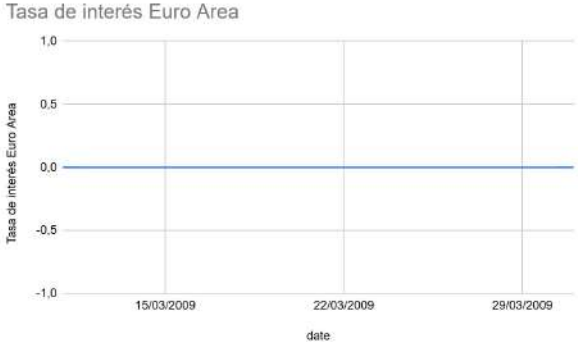


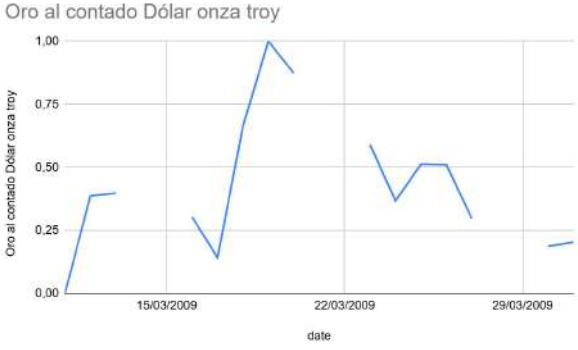
Evento 44 (11/03/2009 - 31/03/2009)

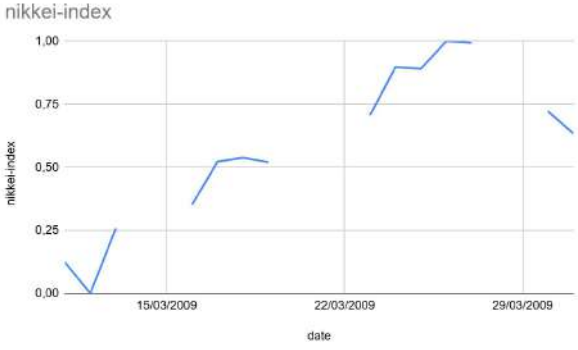




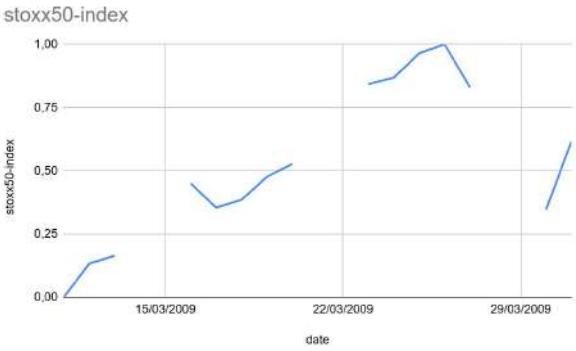


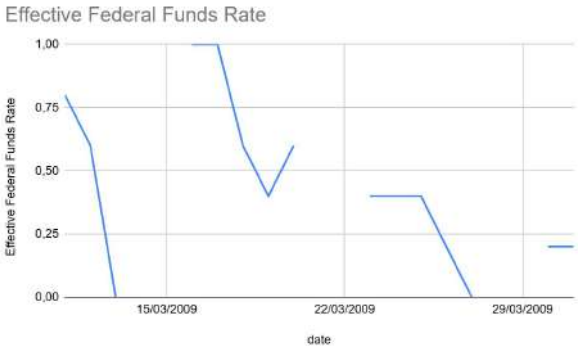


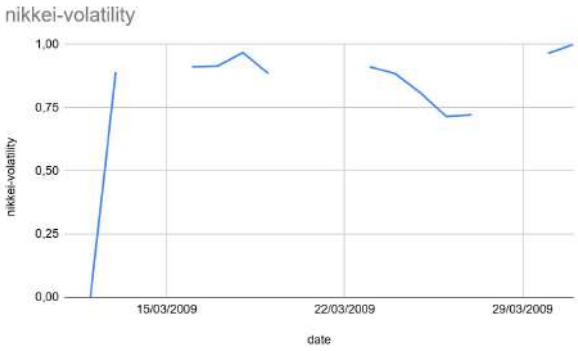




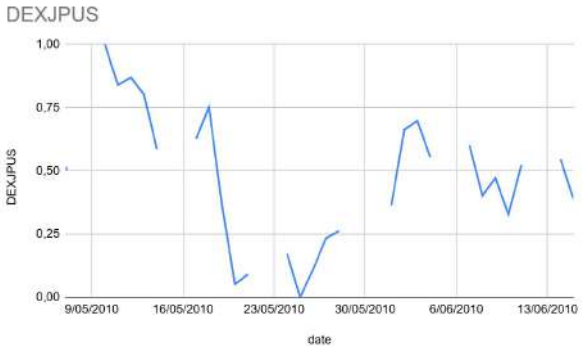
Evento 44 (11/03/2009 - 31/03/2009)

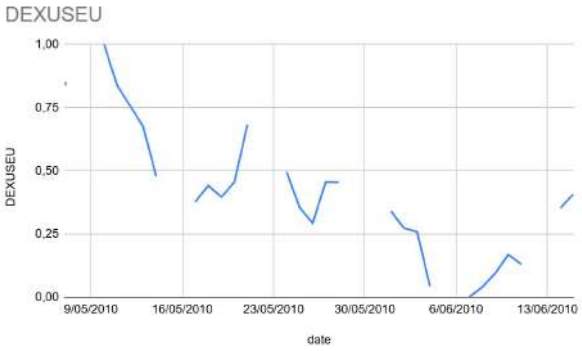


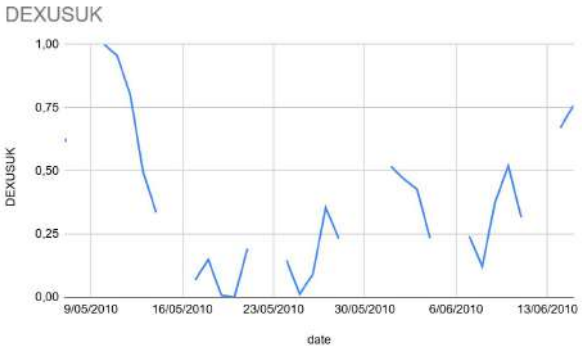


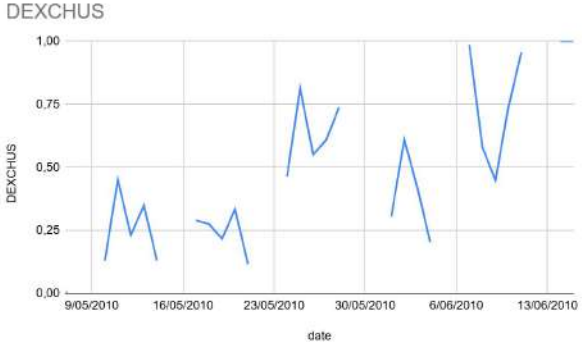


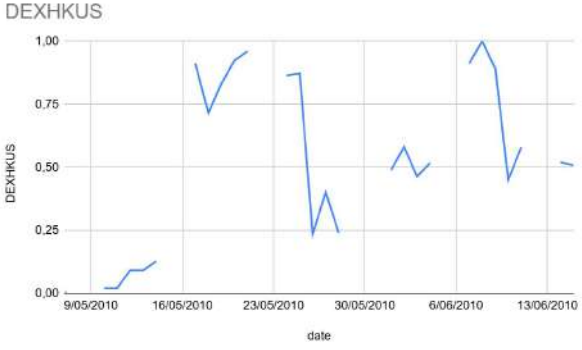
Evento 45 (7/05/2010 - 15/06/2010)

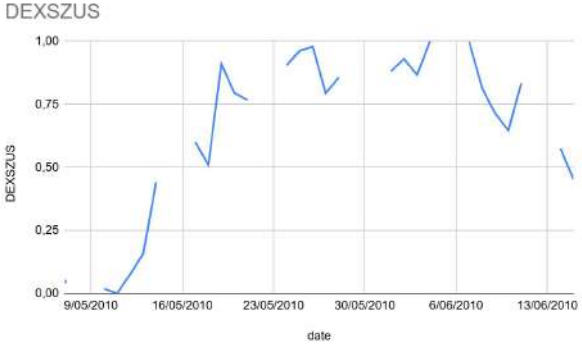




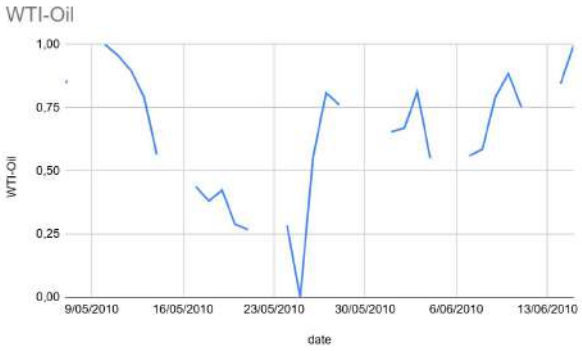


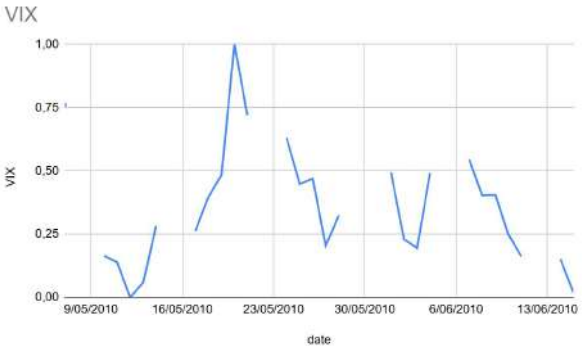


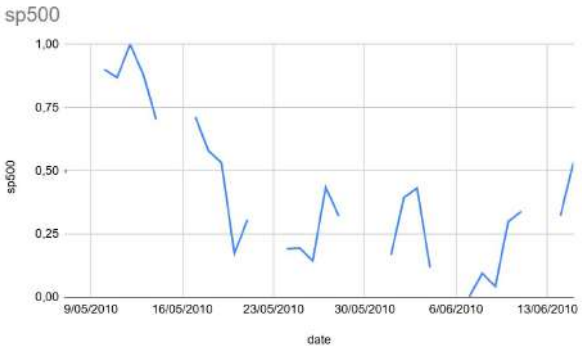


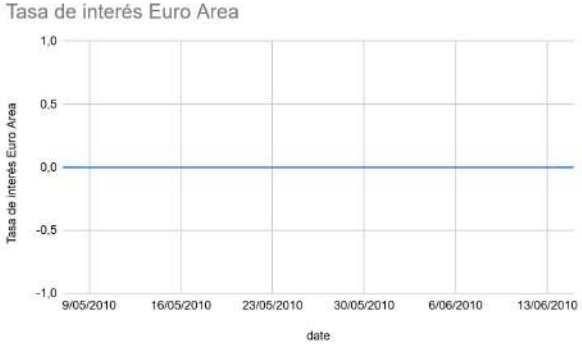


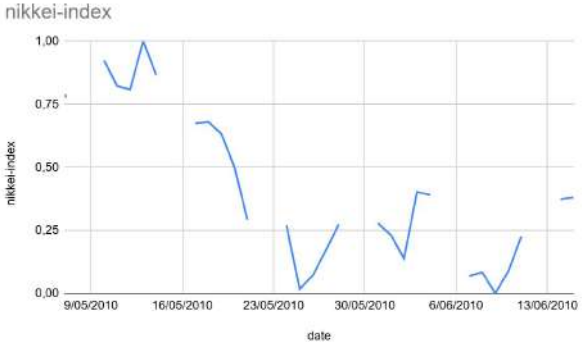
Evento 45 (7/05/2010 - 15/06/2010)



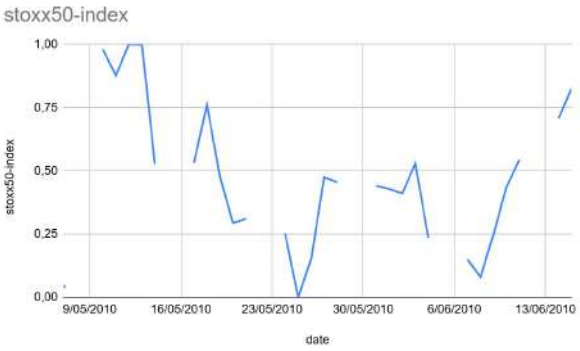


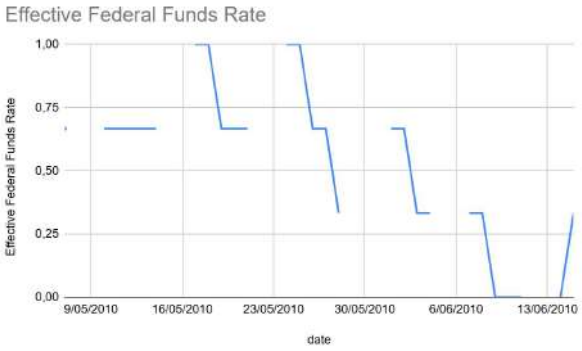


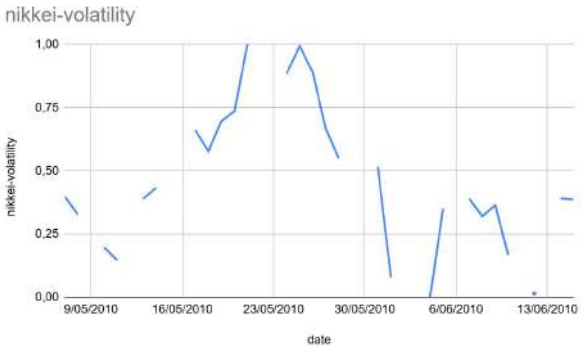




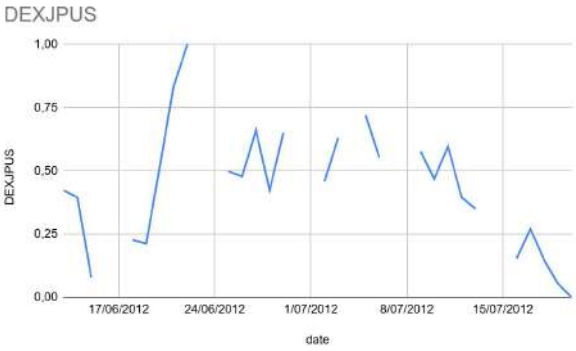
Evento 45 (7/05/2010 - 15/06/2010)

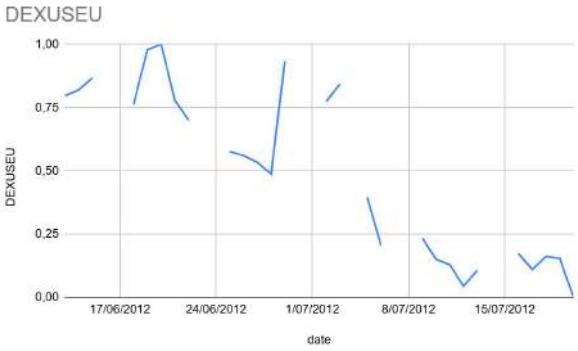


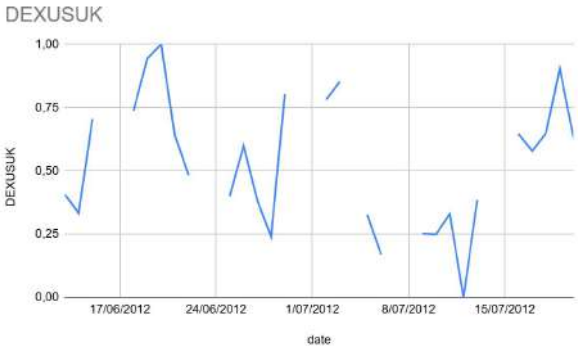


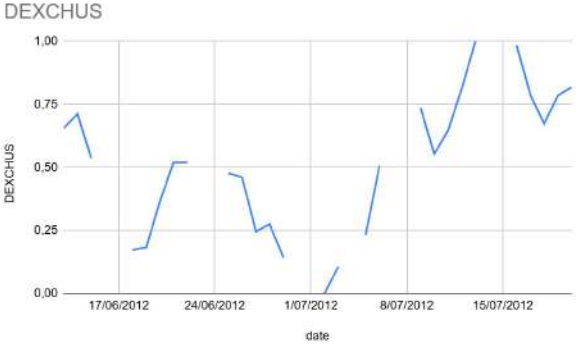


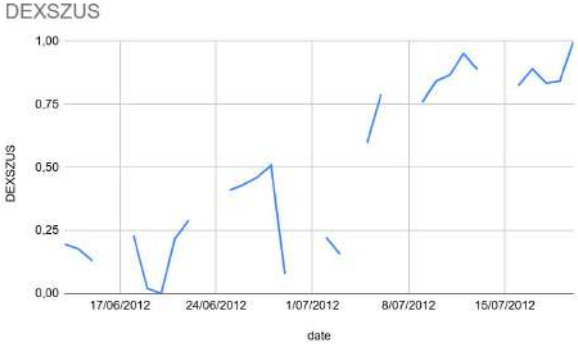
Evento 46 (13/06/2012 - 20/07/2012)



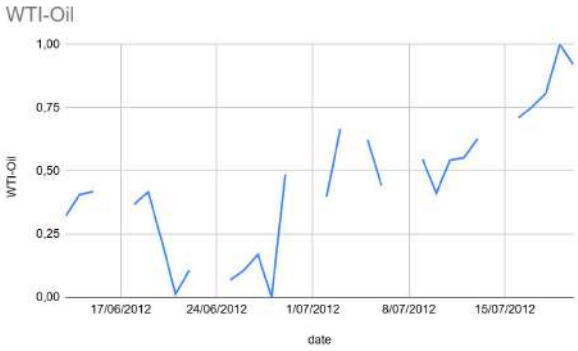


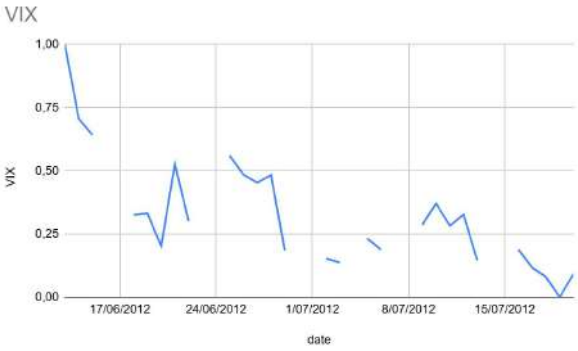


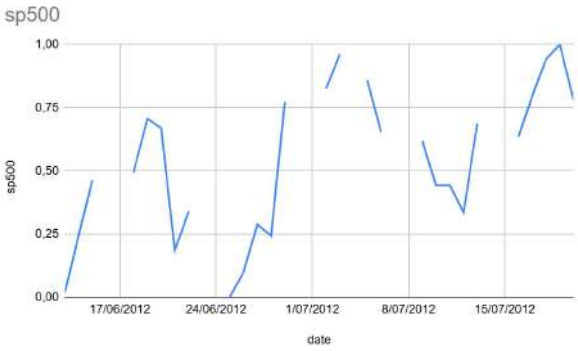


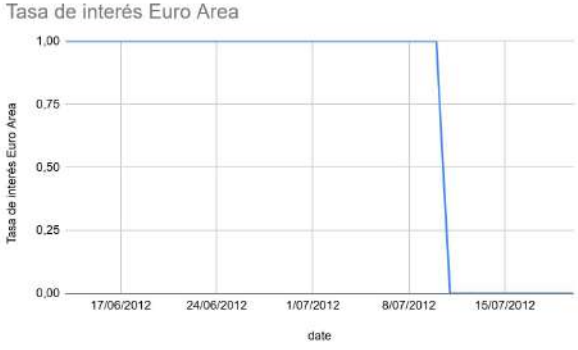


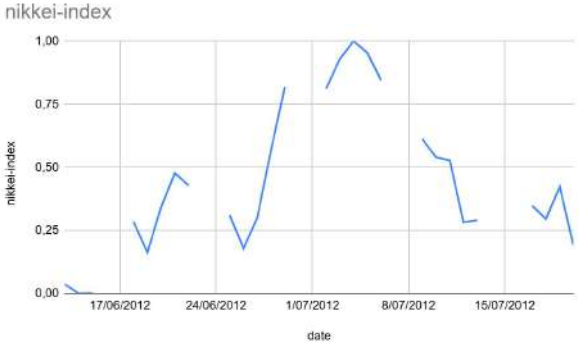
Evento 46 (13/06/2012 - 20/07/2012)



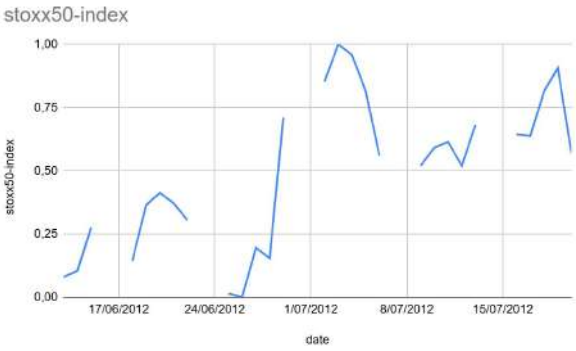


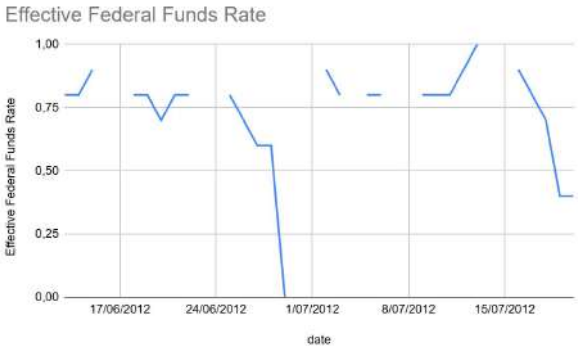


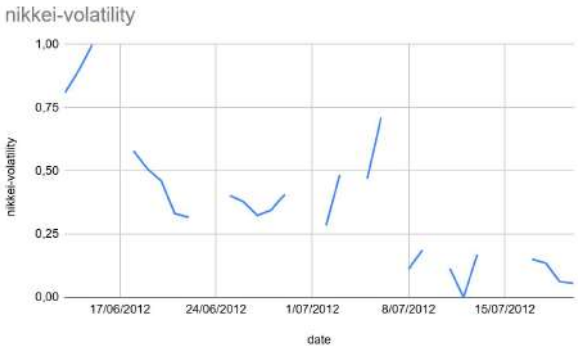




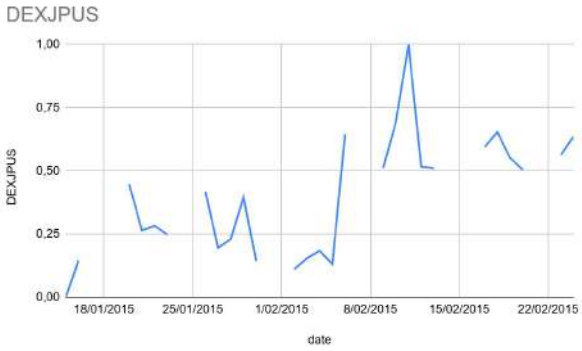
Evento 46 (13/06/2012 - 20/07/2012)

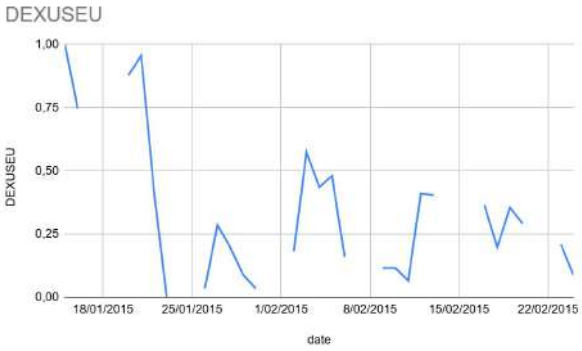


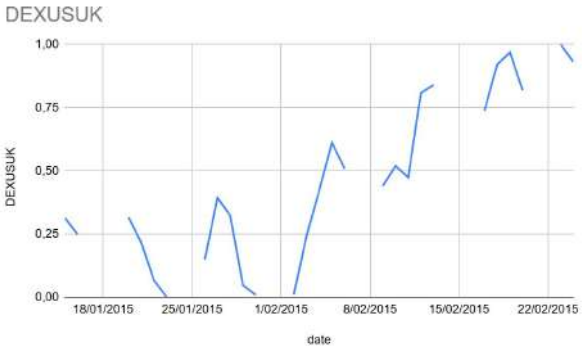


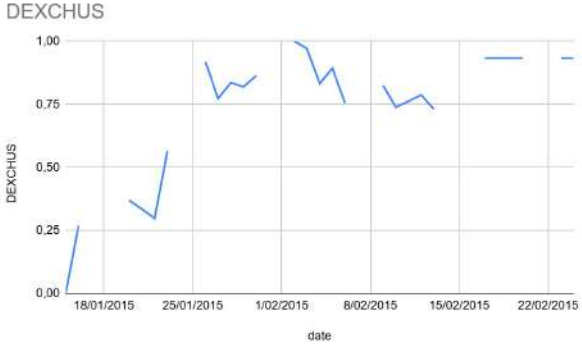


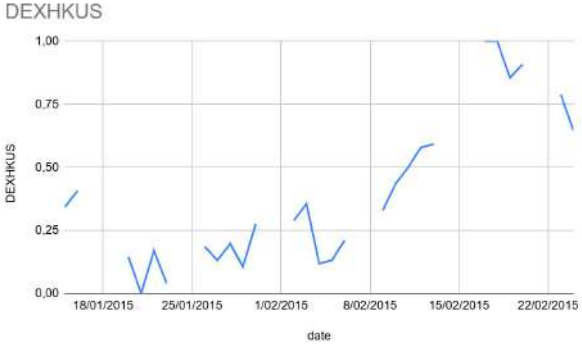
Evento 47 (15/01/2015 - 24/02/2015)

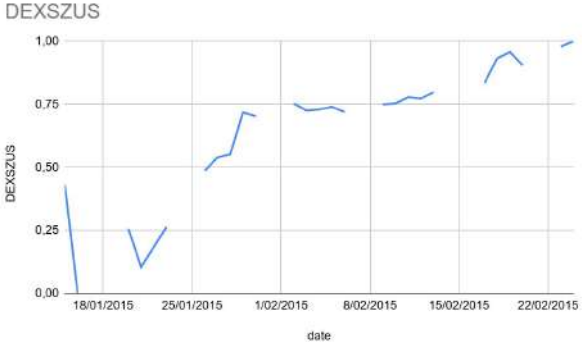




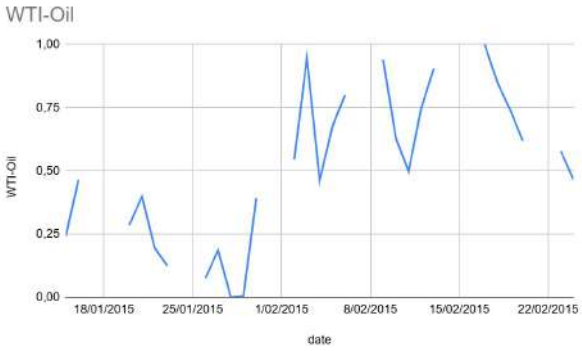


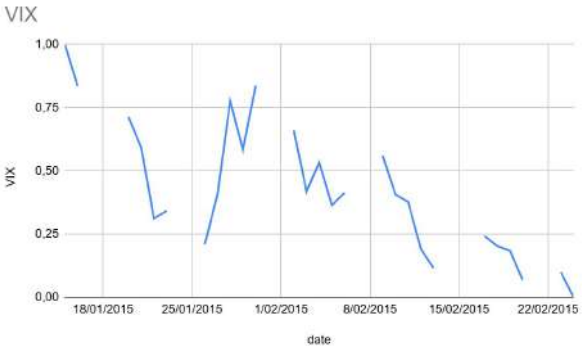


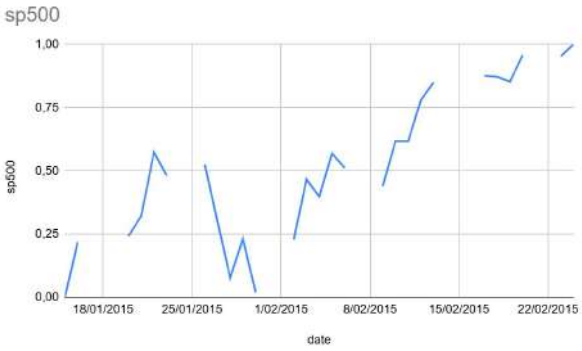


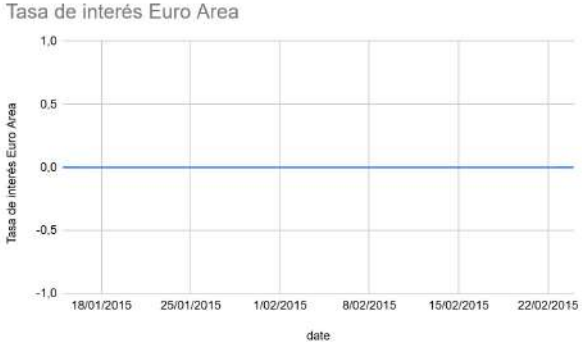


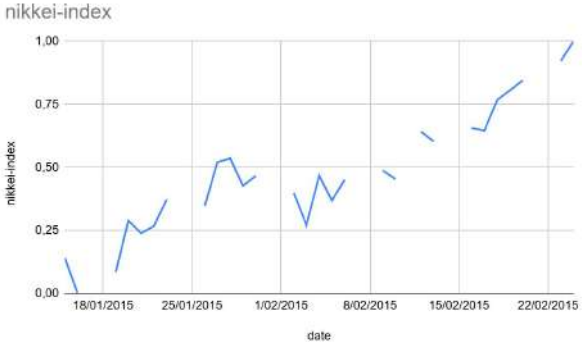
Evento 47 (15/01/2015 - 24/02/2015)



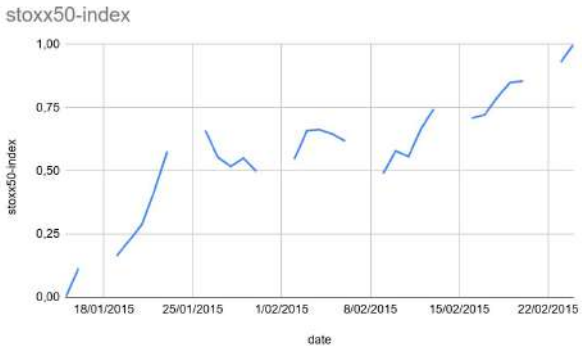


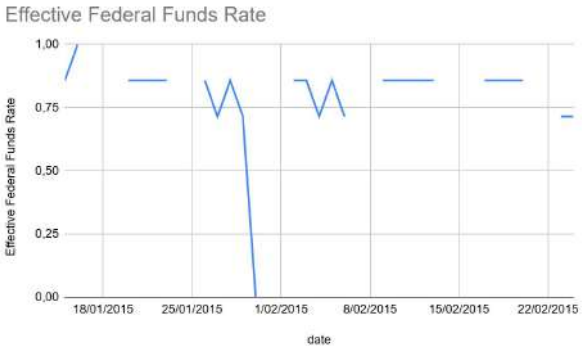


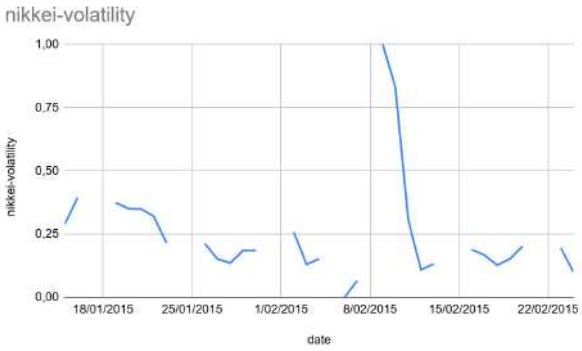


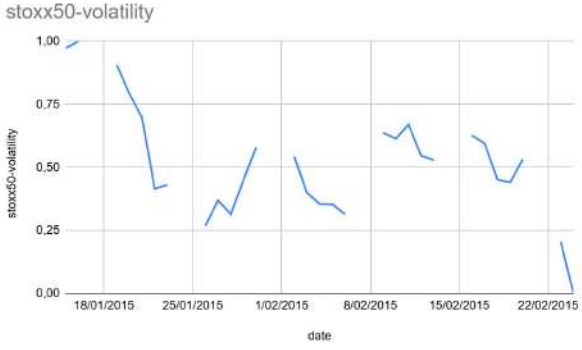


Evento 47 (15/01/2015 - 24/02/2015)

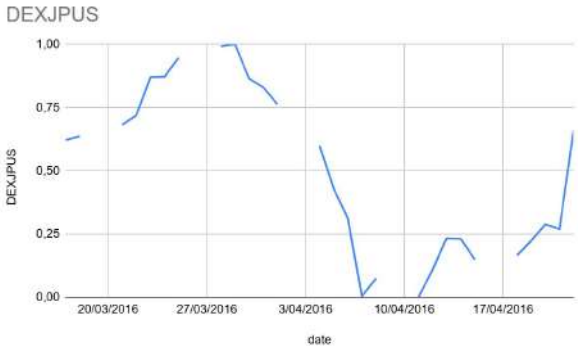


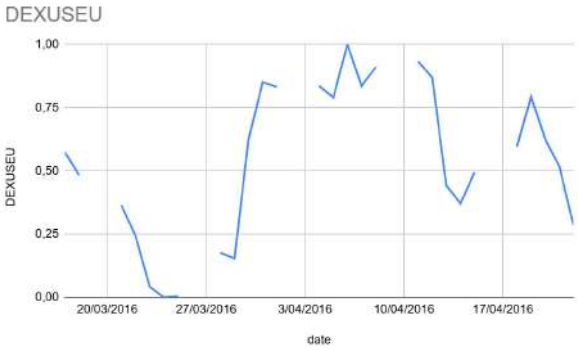


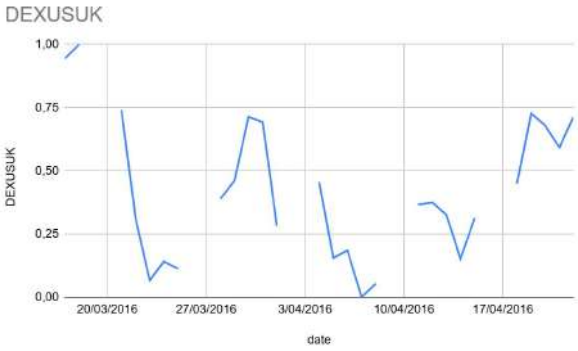


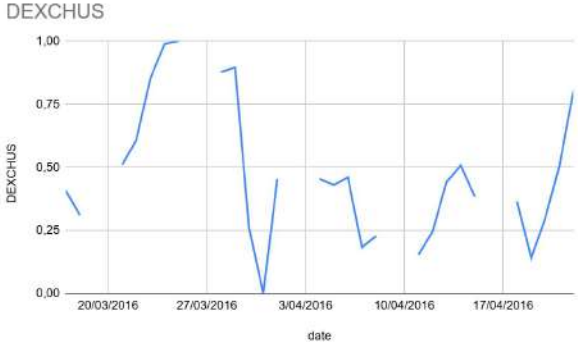


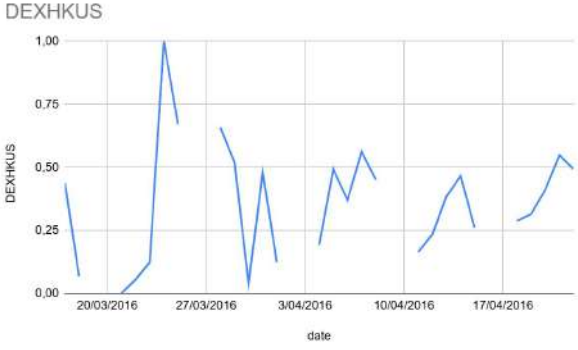
Evento 48 (17/03/2016 - 22/04/2016)

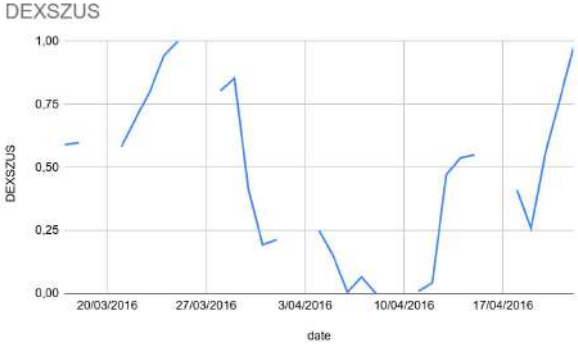




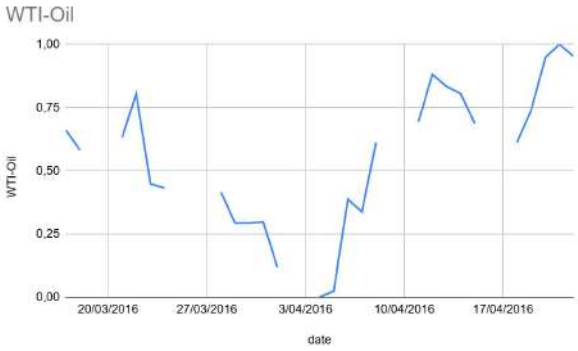


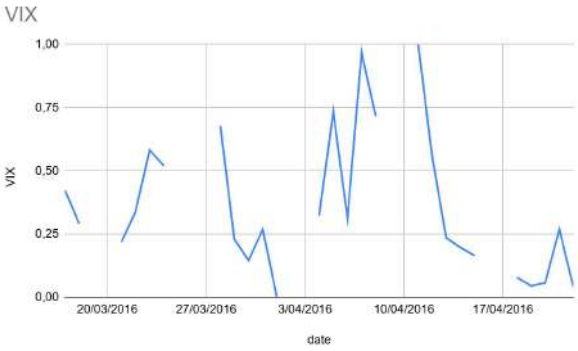


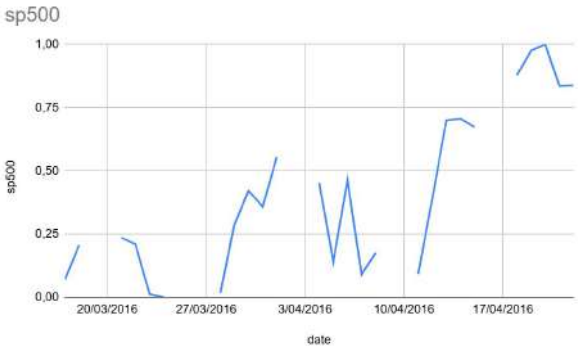


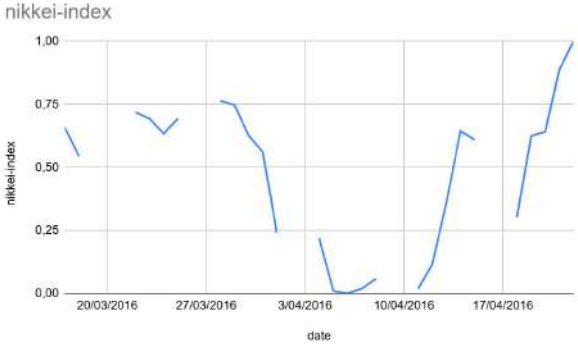


Evento 48 (17/03/2016 - 22/04/2016)

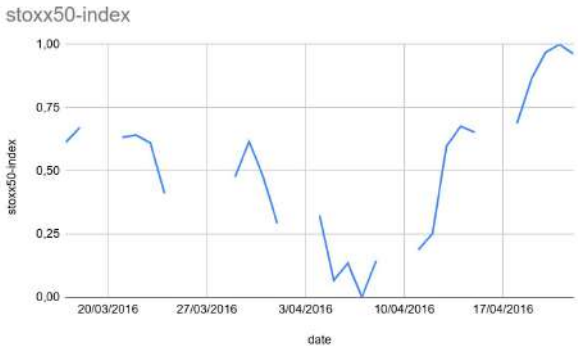


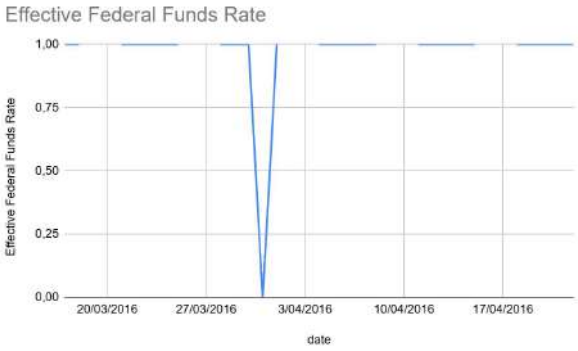


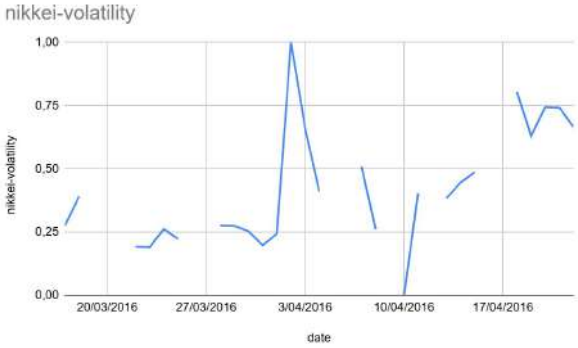


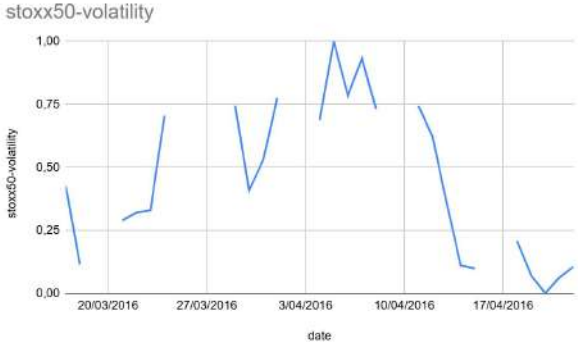


Evento 48 (17/03/2016 - 22/04/2016)

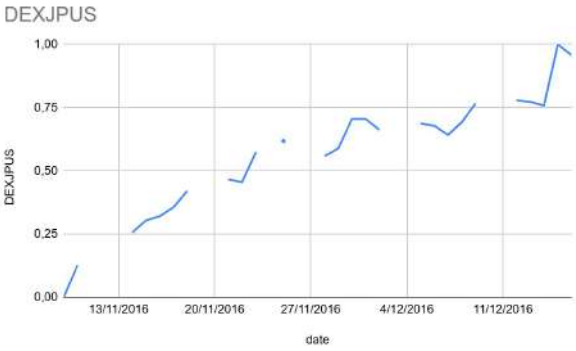


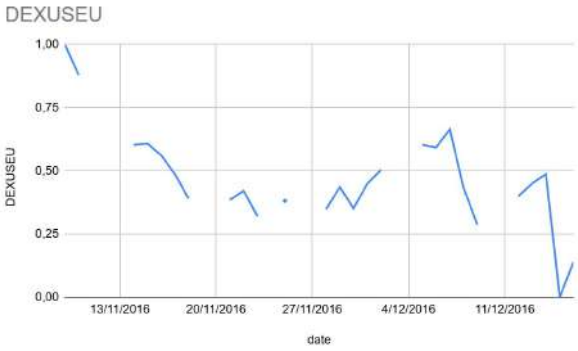


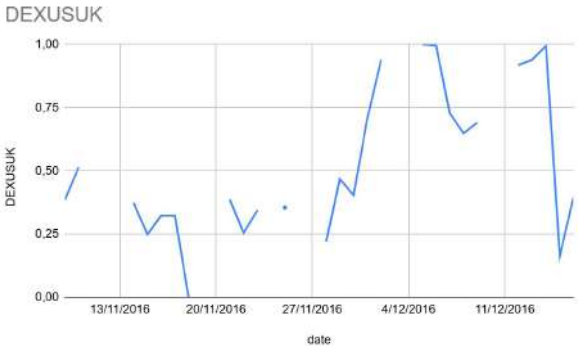


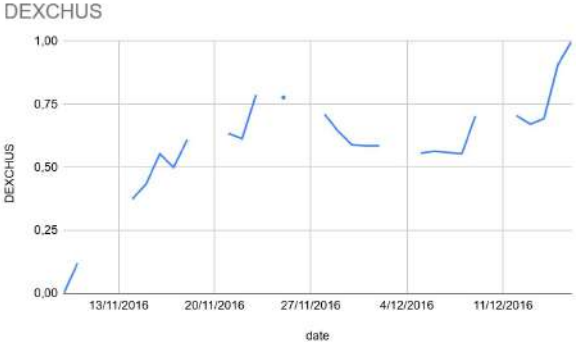


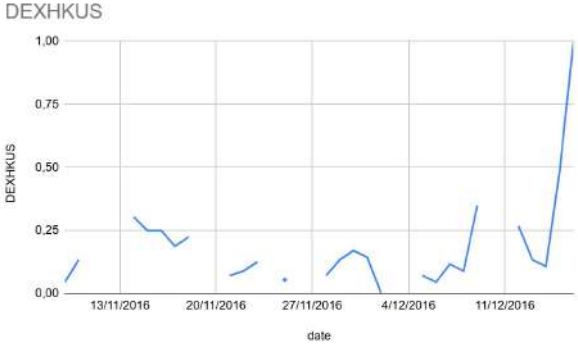
Evento 49 (9/11/2016 - 16/12/2016)

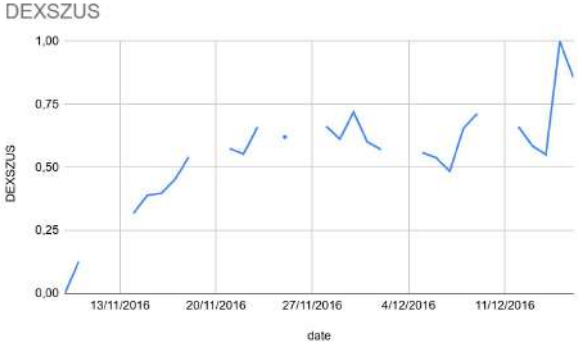




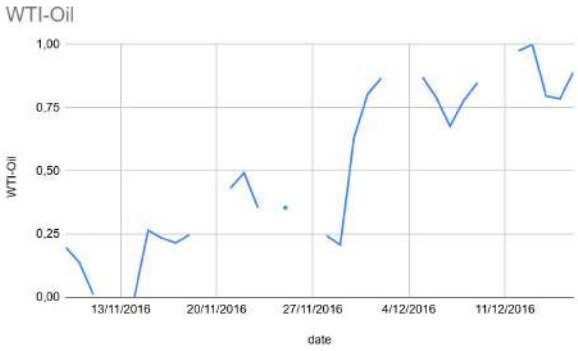


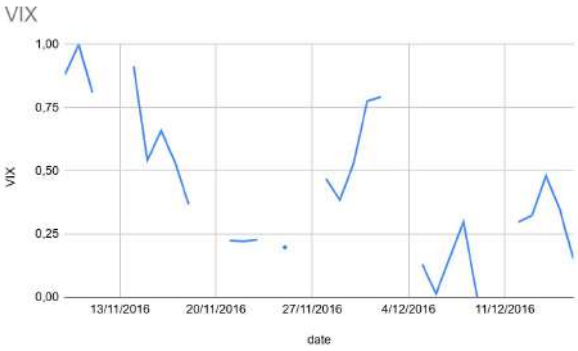


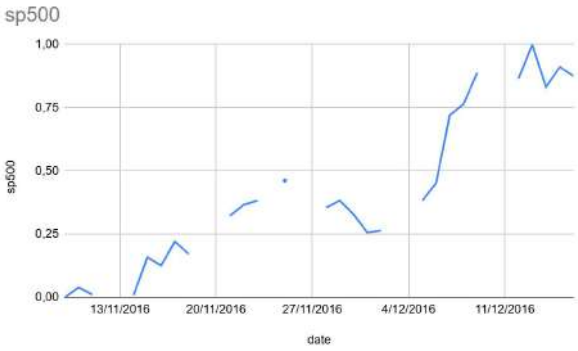


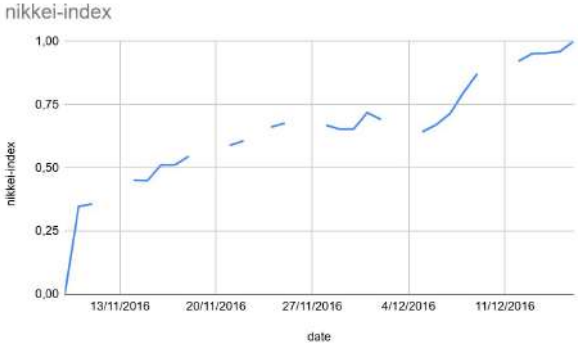


Evento 49 (9/11/2016 - 16/12/2016)

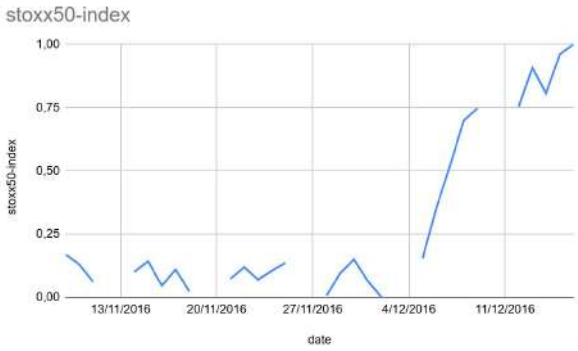


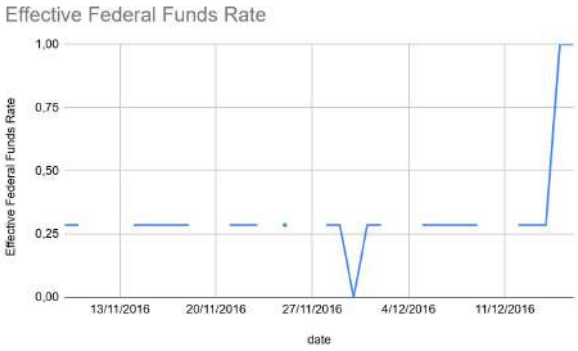


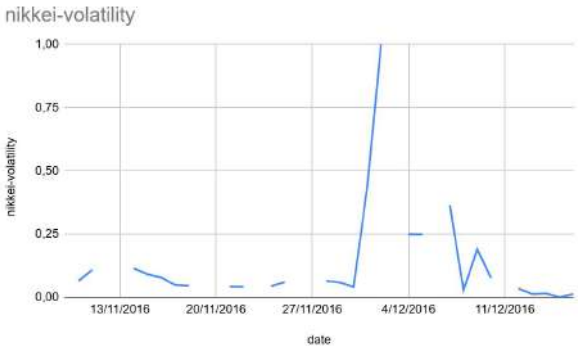


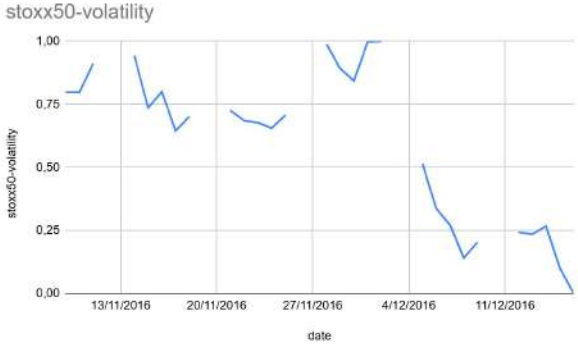


Evento 49 (9/11/2016 - 16/12/2016)

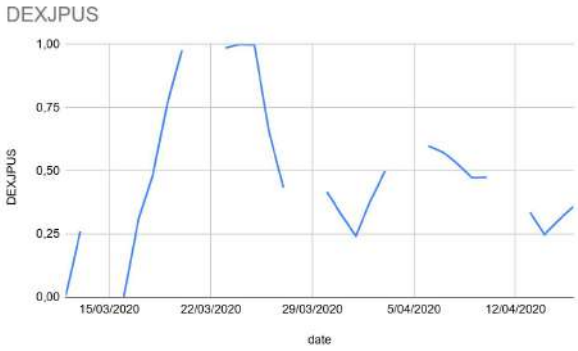


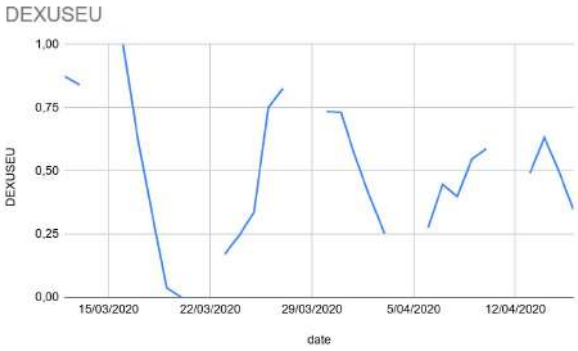


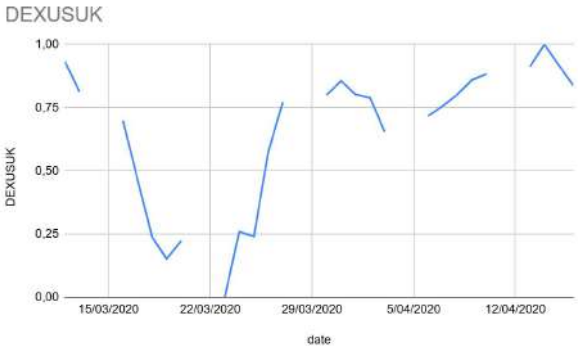


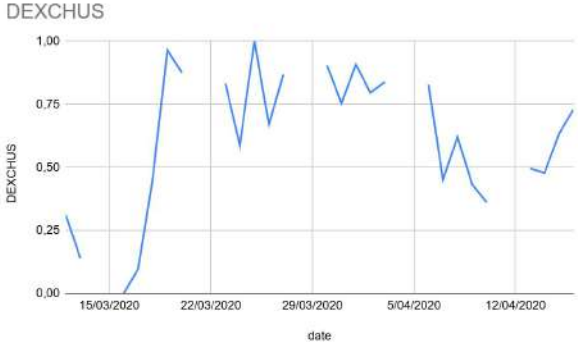


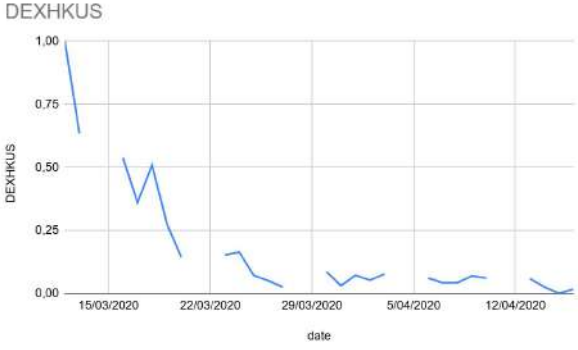
Evento 50 (12/03/2020 - 16/04/2020)

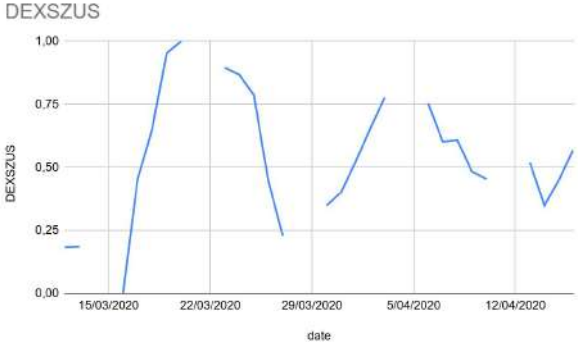




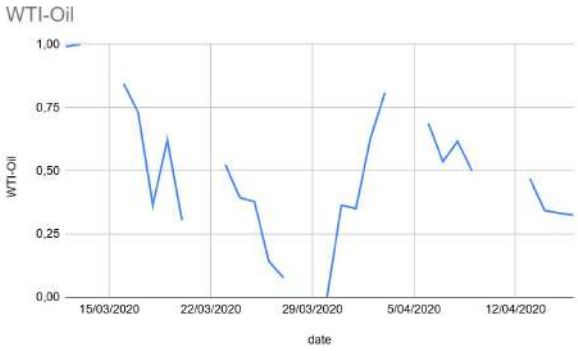


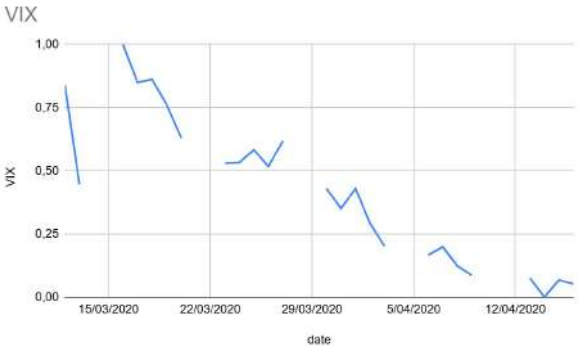


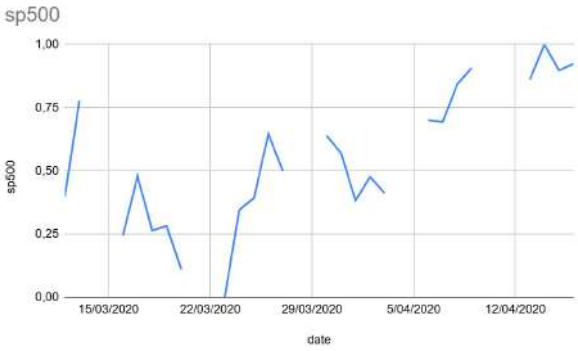


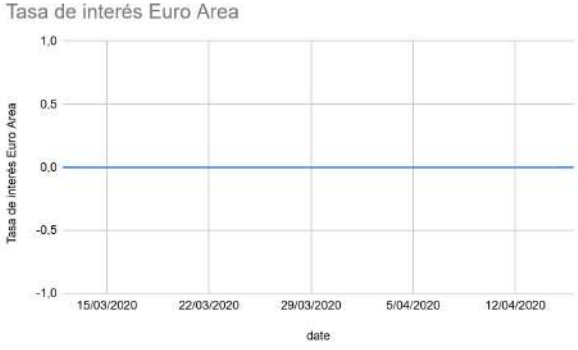


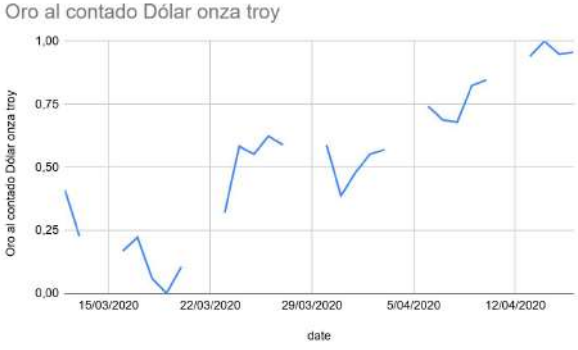
Evento 50 (12/03/2020 - 16/04/2020)

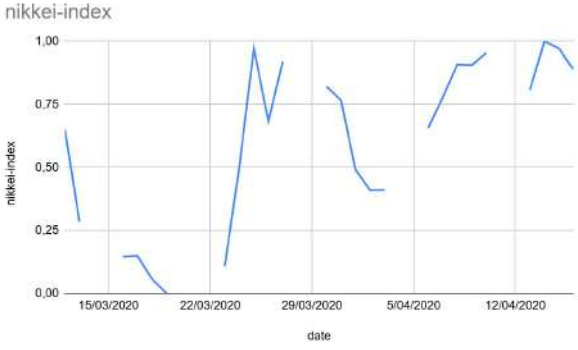




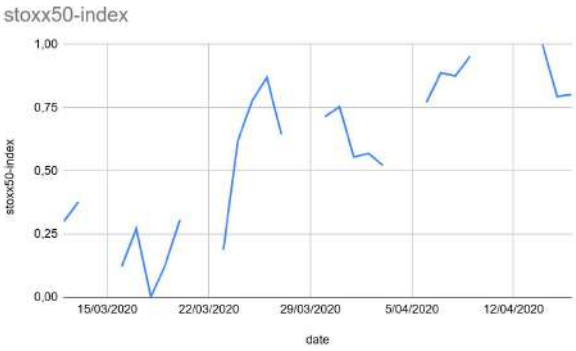


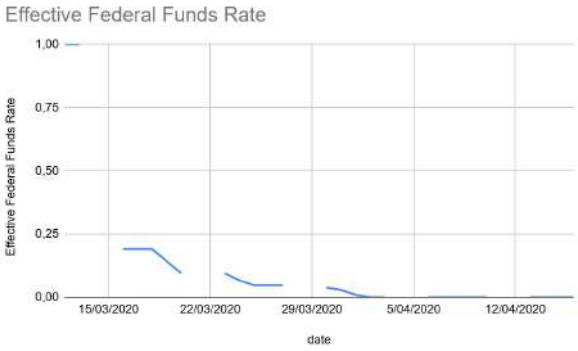


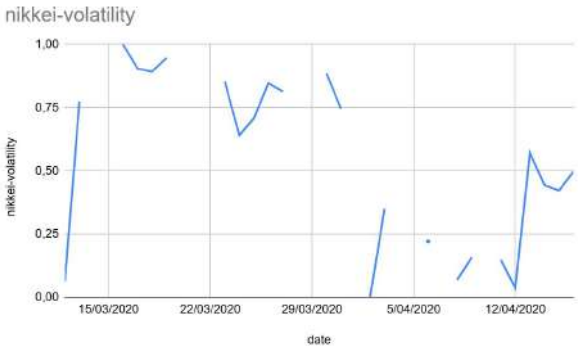


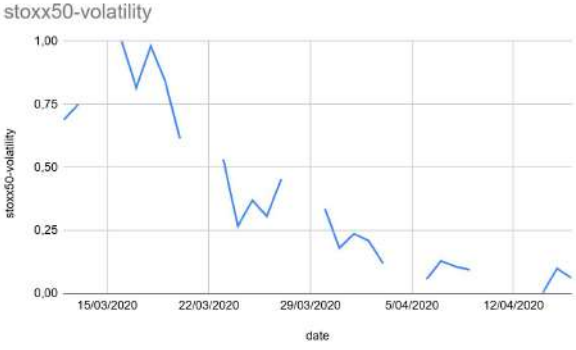


Evento 50 (12/03/2020 - 16/04/2020)









Links a los Archivos Importantes

- Estadísticos: tracker-pasantias-oct25-feb26/docs/Estadísticos.csv at main · arturos-inait/tracker-pasantias-oct25-feb26

- Eventos originales: tracker-pasantias-oct25-feb26/docs/Eventos-Originales.csv at main · arturos-inait/tracker-pasantias-oct25-feb26

- Eventos normalizados: tracker-pasantias-oct25-feb26/docs/normalizacion-individual-eventos.csv at main · arturos-inait/tracker-pasantias-oct25-feb26

Referencias

Japanese Yen to U.S. Dollar Spot Exchange Rate (DEXJPUS) | FRED | St. Louis Fed

U.S. Dollars to Euro Spot Exchange Rate (DEXUSEU) | FRED | St. Louis Fed

U.S. Dollars to U.K. Pound Sterling Spot Exchange Rate (DEXUSUK) | FRED | St. Louis Fed

Chinese Yuan Renminbi to U.S. Dollar Spot Exchange Rate (DEXCHUS) | FRED | St. Louis Fed

Hong Kong Dollars to U.S. Dollar Spot Exchange Rate (DEXHKUS) | FRED | St. Louis Fed

Swiss Francs to U.S. Dollar Spot Exchange Rate (DEXSZUS) | FRED | St. Louis Fed

Bank of England | Database

Exchange rate indices, October 2025

Cushing, OK WTI Spot Price FOB (Dollars per Barrel)

Referencias

Crude Oil Prices: West Texas Intermediate (WTI) - Cushing, Oklahoma (DCOILWTICO) | FRED | St. Louis Fed

Global price of Dubai Crude (POILDUBUSDM) | FRED | St. Louis Fed

Global price of Brent Crude (POILBREUSDM) | FRED | St. Louis Fed

Gold Spot Prices & Market History | World Gold Council

XAU USD Historical Data - Investing.com

STOXX 50 Volatility VSTOXX EUR Historical Data (V2TX) - Investing.com

CBOE Volatility Index: VIX (VIXCLS) | FRED | St. Louis Fed

Nikkei Volatility Index Today (JNIVE) - Investing.com

S&P 500 Index (SPX) - Investing.com

Índice Nikkei Volatility (JNIVE) | Cotización Nikkei Volatility - Investing.com

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SX5E 5.681,67 (▼0,63 %) EURO STOXX 50 | Google Finance

University of Michigan: Consumer Sentiment (UMCSENT) | FRED | St. Louis Fed

Effective Federal Funds Rate (EFFR) | FRED | St. Louis Fed

ECB Deposit Facility Rate for Euro Area (ECBDFR) | FRED | St. Louis Fed